

PI: Dail, Robin Britt	Title: Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants	
Received: 07/01/2024	Opportunity: PA-21-201	Council: 01/2025
Competition ID: FORMS-H	FOA Title: Research on Autism Spectrum Disorders (R01 Clinical Trial Optional)	
1R01MH138028-01A1	Dual: HD	Accession Number: 5012207
IPF: 1524302	Organization: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA	
Former Number: 1R01MH138028-01	Department: Biobehavioral Health & Nursing Science	
IRG/SRG: ZRG1 BP-D (02)M	AIDS: N	Expedited: N
<u>Subtotal Direct Costs</u> <u>(excludes consortium F&A)</u> Year 1: 499,899 Year 2: 499,842 Year 3: 499,434 Year 4: 499,892 Year 5: 499,468	Animals: N Humans: Y Clinical Trial: N Current HS Code: 30 HESC: N HFT: N Special Topics: Data Management Sharing	New Investigator: N Early Stage Investigator: N
<i>Senior/Key Personnel:</i>	<i>Organization:</i>	<i>Role Category:</i>
Robin Dail Ph.D	UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA	PD/PI
Kayla Everhart Ph.D	UNIVERSITY OF SOUTH CAROLINA	Co-Investigator
John Richards Ph.D	University of South Carolina at Columbia	Co-Investigator
Christian O'Reilly Ph.D	University of South Carolina at Columbia	Co-Investigator
Raviprasad Rao M.D.	Prisma Health Midlands	Co-Investigator
Jessica Bradshaw Ph.D	University of South Carolina at Columbia	MPI

APPLICATION FOR FEDERAL ASSISTANCE
SF 424 (R&R)

3. DATE RECEIVED BY STATE		State Application Identifier
1. TYPE OF SUBMISSION*		4.a. Federal Identifier MH138028
☐ Pre-application ☒ Application ☐ Changed/Corrected Application		b. Agency Routing Number
2. DATE SUBMITTED	Application Identifier	c. Previous Grants.gov Tracking Number
5. APPLICANT INFORMATION UEI*: J22LNTMEDP73		
Legal Name*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA Department: SPONSORED AWARDS MANAGEMENT Division: OFFICE OF THE VICE PRESIDENT FOR RESEARCH Street1*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA Street2: 1600 Hampton Street, Suite 414 City*: COLUMBIA County: RICHLAND State*: SC: South Carolina Province: Country*: USA: UNITED STATES ZIP / Postal Code*: 292080001		
Person to be contacted on matters involving this application Prefix: First Name*: Kristie Middle Name: Last Name*: Marshall Suffix: Position/Title: Sponsored Programs Administrator Street1*: 1600 Hampton Street, Suite 414 Street2: City*: Columbia County: Richland State*: SC: South Carolina Province: Country*: USA: UNITED STATES ZIP / Postal Code*: 29208-0001 Phone Number*: 803-777-5260 Fax Number: 803-777-4136 Email: kristiem@mailbox.sc.edu		
6. EMPLOYER IDENTIFICATION NUMBER (EIN) or (TIN)*		1576001153A1
7. TYPE OF APPLICANT*		H: Public/State Controlled Institution of Higher Education
Other (Specify): Small Business Organization Type ☐ Women Owned ☐ Socially and Economically Disadvantaged		
8. TYPE OF APPLICATION*		If Revision, mark appropriate box(es).
☐ New ☒ Resubmission ☐ Renewal ☐ Continuation ☐ Revision		☐ A. Increase Award ☐ B. Decrease Award ☐ C. Increase Duration ☐ D. Decrease Duration ☐ E. Other (specify) :
Is this application being submitted to other agencies?* ☐ Yes ☒ No What other Agencies?		
9. NAME OF FEDERAL AGENCY* National Institutes of Health		10. CATALOG OF FEDERAL DOMESTIC ASSISTANCE NUMBER TITLE:
11. DESCRIPTIVE TITLE OF APPLICANT'S PROJECT* Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants		
12. PROPOSED PROJECT Start Date* Ending Date* 04/01/2025 03/31/2030		13. CONGRESSIONAL DISTRICTS OF APPLICANT SC-006

14. PROJECT DIRECTOR/PRINCIPAL INVESTIGATOR CONTACT INFORMATION

Prefix: Dr. First Name*: Robin Middle Name: Britt Last Name*: Dail Suffix: Ph.D
Position/Title: PhD Director & Health Sciences Endowed Prof
Organization Name*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA
Department: Biobehavioral Health & Nursing Science
Division: College of Nursing
Street1*: 1601 Greene Street
Street2:
City*: Columbia
County: Richland
State*: SC: South Carolina
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 292084001
Phone Number*: 803-576-6187 Fax Number: Email*: rdail@mailbox.sc.edu

15. ESTIMATED PROJECT FUNDING

a. Total Federal Funds Requested* \$3,671,756.00
b. Total Non-Federal Funds* \$0.00
c. Total Federal & Non-Federal Funds* \$3,671,756.00
d. Estimated Program Income* \$0.00

16. IS APPLICATION SUBJECT TO REVIEW BY STATE EXECUTIVE ORDER 12372 PROCESS?*

a. YES ☐ THIS PREAPPLICATION/APPLICATION WAS MADE AVAILABLE TO THE STATE EXECUTIVE ORDER 12372 PROCESS FOR REVIEW ON:
DATE:
b. NO ☒ PROGRAM IS NOT COVERED BY E.O. 12372; OR
☐ PROGRAM HAS NOT BEEN SELECTED BY STATE FOR REVIEW

17. By signing this application, I certify (1) to the statements contained in the list of certifications* and (2) that the statements herein are true, complete and accurate to the best of my knowledge. I also provide the required assurances * and agree to comply with any resulting terms if I accept an award. I am aware that any false, fictitious, or fraudulent statements or claims may subject me to criminal, civil, or administrative penalties. (U.S. Code, Title 18, Section 1001)

☒ I agree*

* The list of certifications and assurances, or an Internet site where you may obtain this list, is contained in the announcement or agency specific instructions.

18. SFLLL or OTHER EXPLANATORY DOCUMENTATION

File Name:

19. AUTHORIZED REPRESENTATIVE

Prefix: First Name*: Kristie Middle Name: Last Name*: Marshall Suffix:
Position/Title*: Sponsored Programs Administrator
Organization Name*: University of South Carolina at Columbia
Department: Sponsored Awards Management
Division: Office of the Vice President for Research
Street1*: 1600 Hampton Street, Suite 414
Street2:
City*: Columbia
County: Richland
State*: SC: South Carolina
Province:
Country*: USA: UNITED STATES
ZIP / Postal Code*: 29208-0001
Phone Number*: 803-777-5260 Fax Number: 803-777-4136 Email*: kristiem@mailbox.sc.edu

Signature of Authorized Representative*

Jessica Stucker

Date Signed*

07/01/2024

20. PRE-APPLICATION File Name:**21. COVER LETTER ATTACHMENT** File Name: Dail_Cover_Letter.pdf

424 R&R and PHS-398 Specific

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Project/Performance Site Location(s)**Project/Performance Site Primary Location**

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA
UEI: J22LNTMEDP73
Street1*: UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA
Street2: 1600 Hampton Street, Suite 414
City*: COLUMBIA
County: RICHLAND
State*: SC: South Carolina
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 292080001
Project/Performance Site Congressional District*: SC-006

Project/Performance Site Location 1

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: PMG Services, Inc.
UEI: HGG1PLL88294
Street1*: 1301 Concord Terrace
Street2:
City*: Sunrise
County:
State*: FL: Florida
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 33323-0000
Project/Performance Site Congressional District*: FL-023

Project/Performance Site Location 2

☐ I am submitting an application as an individual, and not on behalf of a company, state, local or tribal government, academia, or other type of organization.

Organization Name: Prisma Health
UEI: KQ3AX9AHQY76
Street1*: 3 Richland Medical Park
Street2:
City*: Columbia
County: Richl
State*: SC: South Carolina
Province:
Country*: USA: UNITED STATES
Zip / Postal Code*: 29203-0000
Project/Performance Site Congressional District*: SC-006

Additional Location(s) File Name:

RESEARCH & RELATED Other Project Information**1. Are Human Subjects Involved?*** ☒ Yes ☐ No

1.a. If YES to Human Subjects

Is the Project Exempt from Federal regulations? ☐ Yes ☒ NoIf YES, check appropriate exemption number: ☐ 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 7 ☐ 8If NO, is the IRB review Pending? ☒ Yes ☐ No

IRB Approval Date:

Human Subject Assurance Number 00000404

2. Are Vertebrate Animals Used?* ☐ Yes ☒ No

2.a. If YES to Vertebrate Animals

Is the IACUC review Pending? ☐ Yes ☐ No

IACUC Approval Date:

Animal Welfare Assurance Number

3. Is proprietary/privileged information included in the application?* ☐ Yes ☒ No**4.a. Does this project have an actual or potential impact - positive or negative - on the environment?*** ☐ Yes ☒ No

4.b. If yes, please explain:

4.c. If this project has an actual or potential impact on the environment, has an exemption been authorized or an environmental assessment (EA) or environmental impact statement (EIS) been performed? ☐ Yes ☐ No

4.d. If yes, please explain:

5. Is the research performance site designated, or eligible to be designated, as a historic place?* ☐ Yes ☒ No

5.a. If yes, please explain:

6. Does this project involve activities outside the United States or partnership with international collaborators?* ☐ Yes ☒ No

6.a. If yes, identify countries:

6.b. Optional Explanation:

	Filename
7. Project Summary/Abstract*	Dail_Project_Summary_Final.pdf
8. Project Narrative*	Dail_Project_Narrative_Final.pdf
9. Bibliography & References Cited	Dail_References.pdf
10.Facilities & Other Resources	Dail_Facilities.pdf
11.Equipment	Dail_Equipment.pdf

Project Summary/Abstract

Autonomic nervous system (ANS) development begins in the prenatal period and is critical for a successful transition to extrauterine life. ANS supports vital functions, like cardiac activity and thermal regulation, as well as higher-level cortical processes from birth through adulthood. Adverse neonatal conditions can impair or alter ANS function and ANS dysfunction has been associated with several neurodevelopmental impairments (NDIs), including autism spectrum disorder (ASD). The rate of ASD in infants born preterm is more than triple that of the general population, with reports of a 10-fold increase for very preterm (VPT) infants. Yet, the etiological link between preterm birth and the development of ASD remains unknown. The goals of this prospective, longitudinal study are 1) to evaluate preterm ANS dysfunction as an ASD-specific etiological mechanism in VPT infants, using complementary measures of heart rate and thermal gradients 2) to examine whether and how neonatal morbidities (severe brain injury, infection, bronchopulmonary dysplasia, retinopathy of prematurity) moderate the association between ANS dysfunction and ASD/NDI outcomes and 3) to comprehensively document emerging ASD and NDI features across the first three years of life for VPT infants, who are at increased likelihood for both ASD and NDI. This study will quantify ANS dysfunction for VPT infants across the NICU hospitalization to compare with the emergence of ASD-specific and NDI features across the first three years (months 12, 24, 36). Identification of infants who are at heightened developmental risk prior to NICU discharge is essential to guide very early intervention efforts and improve childhood outcomes.

Project Narrative

The current study will determine the role of autonomic nervous system dysfunction during NICU hospitalization of very preterm infants in the later emergence of autism spectrum disorder (ASD) and neurodevelopmental impairment (NDI). This work will inform models for earlier identification of ASD and NDI within the first year of life in very preterm infants to identify children most in need of very early intervention, for improvement in long-term outcomes.

Facilities & Other Resources

The University of South Carolina (USC) is ranked by the Carnegie Foundation as being in the highest tier of research institutions (“very high research activity”). MPI Dr. Dail is the Health Sciences Endowed Professor at the University of South Carolina (USC), College of Nursing (CON) and MPI Dr. Bradshaw is an Associate Professor in Psychology, College of Arts in Sciences. USC and each College carry a large number of supports in place to ensure the success of highly productive investigators. In addition, significant resources are available via the MPIs’ positions as founding director of the Perinatal, Pediatrics, and Family Research Center (MPI Dail), Carolina Autism and Neurodevelopment (CAN) Research Center executive committee (MPI Bradshaw), and Institute for Mind and Brain faculty affiliates. The resources to complete this project are substantial and contribute directly to the feasibility of completing the research goals of this proposal. These resources are outlined below regarding Scientific Environment, Recruitment/Enrollment Site Resources, Institutional Investment, and Facilities, College Facilities & Resources, and University Resources.

SCIENTIFIC ENVIRONMENT

Dail Research Lab, PI: Dail

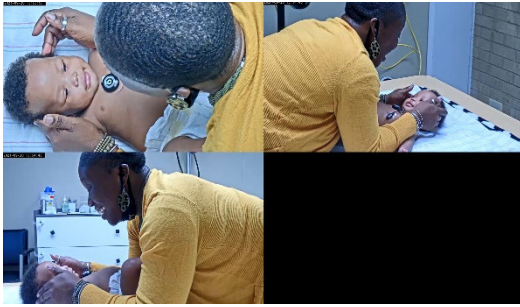
MPI Dr. Dail maintains a dedicated research lab (Dail Research Lab) on the 6th floor of the College of Nursing. The lab is equipped with three computer stations, each with dual monitors. The research coordinator space for Dail’s present R01 study (R01NR017872) with active funding (09/26/2018-06/30/2024) is located in the research lab with computer and dual monitors, equipment storage, and an instrument testing area. The lab also has conferencing space with video projection, microphone, and speakers to use for study meetings and education for research personnel in Dr. Dail’s and colleagues’ studies. Dr. Dail has research funds from start-up funds and industry studies to facilitate education of doctoral, masters, and undergraduate students conducting research projects with Dr. Dail. Several students are working on Dr. Dail’s research studies and the lab provides a rich environment to experience all aspects of study setup and methods, instrument testing, study team meetings, and data analysis. Dr. Everhart, Co-I, is conducting research out of the Dail research as well, and currently is conducting a randomized clinical trial at the Prisma NICU (study site for proposed study). The Project Coordinator for the proposed study will work out of the Dail Research Lab and will have dedicated space with a desk, computer and monitors with two locking file cabinets.

The Early Social Development Lab, PI: Bradshaw

MPI Dr. Bradshaw’s Early Social Development Lab is located in the **Institute for Mind and Brain (IMB)**, an interdisciplinary institute based in the Psychology Department focused on understanding processes in the human brain, health, and disease. Dr. Bradshaw is an invited faculty affiliate of the IMB. The IMB was created with a \$1 million renovation to a historic 13,675 square foot building on campus. This space houses approximately 5 faculty members from the Department of Psychology, along with their staff, postdocs, students, and equipment. The IMB has several meeting rooms equipped with large LCD TV screens for presentations and training purposes and offers airy, appealing spaces to facilitate discussions and promote collaborations across disciplines. The IMB hosts a monthly colloquium, which attracts internationally renowned psychologists and cognitive scientists. There is also free, easily accessible parking, ideal for families and infants with strollers.



Dr. Bradshaw has five experimental testing rooms within the IMB in addition to a personal office and 9 staff/student offices with 18 total workstations for undergraduate students, graduate students, research staff, postdocs, and research scientists. Ample room for data storage is also available. The primary experimental room is equipped with a **Noldus mounted 3-angle camera system** and acoustic treatment. It is adjacent to a control



room that houses the video recording hardware and software. The second experimental testing room includes an **SR Eye Link Eye Tracker** and **six telemetry-based ECG waveform recorders (CamnTech ActiHeart)** outfitted for data collection with infants and toddlers. This equipment is accompanied by eye-tracking and heart-rate data analysis software, including SR Experiment Builder, SR DataViewer, EPrime, and CardioEdit. There is also a family-friendly waiting area that contains comfortable, family-friendly seating, a changing table, and a variety of toys.

In addition, lab space includes tables/chairs for workstations, infant and child-friendly tables/chairs for assessments, clinical assessment kits and protocols, 10 computers, three printers, locked filing cabinets, and bookshelves. Research assistant and student workstations are connected via high-speed local area network (100MB) and interface with the University's internet connection. Other equipment includes multiple external hard drives, five video cameras with tripods for remote data collection, and clinical assessments and protocols (NICU Network Neurobehavioral Scales (NNNS), Test of Infant Motor Performance, Carey Temperament Scales, Communication and Symbolic Behavior Scales, Preschool Language Scale, Language Use Inventory, Autism Diagnostic Observation Schedule, Bayley Scales of Infant Development, Mullen Scales of Early Learning, Differential Ability Scales). Dr. Bradshaw's personal office is equipped with a personal computer, printer/scanner, locked filing cabinet, bookshelves, and a table with chairs to meet with collaborators and staff. Computer software includes Microsoft Office, Datavyu, SPSS, SAS, and Matlab. Of note, Dr. Bradshaw has significant institutional, discretionary funds to continue building her already well-equipped laboratory space.

As an Associate Professor in the Department of Psychology at the USC in Columbia, the PI has an extremely supportive scientific environment in which to conduct research. One of the primary resources is a strong and extensive network of collaborators and colleagues within the department and across the state of South Carolina. In addition, Dr. Bradshaw has developed strong collaborations with all co-investigators on this R01 proposal.

Secure Data Storage. Keyed entry is required to enter the PI's lab. Three large, five-drawer locked cabinets are available to store assessment protocols in a dedicated administrative office that requires additional lock and key access. The lab also has 100 TB of available space on a secure networked database that only research personnel can access via password. This space will be available to store and back-up video recordings so that no hard copies of the videos (e.g., SD cards) will be created or stored. Data are also routinely backed up on a secure network.

Undergraduate and graduate paid and unpaid research assistantships are also a significant source of scientific support for research programs at the USC. Significant university research infrastructure (described in more detail below) provides several mechanisms by which undergraduate and graduate-level students can contribute to research. Relevant to the current proposal, Dr. Bradshaw has recruited four clinical and school psychology doctoral students and over 20 undergraduate research assistants who are currently working in her lab. The doctoral students are highly clinically trained (ADOS-2, NNNS research reliability) and contribute significantly to ongoing research operations. Graduate and undergraduate students will contribute to behavioral coding of attention and the heart-defined attention processing pipeline, as well as to the development of co-authored manuscripts and conference presentations.

The O'Reilly Lab at the Artificial Intelligence Institute (AIISC), PI: O'Reilly

Co-I, Christian O'Reilly, is an Assistant Professor in Computer Science and Engineering and the Artificial Intelligence Institute (AIISC). To bolster collaboration and interactions across students and faculty, faculty of the AIISC share their space and infrastructure in common "laboratories" rather than having separate physical space for their laboratories. As such, the equipment available for Co-I Dr. O'Reilly are as listed for the AIISC. The Co-I

has a Mac OS computer networked to the USC system for word-processing, imaging/graphics, bibliographical and statistical software, plus wireless internet access to an extensive collection of electronic databases and journals, via the entire USC Library system. This computer is set up for complex LFP analyses using Python and its rich scientific (Numpy, Scipy, Scikit-learn), statistical (StatsModels, Pandas, XArray, rpy2, Pyro), graphic (Matplotlib, Seaborn), AI/ML (PyTorch, Keras, TensorFlow), neuroimaging (NiBabel, MNE-Python), and computational neuroscience (Brian2, LFPy) software stack. For more intensive applications, the Co-I has a 24/7 access to high-performance computing (HPC) and associate technical support in the AIISC. This HPC infrastructure includes:

- Multiple GPU clusters for deep learning and GPU intensive computations: (a) eight A100 NVidia "Multi-Instance GPU" (usable as 56 GPUs) with 1TB of RAM; Performance: five petaFLOPS AI, 10 petaOPS INT8; (b) Nvidia V100 GPU with 32GB memory and 6,000 GPU cores for compute-intensive calculations and three M-6000 GPUs each with 24 GB memory and 3,072 GPU cores for computational research. Reserved budget is available for the procurement (if and when needed) of an additional dedicated node on this cluster for data-intensive analyses being performed by members of Dr. O'Reilly's Laboratory.
- High-Performance Computing (HPC) cluster, Hyperion, which includes 6,760 compute cores that include powerful Graphical Processing Unit (GPU) and Big Data nodes connected by a very high-speed 100 Gigabit/second network.
- The OpenPOWER server powered by an IBM Power-8 CPU supporting up to 160 concurrent threads. Additionally, high-end GPUs communicate with the CPU over high-speed NV-LINK connections that significantly increases throughput for GPU-intensive application.

RECRUITMENT/ENROLLMENT: NICU Study Site

Prisma Health Richland's Children's Hospital Neonatal Intensive Care Unit (NICU). Dr. Ravi Rao (Co-I on the proposed study) is the Medical Director and Neonatologist at Prisma Health, and has pledged significant support for this project (see *Letters of Support*). Dr. Rao is employed by Pediatrix, a physician-led health care organization that partners with hospitals to offer clinical services (www.pediatrix.com). The Birthplace at Prisma Health Richland features a 70-bed level III neonatal intensive care unit which is designed for care of infants of all gestational ages. This unit is the Midlands regional perinatal center, caring for infants from across the 16-county Midlands Perinatal Region. Prisma Health is the largest and most comprehensive integrated health care system in the South Carolina Midlands region and serves 15 counties in South Carolina. The system is made up of Prisma Health Baptist (413 beds), Prisma Parkridge (76 beds), Prisma Health Children's Hospital (163 beds at Prisma Health Richland), Prisma Heart Hospital (124 beds at Prisma Health Richland), Prisma Health Richland (649 beds), and Prisma Health Tuomey (301 beds). This not-for-profit system is locally owned and consists of six Joint Commission-accredited acute-care hospitals. Prisma Health is supported by a network of physicians and advanced practice providers in the Prisma Health-USC Medical Group and the Prisma Health Quality Collaborative, as well as two 501(c)(3) foundations. Annually, Prisma Health provides care for more than one million patients and has collectively over 7,000 infant births. In addition to its seven acute care hospitals, Prisma Health has an expansive physician network and dozens of affiliated clinics and specialty care practices. Prisma Health is a teaching hospital with 24 residency and fellowship programs affiliated with the USC School of Medicine. The mission of the **Prisma Health Research Division** is to improve the health and quality of life of all the individuals served by providing patients with a supportive and safe environment to participate in cutting edge research; researchers with a framework of quality, compliant, and ethical services for the development, coordination, and oversight of their research; and sponsors the assurance that Prisma Health is a partner in research. The Research Division is composed of a clinical trials division, institutional review board (IRB) administration, research compliance division, and resident research division. The Research Division hosts bi-monthly meetings for those interested in clinical research, including investigators, research staff professionals, and IRB members. Prisma Health Research Division has full accreditation by the Association for the Accreditation of Human Research Protection Programs, Inc. The Clinical Trials department provides project management and centralized budget, planning, and contract processes. The department provides seamless scheduling, case report form completion, and submissions to the local IRB. The department also provides services ranging from multi-center study coordination to assistance with setting up data management systems and patient recruitment.

The Research Compliance department coordinates research contracting and subaward review, establishes appropriate research billing and coding mechanisms, provides biostatistical support for protocol development and data analysis, offers grant support services, administers periodic audits to those conducting research, and administers the Prisma Health Administrative Review (PHARR) process. This process provides a standard method for research compliance issues to be assessed. Items to be reviewed include conflict of interests, scientific validity of projects, research/subaward contracts, and financial feasibility.

INSTITUTIONAL INVESTMENT

The University, College of Arts and Sciences, and Department of Psychology have demonstrated **a great deal of investment in the MPI's success** through a variety of resources spanning resources, career enrichment programs, formalized department mentorship, logistical support, and financial support. Each new faculty member in the department is provided a formal mentor to assist with starting their research program and demystifying administrative processes within the department. In regard to financial support, the startup package for Dr. Bradshaw was quite generous and the **MPI recently received an institutional funding package that more than quadrupled the size of her original startup**. These funds represent significant institutional investment that can support the science of the proposed project. This investment will allow the MPI to support graduate students, purchase equipment, and travel to scientific meetings to develop collaborations. In addition, the Department of Psychology is very supportive of protected research time for all faculty, especially Early Stage Investigators. There are a number of departmental **Research Incentives** designed to decrease teaching and increase support for grant-funded faculty. Investigators with grant funding in our department routinely teach only 1-2 classes *per year*. Thus, the PI will have significant protected research time. The Psychology Department has significant administrative resources that assist with finance management, travel planning, grants pre- and post-award. The MPI has unlimited access to these resources to assist with the proposed project.

USC Psychology Students in the Undergraduate Program and Clinical-Community Doctoral Program. Another valuable resource to the PI through the department is the availability of undergraduate and graduate students. With nearly 1,000 undergraduate psychology majors at the USC, a large number of students seek out research experiences with faculty holding interests common to theirs. The PI routinely receives dozens of applications (>40) after each advertisement for research assistants. The PI and lab staff currently oversee ~20 undergraduate students who assist with data collection, data management, behavioral coding, literature reviews, and general administrative tasks. With regard to graduate students, the department provides a full stipend (\$18,250 academic year) and covers tuition (\$13,650) for every graduate student in the program. In addition, the department prioritizes Early Stage Investigators for selection of students and funding packages. This results in very high-qualified candidates and the ability for faculty to be selective in admission selections to enroll the highest qualified applicants with interests lined up to respective faculty. The department uses a mentorship model and has a primary goal of preparing students in research and only accepts Ph.D. level students. The PI currently has four students funded under an **NIH T32 training program**.

The University, College of Nursing, and Department of Biobehavioral Health & Nursing Science have supported MPI Dail's success since she arrived in 2017, after spending 10 years at Duke University School of Nursing. Dr. Dail received generous start-up funding and was provided research space in the College of Nursing in 2018, upon securing NIH grant (R01NR017872) to study premature infants. In 2022, **Dr. Dail was supported to start the Perinatal, Pediatrics, & Family Research Center to grow maternal/child research** within the College of Nursing faculty and across the USC campus. Dean Jeannette Andrews offers generous support to all new Faculty Researchers, through start up packages and release time for research. All faculty are assigned a research mentor and enjoy support through the Office of Research in the College for Mock Grant Reviews prior to grant submissions. Research faculty receive ongoing release time of 40% effort, which reduces teaching to 1-2 courses per year and supports research development. Research faculty receive a portion of their grant indirect funds returned to incentive accounts which can be used to purchase research equipment, hire graduate research assistants, and support their ongoing development. There are other **Research Incentives** provided by the College of Nursing for faculty, upon successful research funding and dissemination. MPI Dail was awarded the **Health Sciences Endowed Professor** in 2022, which also offers research support and incentives. Dr. Dail is a

senior mentor of faculty and students, and her position within the College of Nursing is devoted to Research, Directing the PhD Program, and Mentoring.

MPI Dr. Dail is the Director of the Nursing Science PhD Program, which draws applications from potential students across the nation. Students are supported with support for tuition for their core courses, through a Dean's Scholarship. The PhD program has 30-35 students, with 10-12 admitted annually. Each student works with a Faculty Researcher throughout their program, and during the first two years, work with their mentors as graduate research assistants. Currently, MPI Dail is working with three PhD students, and has graduated 2 former students in the last 3 years, one of which is Dr. Kayla Everhart (Co-I) who was hired as faculty upon her graduation in 2020. Co-I Everhart was MPI Dail's Research Coordinator on Dail's NIH funded grant, then upon graduation became a Co-I on NIH funded R01NR017872. Dr. Everhart has worked with Drs. Dail & Bradshaw on infant research for the last 2 years. Dr. Dail currently is the Sponsor for her PhD student, Karen Warren's F31 funded dissertation, which is being conducted in the proposed study site NICU at Prisma Health. The College of Nursing and the PhD Nursing Science program is committed to supporting collaborative research with PhD students and mentoring the next Nurse Scientists to support improving health outcomes for generations to come.

Commitment to Supporting Investigators by the Office of the Vice President for Research. This office at the USC plays a primary role in strategic development of the research infrastructure of the University in order to help build the institution's existing research base and provide support and assistance for faculty and students conducting research throughout the University. There is a particular emphasis on Early-Stage Investigators. The office supports faculty in securing funding for their research and scholarly activities. The Office enhances the teaching and learning environment at the University by integrating research into the educational experience of undergraduate and graduate students, ensuring the integrity of research, providing assurance to governmental and private funding agencies and to the public that USC research is conducted in accordance with the law and the highest ethical standards, and promoting interdisciplinary research as an important component of economic development in the Columbia region and across the state of South Carolina. Specifically, the office offers:

- Grantsmanship and research coursework for faculty, such as the "R01's and Beyond" and "National Institutes of Health Peer Review" series.
- Propel Research Mentorship Program-provides new and early stage faculty researchers nine months of intensive mentorship, education and editing support as the plan, draft and finalize a federal grant submission. MPI Dail is a mentor for the Propel program.
- Several research support offices including: Sponsored Awards Management (pre-award), Office of Research Compliance, Information and Technology Data Management, and Office of Undergraduate Research (centralized resource for engaging undergraduates in research)
- Research Cyberinfrastructure works with researchers to improve project performance and secure computing resources.
- Advanced Support Programs for Innovative Research Excellence (ASPIRE) Funding Program allows faculty the opportunity to compete for funding to begin a promising new research endeavor for later external funding, build up a collaborative, interdisciplinary research project, and/or invest in the university's research infrastructure. Awards are given annually and range from \$15,000- \$100,000.
- Breakthrough Faculty Awards recognize relatively early-career assistant and associate professors who demonstrate considerable contributions to their field in terms of research and scholarly activity while at USC. This recognition can contribute to recipients' career advancement and retention and helps the university attract the best and brightest young faculty to join USC.
- Governor's Awards for Excellence in Science is an internal selection process to nominate a candidate for the Governor's Awards for Excellence in Science, helping to ensure that USC is as competitive as possible in this prestigious statewide awards program.
- The Research Initiative for Summer Engagement, or RISE, program supports summer research for faculty members at USC. This program employs a competitive application process to provide money for summer salary, research supplies, travel related to research, and undergraduate student support to bolster scholarship throughout USC.
- Support for Minority Advancement in Research Training aims to enhance the training of under-represented minority students by providing funds for USC faculty to mentor minority undergraduate students through summer research projects.

Perinatal, Pediatrics, & Family Research Center (PPFRC) MPI Dail is the **founding director** of the Perinatal, Pediatrics, and Family Research Center (PPFRC), University of South Carolina, College of Nursing. The Mission of the PPFRC is to improve health and wellbeing for families in SC and beyond through research, innovation, leadership and excellence through collaborative, interdisciplinary research by the members of the PPFRC. The main Goals for the PPFRC are to 1) Decrease morbidity and mortality for women in SC and beyond, before and after childbirth, 2) Decrease morbidity and mortality for premature infants and children born in SC from birth to 2 years old, 3) Facilitate the health and wellbeing of families in SC during the child-bearing years with increased access to equitable, quality healthcare through research. Currently, the center has 11 Core Faculty members (Drs. Dail & Everhart (Co-I) are Core members) from the USC College of Nursing, 2 Associate Faculty members, including Dr. Bradshaw (MPI) from the USC Campus, and 10 Affiliate Members from across the United States, as well as 4-8 student affiliates. The PPFRC sponsors faculty speakers across the nation to present their research each month to the PPFRC. These efforts are designed to expand group research knowledge and expertise and to foster research collaboration amongst group members.

The Carolina Autism and Neurodevelopment (CAN) Research Center. The CAN Research Center is a \$4.8 million initiative, funded by the USC Office of the Provost, established to promote research on neurodevelopmental disorders from 2019-2024. The CAN Research Center, led by co-founders Dr. Jane Roberts and Dr. Jeff Twiss, brings together interdisciplinary faculty members from more than a dozen disciplines across campus to collaborate and advance basic, clinical, and intervention research and education in of autism and neurodevelopment. CAN awarded pilot funding to MPIs Dail/Bradshaw and Co-I O'Reilly, which produced the preliminary data to support the proposed project and is thus very supportive of our research team and this grant. The investigative team is heavily involved in CAN with access to significant infrastructure including pilot grant funds, the neurodevelopment research database, interdisciplinary faculty with expertise in neurodevelopmental disorders, quarterly seminar series, and structured meetings to foster collaboration, as detailed below:

sPromotion of Grant Development. CAN established a pilot grant program designed to cultivate interdisciplinary research and precipitate consequential interactions between research fields. Each year, two pilot grants, funded at \$40,000, are awarded following an NIH-style grant review by a grant review committee. Furthermore, CAN hosts a monthly peer grant review workshop, during which faculty members working on grant proposals share their work with CAN members to solicit feedback.

Visiting Scientist Seminar Series. Each year, the CAN Research Center hosts a Visiting Scientist Seminar Series to bring together scholars across the country to share their research with the USC community. The seminars are promoted as community events, and practitioners and policymakers from across the state were invited with great attendance. The seminars also highlight the important work of the university to the broader community and serves their needs through professional development (e.g., offering attendees CEU's for selected events). During their time at USC, in addition to leading a seminar, visiting scholars meet one-on-one with interested CAN faculty and trainees, as well as participate in group discussions (e.g., "Biomarkers, Screening, and the Path to Intervention"). Recent visiting scholars include Dr. Lawrence Scahill, who spoke on "Measuring Anxiety in Youth with Autism Spectrum Disorder," and Dr. Diana Robins, who gave a seminar entitled "Early Detection of Autism."

Research Engagement. To foster the burgeoning autism and neurodevelopment research community at USC, the CAN Research Center hosts several endeavors in both formal and informal settings to facilitate research discussion and to share ongoing work CAN faculty and trainees.

- **Annual Retreat.** Each August, CAN hosts a full-day retreat for all affiliated faculty and trainees. At this retreat, CAN goals, objectives, and plans for the upcoming academic year are discussed. Small-group and large-group brainstorming sessions are used to generate new strategies for research engagement, community engagement, and trainee opportunities
- **Research Roundup.** CAN hosts an annual Research Roundup, a half-day event held at the Institute for Mind and Brain, to facilitate networking and disseminate ongoing research efforts at all academic levels. Previous roundup features included flash talks from CAN pilot award recipients, as well as students and trainees, and a science cluster conversation hour to connect researchers with similar interests and support collaborative efforts.
- **Science Meetings.** Once per month, CAN hosts a Science Meeting, during which time two CAN faculty members or trainees presents on their research. Presentations at each meeting are

thematically linked (e.g., Autism treatment; Down syndrome) to encourage translational and interdisciplinary connections between scientists.

- **First Thursdays with CAN.** CAN hosts a social gathering on the first Thursday of every month at the Graduate Hotel, located a 5-minute walk from the USC campus. The First Thursday event, first held in spring of 2022 brings together faculty, trainees, and research staff from across the university to network and chat about neuroscience and academia in an informal setting.

Community Outreach and Engagement. The CAN Research Center, in collaboration with its affiliated labs, hosts events and workshops to engage the community, promote the work of CAN, and recruit individuals to the CAN research registry.

- **Outreach Events.** CAN engages with local organizations and businesses to setup CAN-sponsored tables/booths which provide faculty the opportunity to share ongoing research efforts at USC and register families for a research listserv. CAN has attended events such as (1) Soda City Market, a weekly free market in Columbia attended by ~5,000 each week, (2) the Columbia Fireflies Autism Awareness Night, (3) the SC State Museum Sensory Friendly Morning, and (4) the Family Connections Buddy Walk. CAN also independently hosts an annual free “Autism’s Got Talent” show, during which participants have three minutes to show off their special skill, ability, achievement, or creative expression. Further, CAN will host two outreach events per year, with the inaugural event being an Autism Family Fair at EdVenture Children’s Museum. This free, sensory-friendly event, featuring a juggler, magician, balloon twister, popcorn, snowballs, provided children with autism and special needs the opportunity to explore EdVenture in a more adapted environment, while families were able to engage with autism service providers and research groups. Approximately 500 people attended this event, allowing for direct engagement between USC neuroscience researchers and the autism community in a relaxed environment.
- **Community Services.** The ESD Lab (led by MPI Bradshaw) established a free Mindfulness Group for parents of children with autism. Every week, for six weeks, the mindfulness group meets, incorporating mindfulness, Acceptance and Commitment Therapy, and self-compassion practices to support parents’ coping and wellbeing.
- **The CAN Research Registry** is a database of contact information of families with children with special needs and typically developing children that is accessible to faculty who are conducting research in the area of autism and neurodevelopment. Families in this database have expressed interest in being contacted about research studies and receive periodic updates about new study openings within CAN. This database currently includes over 2,000 families, with approximately 5-10 families being added each week. The aforementioned events, as well as ongoing collaboration with local pediatricians, OBGYNs, and community partners, have been instrumental in recruiting new families for the research registry.

CAN Trainee Committee. A CAN Trainee Committee was established in Fall 2022, led by a group of four students across departments and colleges, to increase participation of USC postdoctoral fellows, graduate and undergraduate researchers, and research staff in building our community. Initiatives set forth by the CAN Trainee Committee include informal lab rotations for trainees, semi-structured social events with principal investigators for networking, and panels between trainees and undergraduate students.

COLLEGE FACILITIES & RESOURCES

The College of Nursing

MPI Dail’s and Co-I Everhart’s home department is in the College of Nursing is located on USC’s Columbia campus, housed in the Williams-Brice building, a six-story facility that has been remodeled over time to maintain state-of-the-art technology to support the College’s mission. The College is located next to the College of Liberal Arts and is in close proximity to other colleges on the main campus and the library. The College of Nursing is one of five health-related colleges including pharmacy, social work, public health, and medicine. The faculty has a philosophical commitment to collaborate with other health disciplines. This commitment has been fulfilled through interdisciplinary collaboration in health planning, grant writing, delivering services, and developing educational programs. The College of Nursing offers both graduate and undergraduate programs. Graduate programs include two doctoral programs, the Doctor of Nursing Practice (DNP) degree and the Doctor of Philosophy in Nursing Science degree, and a Master of Science in Nursing (MSN) degree. The College offers a

four-year undergraduate program that leads to the Bachelor of Science in Nursing (BSN) degree with an RN/BSN option. A combined BSN/MSN Education Plan is available for both BSN and RN/BSN.

College of Nursing Research Resources. The College of Nursing has significant infrastructure to support research operations. There are four video conferencing rooms, three Polycom rooms and one setup for a variety of web conferencing protocols. This technology affords the ability to offer a wide range of video and data collaboration functions. Web collaboration is possible through the use of Adobe Connect and Gotomeeting which can provide unlimited on-demand web meetings. Faculty and staff can create virtual conference rooms from any web-connected location. The College also provides the Technology Resource Center (TRC) for faculty and student use. The TRC provides 52 general use computer workstations and provides AV and portable computer resources to the faculty. The TRC is equipped with a laser printer, nursing computer applications, web access, statistical software packages, including SAS (9.4) and SPSS, and access to RedCap database. The College of Nursing Office of Research (OR) is under the leadership of an Associate Dean for Research and staffed with two program coordinators, a biostatistician, and graduate assistants. The OR assists faculty with research proposal development and submission including budget preparation. The OR also provides assistance to faculty with dissemination activities such as abstract, manuscript, and poster preparation and provides information on funding opportunities. A full-time doctorally prepared statistician and part-time graduate assistants augment the research capabilities of faculty and graduate students who are conducting research, preparing proposals, and/or developing manuscripts. Faculty have access to retrieval systems containing national census and health data within the University computer system. A variety of packaged computer software (statistical, graphics, word processing, and spreadsheets) are available for faculty use. The campus also provides access to several supercomputer cores for major processing applications.

The College of Arts and Sciences

MPI Bradshaw's home department of Psychology is housed within the College of Arts and Sciences. The Psychology department is located in the Barnwell and Hamilton buildings which are five stories and 110,000 square feet. The graduate psychology programs offer both the Master of Arts and the Doctor of Philosophy degree in one of four emphases: experimental psychology, quantitative psychology, clinical-community psychology, and school psychology. In addition to the four graduate degree programs, the Department of Psychology has developed research focus areas in Developmental Cognitive Neuroscience, Prevention Science (youth, family, school, and community domains), and Child and Family Outcomes. Students in any of the graduate degree programs can interface with any of the research areas. The Department of Psychology Information Technology Services (IT Services) provides complete file and print services, internet connectivity, electronic mail, webmail, and web publishing support to the Early Social Development and Intervention Lab. The IT staff maintains all computer, web, and network services and supports a standard suite of software, as well as advanced statistical software, database management, web publishing, and print publishing.

The Psychological Services Center (PSC) is a service, training, and research facility of the Psychology Department at the USC. The PSC is a nonprofit facility established in 1965 in order to provide assessment and treatment services to the USC community and the Columbia Metropolitan area. This facility has 4,700 square feet of professional office space centrally located in downtown Columbia. The space has a large two-level parking garage, ample handicapped parking spaces, is clearly visible from the street, and is located on a city bus line to provide convenient access. Thus, the facility serves as a natural focal point for face-to-face interactions between the Department of Psychology and the local community. The space features the PSC, a Psychology Research Suite, and shared facilities for these two operations. The PSC is a 2,100 square foot outpatient mental health clinic staffed by the faculty and graduate students of the Psychology Department. In addition to serving as an outpatient mental health clinic, the PSC is used for clinical research projects. The PSC is a source of patient populations for research and also provides opportunities to evaluate assessment and intervention practices conducted within the clinic. The PSC features eight meeting rooms (~130 sq. ft. each) for individuals, couples, or families receiving services through the center, a conference room for group meetings (~230 sq. ft.), a reception area (~380 sq. ft.), and three offices for core PSC staff (~145 sq. ft. each). The Psychology Research Suite is a 750 square foot facility adjacent to the PSC which features two meeting rooms for individuals or small groups (~130 sq. ft.), as well as a conference room (~200 sq. ft.) and reception area (~300 sq. ft.). Shared facilities include a large conference room (~600 sq. ft.), an information technology and computing lab (~240 sq. ft.), and a flexible use multi-purpose room (~850 sq. ft.). All meeting and conference rooms in the center are equipped

with desktop Pentium 4 Personal Computers linked to the College of Arts and Sciences computer services. The computing lab has an additional four Pentium 4 computers, plus photocopying, scanning, and fax equipment. In addition, all conference and meeting rooms have built-in digital audio-video equipment (Axis Communications 211A network cameras with microphones) linked and secured via a dedicated server with audio-video recording software to support observational studies of behavior. There is extra capacity built into the recording system so that it can accommodate additional cameras as needed for special projects. All of the Ethernet-based equipment for the facility is connected directly to the University backbone via a T1 line.

The College of Engineering and Computing

Co-I Christian O'Reilly's home college is the College of Engineering and Computing (CEC). Based at USC's main campus in Columbia, CEC is ranked by both *U.S. News and World Report* and the National Research Council. The first Engineering classes were taught at the USC in 1848, and engineering degrees have been offered since 1894. The CEC was established in 1909 and currently has an enrollment of more than 3300 students, including 423 full-time graduate students and 2545 full-time undergraduates. The College currently employs 124 tenure, research, and clinical track faculty members. The CEC is located in the Swearingen Engineering Center, which is one of the largest engineering facilities in the Southeast. The USC College of Engineering and Computing's mission is to become a leader in technology innovation, engineering and computing education, and entrepreneurship. The CEC provides support to the Artificial Intelligence Institute of South Carolina.

Artificial Intelligence Institute (AIISC) is a university-wide AI Institute where Dr. O'Reilly is located. The AIISC is a major new initiative at USC. It is located on the top floor of the Science & Technology Building on main campus. Since its inception in 2019, AIISC has grown to around 35 faculty and graduate students in Fall 2021. This growth continues as the institute has embarked on hiring five additional core AI and five new interdisciplinary faculty members at all ranks. These faculty are complemented by existing faculty collaborators in Computer Science and Engineering department, as well as other collaborating centers and faculty members engaged in interdisciplinary research. Faculty at AIISC conduct research on a broad array of AI topics, providing a rich environment supportive of the proposed work. Further, the AIISC has identified the intersection of neuroscience and AI as a priority target for the development of the institute and, in a joint effort with the CAN Center of Excellence, has recently hired two new faculty (including Dr. O'Reilly) in that area.

UNIVERSITY RESOURCES

The University of South Carolina in Columbia is the state's flagship research university and is ranked by the Carnegie Foundation as being in the highest tier of research institutions ("very high research activity") in the United States. It is located in Columbia, the capital city and geographical center of the state, and is adjacent to the state government complex and the downtown business and cultural districts. The Greater Columbia area has a population of more than 600,000 and is easily accessible by public transportation and interstate highways. The University enrolls 16,000 undergraduate students and 8,400 graduate and professional students. It offers more than 350 major degree programs, including baccalaureate degrees in 126 areas, master's degrees in 172 areas, doctoral degrees in 74 areas, and professional doctorates in business, education, engineering, law, medicine, music, pharmacy, and public health. More than two dozen programs are nationally and internationally ranked.

USC Honors College. Additionally, the USC is home to the South Carolina Honors College, which has been ranked the nation's top public honors college since 2012. The Honors College strongly encourages student research participation and provides funds for undergraduates to engage in research (up to \$10,000 per student). This is an incredible resource for the MPIs, whose lab models incorporate undergraduate research.

South Carolina Autism and Neurodevelopmental Disorders (SCAND) Consortium is a network of professionals in various fields who are united by their mission to understand neurodevelopmental and autism disorders to provide support for families affected by these disorders throughout the state of South Carolina. The mission of SCAND is to drive research excellence in South Carolina for clinical, translational, and mechanistic

studies aimed at understanding and treating neurodevelopmental and autism spectrum disorders through a state-wide, multi-disciplinary research consortium. SCAND holds an annual research symposium, bringing in internationally recognized experts in ASD, such as Matthew State (UCSF) and Marsha Mailick (U Wisconsin-Madison). The first annual SCAND Symposium was held in 2017, with 110 people in attendance from 14 institutions and organizations, including state legislators, throughout South Carolina. MPI Dr. Bradshaw is an affiliate of this consortium.

The SmartState Center for Childhood Neurotherapeutics, supported by the Office of the Vice President for Research, is an initiative that has attracted more than \$1.4 billion in investments. The SmartState Center for Childhood Neurotherapeutics, led by endowed chairs Dr. Jeffery Twiss and Dr. Manuel Casanova, supports research on the prevention of brain damage in premature infants and curing infant brain diseases through cellular engineering. Dr. Twiss, the Associate Dean for Research, co-founded the SCAND Consortium.

The ACORN Center. The Advancing Chronic Care Outcomes through Research & iNnovation (ACORN Center) is one of South Carolina's SmartState Centers. The SmartState program was established by the state's General Assembly in 2002 with \$180 million of non-tax revenue generated from the South Carolina Education Lottery. SmartState research centers (1) expand the state's knowledge base, (2) create public-private partnerships, (3) support startup firms, and (4) help retain highly talented workers to actively support ongoing development of the knowledge economy. The ACORN Center focuses on innovative, multidisciplinary research to improve health and quality of life among people with chronic conditions. The Center is located within the College of Nursing on the USC Columbia campus and is directed by Dr. Cynthia Corbett, Professor and Endowed Chair. The Center also has a project coordinator, six additional full-time faculty members, and several undergraduate and graduate research assistants. The Center has funds to support pilot and feasibility projects, three offices with workspace, filing cabinet storage, and technology such as digital recorders, iPads, and laptops available for research, as well as space for students that are in active research collaboration with ACORN faculty.

McCausland Center for Brain Imaging, with its new Siemens Magnetom Trio (3 Tesla), Magnetic Resonance Imaging system, will provide USC faculty, as well as collaborative and externally funded researchers, the ability to perform high-resolution, total body imaging studies in Columbia, South Carolina. The McCausland Center was established with funds from the Department of Defense, Center of Economic Excellence, and South Carolina Education Lottery funds, as well as \$1.5 million gifts from both the Palmetto Richland and Greenville Hospital systems, and \$1.75 million in matching funds from Peter J. McCausland, a '71 graduate of the USC.

Statistical Consulting Laboratory (Stat Lab) provides statistical services to projects with government agencies, private industry, and members of the University community working on grants. The Stat Lab also offers free and fee-for-service statistical consulting services for faculty, staff, and students of USC.

University libraries house over 7,000,000 processed items including 2,639,171 volumes, 4,075,000 units in microform, and 17,405 current periodicals. Thomas Cooper Library contains all of the University's library collections in Columbia except those located in the South Carolina Library, Law Library, Mathematics Library, Music Library, Business Library, and Medical Library. Thomas Cooper Library seats approximately 2,500 readers. Included in the seating are more than 900 private locked facilities for faculty and graduate students involved in research. Special areas in the library include the Student Computer Labs, the Science Library, Special Collections, and the Map Library. Access to all library collections is obtained through the USCAN/NOTIS Online Card Catalog. CD ROM stations are available for user searching of multiple databases. New databases are added regularly, and information on them is available in the Reference Department. Medline Search is also available from all libraries. The USC Library system offers access to literature from international sources through the online computer-assisted reference department. The computerized reference service provides bibliographic citations, statistics, and international news from recent literature on a wide range of subjects. This service is particularly strong in the natural and social sciences. Also included are government activities, publications, and grant sources.

Office of Information Technology oversees centralized and distributed computing and telecommunications services for academic, research, and administrative use to meet the needs of USC faculty, staff, and students. The Office of Information Technology provides the campus community with computing, voice, and data communications, networking, data security, video transport, information technology training, web services,

customer support, desktop and server support, installation and maintenance of IT infrastructure, policies and procedures assistance, PC labs, software licensing and distribution, IT planning, applications development and support, and operational systems. The Columbia campus is covered by wireless service.

Office of Sponsored Awards Management (SAM) provides support to faculty and staff for the development of research grant applications and other external funding requests. SAM is responsible for the initial review of all proposals to ensure that they conform to regulations and requirements of the State, University, and sponsor. Upon receipt of an external award, SAM initiates the processes that establish an account with the University and works with the investigator(s), University administrators, and the sponsoring organization to ensure that the project complies with all grant or contractual agreements. SAM also serves as an interface between investigators and the accounting departments at the University.

Contract and Grant Accounting at the University ensures proper fiscal management and reporting of contract and grant funds. Contract and Grant Accounting works closely with SAM to coordinate pre- and post-award management of grant proposals and grants, and to facilitate access by the Principal Investigators to their grant accounts.

The Office of Research Compliance, a division of the SAM, monitors the human subjects review process through the University's IRB. The IRB has an outstanding record of compliance with federal regulations and of providing service to faculty involved in research. All proposal submissions are managed electronically through the USCeRA system.

USC's Research Foundation is an element in the University's strategy for research growth, supporting faculty efforts and helping faculty with their research goals. The primary objectives of the Research Foundation are to administer proposals and awards more efficiently, assist faculty more effectively in competing for external funding, and maximize the University's returns on its research efforts. The foundation provides greater flexibility and competitiveness in implementing research grants and contracts, building infrastructure, and developing intellectual property management responsibilities.

University Computer Services include an IBM 2066-OX2 Enterprise Server and associated storage arrays, libraries, tape drives, and optical scanners. The University's Computer Services Division provides a full range of software, training, and technical support to faculty researchers. The Gambrell Computer Lab, located in the lower level of Gambrell Hall, is the main computing facility of the College of Arts & Sciences. The facility contains four labs, consisting of a mix of PC and Macintosh computers. In addition, there are a number of peripheral devices available (scanning, printing, video importing) to students in the College. The Research Computing Center supports and manages applications on multiple hardware platforms for the advanced computing technology needs for research, academic, and administrative activities within the College of Arts and Sciences.

Equipment

No equipment will be purchased for this study.

RESEARCH & RELATED Senior/Key Person Profile (Expanded)

PROFILE - Project Director/Principal Investigator				
Prefix: Dr.	First Name*: Robin	Middle Name Britt	Last Name*: Dail	Suffix: Ph.D
Position/Title*:	PhD Director & Health Sciences Endowed Prof			
Organization Name*:	UNIVERSITY OF SOUTH CAROLINA AT COLUMBIA			
Department:	Biobehavioral Health & Nursing Science			
Division:	College of Nursing			
Street1*:	1601 Greene Street			
Street2:				
City*:	Columbia			
County:	Richland			
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292084001			
Phone Number*: 803-576-6187		Fax Number:		
E-Mail*: rdail@mailbox.sc.edu				
Credential, e.g., agency login: RKNOBEL				
Project Role*: PD/PI		Other Project Role Category:		
Degree Type: PHD,MSN		Degree Year: 2006,2000		
Attach Biographical Sketch*:		File Name:	Dail_Biosketch_Final.pdf	
Attach Current & Pending Support:		File Name:		

PROFILE - Senior/Key Person				
Prefix: Dr.	First Name*: Jessica	Middle Name	Last Name*: Bradshaw	Suffix: Ph.D
Position/Title*:	Associate Professor			
Organization Name*:	University of South Carolina at Columbia			
Department:	Psychology			
Division:	College of Arts and Sciences			
Street1*:	1800 Gervais St			
Street2:				
City*:	Columbia			
County:	Richland			
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292013504			
Phone Number*: 9094544652		Fax Number:		
E-Mail*: jbradshaw@sc.edu				
Credential, e.g., agency login: UCSBBRADSHAW				
Project Role*: PD/PI		Other Project Role Category:		
Degree Type: PHD,MA,BS		Degree Year: 2015,2012,2003		
Attach Biographical Sketch*:	File Name:	Bradshaw_Biosketch_Final.pdf		
Attach Current & Pending Support:	File Name:			

PROFILE - Senior/Key Person				
Prefix: Dr.	First Name*: Kayla	Middle Name C	Last Name*: Everhart	Suffix: Ph.D
Position/Title*:	Assistant Professor			
Organization Name*:	UNIVERSITY OF SOUTH CAROLINA			
Department:	Biobehavioral Health & Nursing Science			
Division:	College of Nursing			
Street1*:	1601 Greene Street			
Street2:				
City*:	Columbia			
County:	Richland			
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292084001			
Phone Number*: 803-777-8437		Fax Number:		
E-Mail*: everhac@email.sc.edu				
Credential, e.g., agency login: EVERHAKC				
Project Role*: Co-Investigator		Other Project Role Category:		
Degree Type: PHD,MSN,BSN		Degree Year: 2021,2017,2010		
Attach Biographical Sketch*:	File Name:	Everhart_Biosketch_Final.pdf		
Attach Current & Pending Support:	File Name:			

PROFILE - Senior/Key Person				
Prefix: Dr.	First Name*: Christian	Middle Name	Last Name*: O'Reilly	Suffix: Ph.D
Position/Title*:	Assistant Professor			
Organization Name*:	University of South Carolina at Columbia			
Department:	Computer Science and Engineering			
Division:	College of Engineering and Computing			
Street1*:	1112 Greene St.			
Street2:	Science & Technology Building			
City*:	Columbia			
County:	Richland			
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292010000			
Phone Number*: 8036266667		Fax Number:		
E-Mail*: christian.oreilly@sc.edu				
Credential, e.g., agency login: COREILLY				
Project Role*: Co-Investigator		Other Project Role Category:		
Degree Type: PhD		Degree Year: 2012		
Attach Biographical Sketch*:		File Name:	Oreilly_Biosketch_Final.pdf	
Attach Current & Pending Support:		File Name:		

PROFILE - Senior/Key Person				
Prefix: Dr.	First Name*: Raviprasad	Middle Name	Last Name*: Rao	Suffix: M.D.
Position/Title*:	Director of Neonatology			
Organization Name*:	Prisma Health Midlands			
Department:				
Division:				
Street1*:	7 Medical Park			
Street2:				
City*:	Columbia			
County:	Richland			
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	29203-6808			
Phone Number*: 803-296-5437		Fax Number:		
E-Mail*: raviprasad.rao@prismahealth.org				
Credential, e.g., agency login: RAVIPRASADRAO				
Project Role*: Co-Investigator		Other Project Role Category:		
Degree Type:		Degree Year:		
Attach Biographical Sketch*:		File Name:	Rao_Biosketch_Final.pdf	
Attach Current & Pending Support:		File Name:		

PROFILE - Senior/Key Person				
Prefix: Dr.	First Name*: John	Middle Name E	Last Name*: Richards	Suffix: Ph.D
Position/Title*:	Professor			
Organization Name*:	University of South Carolina at Columbia			
Department:	Psychology			
Division:				
Street1*:	1512 Pendleton Street			
Street2:				
City*:	COLUMBIA			
County:				
State*:	SC: South Carolina			
Province:				
Country*:	USA: UNITED STATES			
Zip / Postal Code*:	292080000			
Phone Number*:	(803) 777-2079		Fax Number:	(803) 777-9558
E-Mail*:	RICHARDS-JOHN@SC.EDU			
Credential, e.g., agency login:	jrichards			
Project Role*:	Co-Investigator		Other Project Role Category:	
Degree Type:	PHD,MA,BA		Degree Year:	1982,1978,1976
Attach Biographical Sketch*:	File Name:	Richards_Biosketch_Final.pdf		
Attach Current & Pending Support:	File Name:			

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Robin B. (Knobel) Dail PhD, RN, FAAN

eRA COMMONS USER NAME (credential, e.g., agency login): RKNOBEL

POSITION TITLE: Professor & Associate Dean for Faculty Affairs, College of Nursing

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
University of Nevada, Las Vegas, NV	ADN	12/1981	Nursing
East Carolina University, Greenville, NC	NNP	08/1994	Neonatal Nurse Practitioner
East Carolina University, Greenville, NC	MSN	12/2000	Nursing
University of North Carolina at Chapel Hill, Chapel Hill, NC	PhD	08/2006	Nursing

A. Personal Statement

I am well suited to lead the proposed study as MPI with Dr. Bradshaw due to my history of leading 1 to 5-year studies with various designs, including multi-site trials, funded through federal, industry and foundation grants. This natural collaboration between myself and Dr. Bradshaw brings together a truly interdisciplinary team, led by a Nurse Scientist with expertise in prematurity and a Child Psychologist with expertise in ASD, to study a physiological process (ANS dysfunction) that impacts children over their lifespan- both physiologically and behaviorally.

As a nurse scientist with a program of research to reduce morbidity and mortality in preterm infants, I have concentrated over 20 years of studies in 3 main areas: 1) reducing hypothermia in very preterm, 2) study of continuous physiologic measurements (central and peripheral body temperature, perfusion index and heart rate indices) to examine physiological mechanisms leading to morbid conditions and 3) examining autonomic nervous system (ANS) dysfunction and neurodevelopmental outcomes in children born preterm. I am currently ending a NIH funded R01 study to measure continuous body temperature (thermal gradients) and heart rate to predict infection in 352 very preterm infants in 5 NICU sites using ANS dysfunction as a biomarker for infection. This work has led to a collaboration with MPI Dr. Jessica Bradshaw, a Psychologist from Arts and Sciences of USC, to examine preterm infants for early indications of autism spectrum disorder (ASD), based on the ANS dysfunction seen through physiologic measures of body temperature and heart in the NICU and neurodevelopmental follow-up and assessments into early childhood.

Through these studies, I have also collaborated with Dr. Christian O'Reilly (Co-I), Computer Science & Engineering, to examine my R01 data for an algorithm to predict infection, as well as conduct the associations between NICU physiologic data and ASD outcome data in a pilot project. Additionally, I have worked with Co-Is Dr. Kayla Everhart in several neonatal studies since 2018 on decreasing morbidity and mortality. As the next step in our team's collaborative studies, MPI Dr. Bradshaw and I look forward to working with Drs. O'Reilly, Everhart and Rao (Prisma NICU) to examine measures for ANS dysfunction in very preterm infants during their NICU hospitalization to compare with longitudinal ASD symptoms and neurodevelopment impairments until 36 months outcome. This innovative and potentially high impact study will contribute greatly to the science of early

identification of ASD and augment early intervention in children born preterm while differentiating preterm non-ASD neurodevelopmental morbidity.

Bradshaw, J., O'Reilly, C., Everhart, K. C., Dixon, E., Vinyard, A., Tavakoli, A. Iskersky, V. Dail, R. B. Neonatal autonomic regulation as a predictor of autism symptoms in very preterm infants. *Journal of Perinatology*, 2024 Mar 29. PMID: 38553604. doi: 10.1038/s41372-024-01942-2. Online ahead of print.

Dail, R. B., Everhart, K., Hardin, J., Chang, W., Kuehn, D., Iskersky, V. Fisher, K. & Murphy, H. J. (2021). Predicting Infection in Very Preterm Infants: A Study Protocol. *Nursing Research*. 70 (2): 142-149. doi: 10.1097/NNR.0000000000000483. PMCID: PMC8044260; PMID: 33630537

Ongoing Projects:

NIH/NINR: 1R01NR017872; Dail (PI); 09/26/2018-06/30/2024

Central & peripheral body temperature in VLBW preterm infants during the neonatal period: Relationship to neonatal infection and necrotizing enterocolitis

The Gerber Foundation; Everhart (PI) Role: Co-Investigator, Senior mentor; 01/01/2022-12/31-2024

Evaluation of a blood warming device for packed red blood cell transfusions to decrease hypothermia in very preterm infants

PCORI; Felder & Jackson (MPIs); Role: Co-Investigator; 05/1/2023-4/31/2025

It Takes a Village: Increasing Capacity for Breast/Chest Feeding Equity among African Americans

R01MH132925; Bradshaw (PI); Role: Co-Investigator: 7/1/2023 – 6/30/2028

The Role of Autonomic Regulation of Attention in the Emergence of ASD

NIH/NINR: FNR020731A; Warren (PI); Role: Sponsor; 7/1/2023-8/30/2024

Exploring the impact of neonatal transport on the health of Black preterm infants and psychosocial challenges faced by parents in relation to social determinants of socioeconomic and racial inequities

Completed Projects:

USC, Carolina Autism and Neurodevelopment Research Center Pilot Award MPI: Dail & Bradshaw

07/01/2021-07/01/2022 Role: MPI

Examining Neonatal Thermal Gradients and Heart Activity as Predictive Risk Factors for Autism Spectrum Disorder in Premature Infants

B. Positions, Scientific Appointments and Honors

Academic Appointments:

2022-present	Health Sciences Endowed Professor & Director of the Perinatal, Pediatric & Family Research Center, University of South Carolina, College of Nursing
2017-2022	Associate Dean for Faculty Affairs & Professor, University of South Carolina, College of Nursing, Columbia, SC
2017-2023	Consulting Associate Professor, Duke University School of Nursing
2014-2017	Associate Professor, Duke University School of Nursing, Durham, NC
2008-2014	Assistant Professor, Duke University School of Nursing, Durham, NC
2006	Assistant Professor, University of North Carolina, Chapel Hill, NC

Hospital Positions Held:

2008-2009	Neonatal Nurse Practitioner, Duke University Hospital
1994-2006	Neonatal Nurse Practitioner, Children's Hospital of Eastern NC, Greenville, NC
1993	Registered Nurse, Neonatal, New Hanover Medical Center, Wilmington, NC
1988-1992	Registered Nurse, Neonatal, Medical College of VA, Richmond, VA
1987-1988	Registered Nurse, Coronary Care Unit, Riverside Hospital, Newport News, VA
1986-1988	Registered Nurse, Maternity, Williamsburg Community Hospital, Williamsburg, VA
1984-1986	Registered Nurse, Neonatal, Landstuhl 2 nd General Hospital, Germany
1981-1983	Registered Nurse, Neonatal, Sunrise Hospital, Las Vegas, NV

Honors:

2022	Student Research Mentor Award, University of South Carolina, College of Nursing
2022	Senior Investigator Award, University of South Carolina, College of Nursing
2022	Sigma International Nurse Researcher Hall of Fame Inductee
2019	University of Nevada, Las Vegas, 2019 School of Nursing Alumna Award
2019	Senior Investigator Award, University of South Carolina, College of Nursing
2019	Best Student Research Mentor, University of South Carolina College of Nursing
2016-2017	Amy V. Cockcroft Nursing Leadership Fellow
2014	Fellow of the American Academy of Nursing
2013	University of North Carolina, Chapel Hill School of Nursing Distinguished Alumna of the Year Award
2010	Inducted into the American Physiological Society
2010-2013	Robert Wood Johnson Foundation Nurse Faculty Scholar
2004-2005	March of Dimes Scholarship for Nursing Graduate Students
2004-2005	Academy of Neonatal Nurses Graduate Nursing Scholarship
2003	Glaxo Scholar
2001-2002	T32 Trainee Predoctoral Fellowship, Interventions to Prevent and Manage Chronic Illness

Special International Responsibilities:

2015	Health Research Board Reviewer, Ireland
2011	NHS Greater Glasgow & Clyde Research Endowment Fund Committee Review
2011	Medical Research Council, UK, Grant Review
2005	International Scientific Advisory Panel, Royal Canadian Nursing Association
2002	Research Practice and Possibilities Inaugural Postgraduate Nursing Conference/International Doctoral Student Research Seminar, Melbourne, AU
1996	Participation in Pediatric Heart Surgeries, Managua, Nicaragua

Special National Responsibilities:

2021-2024	NIH: ICSC Study Section Permanent Member
2020	NIH: ZRGI HDM-H Study Section, Temporary Member
2020	NIH: HSOD Study Section, Temporary Member
2019	NIH: NRCS Study Section, Temporary Member
2019-2022	Council for the Advancement of Nursing, Councilor of Programs (elected)
2017-2019	Council for the Advancement of Nursing, Program Committee Member
2012-2013	Council for Advancement of Nursing Science (CANS) Grant Review
2012-2013	Neonatal Network, Column Editor: Neonatal Lab Values
2014	American Nurses Foundation, Chair for scientific grant review
2013	American Nurses Foundation, Vice-Chair for scientific grant review
2009-2012	American Nurses Foundation- Grant Review
2012	AWHONN Research Advisory Panel, for grant review
2001-2003	Advisory Board Member, Academy of Neonatal Nurses
1998-present	Neonatal Resuscitation Program Instructor

C. Contributions to Science

1. Using polyurethane wrap to reduce heat loss for premature infants in the delivery room. Hypothermia is a major problem after birth in premature infants and has not improved over the last 50 years of research and interventions. Our team conducted a chart review to establish the extent of hypothermia after birth at a southeastern neonatal intensive care unit (NICU) which led to a study designed to test an intervention in 100 infants to decrease heat loss in the delivery room for this vulnerable population. Intervention infants were placed in polyurethane bags, up to their necks and while still wet with amniotic fluid, to envelop the infant and retain body heat. We compared rectal temperatures at NICU admission between wrapped infants and control infants who received standard thermal care in the delivery room. We attempted to increase delivery room temperature for all infants; however, this procedure was not followed and we analyzed delivery room

temperature as a covariate. **Central findings:** We found infants who were wrapped in polyurethane were warmer and not hypothermic ($< 36.5^{\circ}\text{C}$) upon NICU admission compared to infants who received standard thermal care (1). Infants who were in warmer delivery rooms had higher admission temperatures regardless of wrap versus standard care. We also surveyed for thermal practices around the nation and published those data (2) in an effort to change practice. **Influence on science and health:** This RCT contributed to the AAP and AHA changing delivery room resuscitation to include plastic wrap on infants less than 29 weeks gestation after birth in National Resuscitation Standards in 2006. WHO also now recommends to increase delivery room temperatures to approximately 80°F in addition to wrapping infants in plastic wrap.

1. Knobel, R.B., Wimmer, J., & Holbert, D. (2005). Heat loss prevention for preterm infants in the delivery room. *Journal of Perinatology*, 25, 304-308.
2. Knobel, R.B., Vohra, S., & Lehmann, C. (2005). Heat loss prevention in the delivery room for preterm infants: A national survey of Newborn Intensive Care Units. *Journal of Perinatology*. 25, 514-518.

2. Thermoregulation in the first 12 hours of life in extremely low birth weight premature infants. This study (NIH/NINR: 1F31 NR09143-01), measured abdominal and skin temperature and clinical context variables in 10 extremely low birth weight infants over their first 12 hours of life to examine hypothermia and determine if they could exhibit peripheral vasoconstriction in response to hypothermia (decreasing their foot temperature below their abdominal temperature by 2°C to send blood to their core for protection). **Central Findings:** Only one infant was able to exhibit a peripheral vasoconstriction response and more importantly, 7 of 10 infants kept the majority of their foot temperatures higher than their abdominal ones, or a central - peripheral temperature difference (CPTd) $< 0^{\circ}\text{C}$ for much of the 12 hours of study (1). This finding indicates an abnormal thermal pattern and is representative of altered vasomotor tone, or dysregulation of the autonomic nervous system. Findings from this study also indicated that infants' heart rates were within normal parameters if their central body temperature was maintained between 36.8°C - 37°C (2). **Influence on science and health:** This abnormal temperature pattern was reported in the literature very little and had never been studied in depth. Some researchers have advocated for dual temperature measures for standard of care in premature infants and this evidence established that dual temperature monitoring is important and can be indicative of an abnormal temperature pattern, requiring further study. Very premature infants have increased morbidity and mortality due to immature systems and the need for intensive care. They also have inefficient heat production, making them susceptible to hypothermia. If they are unable to control vasomotor tone, it leaves their central core unprotected and susceptible to injury from low perfusion. Our findings of the minimum standard control temperature of 36.8°C for premature infants housed in incubators has become routine in many neonatal units and is frequently cited in standards of care.

3. Knobel, R.B., Holditch-Davis, D., Schwartz, T., & Wimmer, J. (2009). Extremely low birth weight preterm infants lack vasomotor response in relationship to cold body temperatures at birth. *Journal of Perinatology*, 29, 814-21. PMID: PMC2787712
4. Knobel, R.B., Holditch-Davis, D., & Schwartz, T.A. (2010). Optimal body temperature in transitional ELBW infants using heart rate and temperature as indicators. *Journal of Obstetrics, Gynecological & Neonatal Nursing*, 39, 3-14. PMID: PMC3864588
5. Knobel, R. & Holditch-Davis, D. (2007). Thermoregulation and heat loss prevention after birth and during neonatal-intensive-care unit stabilization of ELBW Infants. *Journal of Obstetrics, Gynecological & Neonatal Nursing*, 36, 280-7.

3. Central and peripheral temperature in the first two weeks of life in premature infants. As a next step, I led an interdisciplinary team to examine central and peripheral temperature in 30 preterm infants, less than 1200 grams at birth, for longitudinal patterns of abnormal thermal patterns (CPTd $<0^{\circ}\text{C}$) in relationship to morbidity (1). This study was funded by NIH/NINR: 1R15NR012157-01 and Robert Wood Johnson Foundation #68041 to examine body temperature in very preterm infants over their first 2 weeks of life. This study examined each infant using a within case, observational design and mixed methods. Cases were analyzed to generate hypotheses for future study. **Central Findings:** All infants had periods of time with CPTd $< 0^{\circ}\text{C}$ throughout their first two weeks of life. We also found that infants with a higher percentage of time with CPTd $> 2^{\circ}\text{C}$ cold were associated with having an infection during the first month of life, being born to a mother who smoked during pregnancy, being African American and being male (3). All associations, except for infection, were not statistically significant because this was a small sample for an exploratory case study design;

however, these associations need to be further studied. We also found that increased temperature in the foot was significantly correlated with increased perfusion in the foot (2). This finding indicates that when CPTd <0°C, there is also increased blood flow in the foot, leaving the central body vulnerable to low blood flow and ischemia. **Influence on science and health:** This study gave evidence that continuous monitoring of central and peripheral body temperature might be used as a biomarker to alert clinicians that the infant is in an abnormal circulatory state and that they should suspect possible infection.

6. Knobel, R. B., Levy, J., Katz, L., Guenther, B., & Holditch-Davis, D. (2013). A pilot study to examine maturation of body temperature control in preterm infants. *Journal of Obstetrics, Gynecological & Neonatal Nursing*, 42(5), 562-74. PMID: PMC3783951

7. Knobel-Dail, R. B., Tanaka, D. T., Holditch-Davis, D., White, J. T. (2016). Perfusion index in very low birth weight premature infants during their first two week of life. *Biological Research for Nursing*, E. pub ahead of print. PMID: 27352610. <http://dx.doi.org/10.1177/1099800416656914>. PMID: PMC5942507

8. Knobel-Dail, R. B., Sloane, R., Holditch-Davis, D. & Tanaka, D. T. (2017). Negative temperature differential in preterm infants less than 29 weeks gestational age: Associations with infection and maternal smoking. *Nursing Research*, 66 (6), 442-453. PMID: 29095375. DOI: 10.1097/NNR.0000000000000250 PMID: PMC5679714

4. Dual point body temperature measurement with servo-control in premature infants. This study examined the usability of measuring central and peripheral body temperature and the CPTd using a research software and the Giraffe Omnibed with a within case design and mixed methods. **Central findings:** Nurses liked having dual temperature display with the CPTd and found that this display made them pay more attention to the premature infants' thermal stability. Nurses wanted larger font and alarms associated with abnormal thermal gradients. Two case study results revealed an associate between abnormally large thermal gradients and the onset of infection. **Influence on science and health:** Monitoring abdominal and foot temperature and their difference was feasible under servo-control and augmented nurses' abilities to care for their infants' thermal stability.

9. Dail, R. B., Tavakoli, A. S., Moseley, S. L., Bergstedt, V. B., Kuehn, D., & Chang, W. (2021) Examining the use of dual point temperature display in servo-control with early preterm infants. *Advances in Neonatal Care*. 21 (6). PMID: 33783382 DOI: 10.1097/ANC.0000000000000827

5. Using thermal gradients to predict infection in premature infants. This study is examining infants' longitudinal CPTds in relationship to the onset of infection in 352 preterm infants at five NICUs in North and South Carolina. Each site has a site PI and a research nurse. Infant quantitative data are collected prospectively with temperature dataloggers and HeRO monitors for their first 28 days of life. Data are sent to University of South Carolina electronically and a RedCap database is used to collect clinical outcome data. Enrollment ended in November (NCE due to Covid-19) and the study is currently in analysis phase. (see Ongoing Research Support). A subset of these infants (N=20) were enrolled in a pilot study to **examine ANS dysfunction in the NICU using thermal gradients and HR to compare to early signs of social and language disorders, including autism spectrum disorder (ASD)** in very early infancy, which may lead to earlier diagnoses and interventions. Enrollment in that subset study is complete and infants have completed 12 month outcome assessments.

10. Dail, R. B., Everhart, K., Hardin, J., Chang, W., Kuehn, D., Iskersky, V. Fisher, K. & Murphy, H. J. (2020). Predicting Infection in Very Preterm Infants: A Study Protocol. *Nursing Research*. 70 (2): 142-149. doi: 10.1097/NNR.0000000000000483. PMID: PMC8044260; PMID: 33630537

11. Dail, R.B.; Everhart, K.C.; Iskersky, V.; Chang, W.; Fisher, K.; Warren, K.; Steflink, H.J.; Hardin, J. W. Prenatal and postnatal disparities in very-preterm infants in a study of infections between 2018–2023 in southeastern US., *Tropical Medicine, and Infectious Disease*. 2024 Mar 28;9(4):70. PMID: 38668531 PMID: PMC11054784 DOI: 10.3390/tropicalmed9040070

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Jessica Bradshaw

eRA COMMONS USER NAME: ucsbbradshaw

POSITION TITLE: Associate Professor, Department of Psychology, University of South Carolina

EDUCATION/TRAINING

INSTITUTION AND LOCATION	DEGREE	Completion Date MM/YYYY	FIELD OF STUDY
University of California, San Diego	B.S.	06/2007	Cognitive Science
University of California, Santa Barbara	Ph.D.	09/2015	Clinical Psychology
Emory University School of Medicine, Department of Pediatrics	Postdoctoral	08/2018	Clinical Psychology and Social Neuroscience

A. Personal Statement

My long-term research interests involve methods for 1) quantifying neonatal and infant behavior, 2) identifying diverse developmental pathways that lead to atypical development of social communication and the emergence of autism spectrum disorder (ASD), and 3) translating these basic findings to actionable early detection and intervention strategies. I use a variety of experimental methodologies (electrocardiogram (ECG), behavioral coding, eye tracking) and analytic approaches (non-linear mixed models; structural equation modeling) to evaluate attentional and neurobiological abnormalities and phenotypic variation in the context of early detection and intervention for infants who are at an elevated likelihood for developing ASD (infant siblings of children with ASD and infants born preterm). I have spent much of the last 15 years engaged in and leading prospective, longitudinal research of infant siblings of children with ASD and contributing to NIH awards (PIs: Bradshaw, USC; Klin, Emory; Chawarska, Yale). My current an recently completed NIH-funded studies (R01MH132925; R21DC017252; K23MH120476) utilize prospective, longitudinal designs to study the emergence of ASD in elevated likelihood populations, including preterm infants and siblings of children with ASD. The focus of the proposed study on autonomic nervous system dysfunction as a mechanism for elevated prevalence of ASD in very preterm infants represents a parallel, programmatic extension of my ongoing research program. This project will be led by multiple PIs, myself and Robin Dail. MPI Dail and I have been working closely for the past 3 years collecting preliminary data and developing a theoretical framework for the proposed study. Together with Co-I Christian O'Reilly, we discovered a surprisingly strong association between preterm ANS dysfunction in the NICU and ASD symptoms at outcome (Bradshaw et al., 2024). This finding is a key element of the scientific premise for this study. In collaboration with Co-I John Richards (also a K23 mentor), we are developing trajectories of autonomic regulation of social and nonsocial attention in infancy (Bradshaw, Fu, & Richards, 2024). In addition to my scientific and methodological expertise in heart rate indices of autonomic dysfunction, I am a licensed clinical psychologist with expertise in infants and children with neurodevelopmental disorders and ASD. I am a certified trainer of the Neonatal Neurobehavioral Scale-II (NNS-II) and a research-reliable ADOS examiner. I have extensive training and years of experience in all developmental measures to be used in this study. My scientific and clinical expertise complements that of MPI Robin Dail and our investigative team is in an excellent position to successfully achieve the proposed aims, which will inform new etiological theories of ASD.

Ongoing and recently completed projects that I would like to highlight include:
R01 MH132925
Bradshaw (PI)
7/1/2023-6/30/2028
The Role of Autonomic Regulation of Attention in the Emergence of ASD

K23 MH120476

Bradshaw (PI)

7/19/2019-6/30/2024

Heart Defined Attention in Infancy: Predicting Social Communication and ASD Symptomology

R21 DC017252

Bradshaw (PI)

12/1/19-11/30/23

Neurobehavioral Determinants of Social Communication and Language Impairments in at-risk Infants

R01 HD099295

Feinberg, Hock (MPIs) Role: Co-Investigator

6/17/2020- 2/25/2025

Test of an Innovative, Scalable Support Program for Parents with a Young Child Recently Diagnosed with Autism Spectrum Disorder

Carolina Autism and Neurodevelopment Research Center Pilot Award

Bradshaw/Dail (MPIs)

7/1/21-7/1/22

Examining Neonatal Thermal Gradients and Heart Activity as Predictive Risk Factors for Autism Spectrum Disorder in Premature Infants

Carolina Autism and Neurodevelopment Research Center Pilot Award

O'Reilly (PI), Role: Co-Investigator

6/1/22-6/1/23

Model-Driven Analysis of Autonomic Control in ASD Using ECG

Citations:

1. **Bradshaw**, J., Fu, X., Richards, J. (2024). Infant Sustained Attention Differs by Context and Social Content in the First Two Years of Life. *Developmental Science*. doi: 10.1111/desc.13500.
2. **Bradshaw**, J., O'Reilly, C., Everhart, K. C., Dixon, E., Vinyard, A., Tavakoli, A., & Dail, R. B. (2024). Neonatal Autonomic Regulation as a Predictor of Autism Spectrum Disorder in Very Preterm Infants. *Journal of Perinatology*.
3. O'Reilly, C., Oruganti, S. D. R., Tilwani, D., & **Bradshaw**, J. (2023). Model-Driven Analysis of ECG Using Reinforcement Learning. *Bioengineering*, 10(6), 696. doi: 10.3390/bioengineering10060696 (PMC10295052).

B. Positions, Scientific Appointments, and Honors

Positions and Employment

2022-	Associate Professor, Department of Psychology, University of South Carolina, Columbia, SC
2018-	Assistant Professor, Department of Psychology, University of South Carolina, Columbia, SC
2015-2018	Postdoctoral Research Fellow, Marcus Autism Center, Emory University School of Medicine, Atlanta, GA
2014-2015	Clinical Psychology Intern, Rady Children's Hospital, University of California, San Diego
2010-2014	Graduate Student Researcher, Koegel Autism Center, University of California, Santa Barbara
2009-2010	Lab Manager, Yale Early Social Cognition Lab, Yale Child Study Center, New Haven, CT
2007-2009	Research Assistant, Yale Early Social Cognition Lab, Yale Child Study Center, New Haven, CT

Other Experiences and Professional Memberships

2022-	Elected Board Member, International Congress for Infant Studies
2021	Early Career Reviewer, National Institute of Health, Child Psychopathology Dev. Disabilities Study Section
2021 -	Member, International Congress for Infant Studies Communications Committee
2021 -	Member, Journal of Autism and Neurodevelopmental Disabilities Editorial Board

- 2021 - Member, International Society for Developmental Psychobiology
- 2020 - Member, Baby Siblings Research Consortium
- 2020 - Member, Society for Developmental Psychobiology
- 2020 - Member, Society for Research on Child Development
- 2019- Adhoc Reviewer, Journal of Child Psychology and Psychiatry, Development and Psychopathology, Journal of Neurodevelopmental Disabilities, Autism Research, International Journal of Developmental Disabilities
- 2016- Licensed Psychologist, State of Georgia, PSY004065
- 2016- International Society for Infant Studies
- 2016 - Adhoc Reviewer, Autism
- 2014-2015 Chief Intern, UCSD/VA Psychology Internship Training Program, San Diego, CA
- 2012 - Adhoc Reviewer, Journal of Autism and Developmental Disorders
- 2010 - Member, International Society for Autism Research

Honors

- 2023 McCausland Fellow, USC
- 2021 University of South Carolina Breakthrough Star Award
- 2018 International Society for Autism Research – Travel Award
- 2017 Institute of Education Sciences Summer Research Training Institute Participant
- 2017 American Speech-Language-Hearing Association Lessons for Success Participant
- 2013-2014 Graduate Research Mentorship Program, Graduate Division, UCSB
- 2013 Outstanding Research Award, UCSB
- 2012 Travel Award, Graduate Division, UCSB
- 2011-2012 Autism Science Foundation Predoctoral Fellowship
- 2011-2012 Dean's Fellowship, UCSB
- 2007 BS awarded *cum laude* with *Distinction in Cognitive Science*, University of California, San Diego
- 2007 Cognitive Science Honors Program, UCSD
- 2006 Phi Beta Kappa

C. Contributions to Science

My Contributions to Science are organized into five distinct categories: 1) Infant Autonomic Regulation and the Emergence of ASD, 2) Predicting Neurodevelopmental Outcomes from Attention, Neurobehavior, and Preterm Birth, 3) Longitudinal Modeling to Understand Neurodevelopmental Processes, 4) Early Identification of ASD in Infants and Toddlers, 5) Early Intervention for Infants and Toddlers.

1. Infant Autonomic Regulation and the Emergence of ASD: My K23 was focused on training in physiological underpinnings of attention as a potential biomarker of ASD. Thus, the past five years of my research program have focused on building expertise in the infant autonomic nervous system and heart rate indices of ANS as they pertain to predictive markers of social communication and ASD. My first analysis of heart-defined attention in 3-4-month-old infants evidenced autonomic regulation as a possible indicator of later social-communication skills.^d In collaboration with Co-I Christian O'Reilly, we refined processing of ECG data through computational models, automated data cleaning, and HRI feature detection using machine learning (O'Reilly, Oruganti, Tilwani, & Bradshaw, 2023)^b. Co-I John Richards and I use prospective longitudinal designs to study heart rate-defined sustained attention in early infancy as predictors of ASD^c. These collaborative projects have established the foundation for the preliminary data that serve as a key component of the scientific premise for this study. Together with MPI Robin Dail and Co-I O'Reilly, supported by a pilot grant, we discovered a surprisingly strong association predictive association among preterm ANS dysfunction and ASD outcomes.^a

- a. **Bradshaw**, J., O'Reilly, C., Everhart, K. C., Dixon, E., Vinyard, A., Tavakoli, A., & Dail, R. B. (2024). Neonatal Autonomic Regulation as a Predictor of Autism Spectrum Disorder in Very Preterm Infants. *Journal of Perinatology*, doi: 10.1038/s41372-024-01942-2.
- b. Tilwani, D., **Bradshaw**, J., Sheth, A., & O'Reilly, C. (2023). ECG Recordings as Predictors of Very Early Autism Likelihood: A Machine Learning Approach. *Bioengineering*, 10(7), 827. doi: 10.3390/bioengineering10060696

- c. **Bradshaw, J., Fu, X., Richards, J. (2024).** Infant Sustained Attention Differs by Context and Social Content in the First Two Years of Life. *Developmental Science*. doi: 10.1111/desc.13500.
- d. **Bradshaw, J., & Abney, D. H. (2021).** Infant physiological activity and the early emergence of social communication. *Developmental psychobiology*, 63(7), e22145. (PMID: 8555539)

2. Predicting Neurodevelopmental Outcomes from Attention, Neurobehavior, and Preterm Birth: The role of neonatal attention and neurobehavior in the pathogenesis of ASD and related social and communication impairments is largely unexplored. I have begun to map early infant experience from birth to 6 months across domains of visual and social attention, neurological integrity, and motor development in infants born preterm and at elevated genetic likelihood for ASD. My research in this area suggests that very early biological and behavioral factors, including preterm birth, ANS function, and attention, are key events and processes that form a developmental cascade leading to neurodevelopmental impairment and ASD.

- a. **Bradshaw, J., Fu, X., Yurkovic-Harding, J., & Abney, D. (2023).** Infant embodied attention in context: Feasibility of home-based head-mounted eye tracking in early infancy. *Developmental Cognitive Neuroscience*, 64, 101299. (PMC10522938).
- b. **Bradshaw, J., Shi, D., Hendrix, C., Saulnier, C., Klaiman, C. (2022).** Neonatal Neurobehavior in Infants with Autism Spectrum Disorder. *Developmental Medicine and Child Neurology*, (PMID: 34713902; PMC9209845)
- c. **Bills, S., Johnston, J., Shi, D., Bradshaw, J. (2020).** Social-environmental moderators of neurodevelopmental outcomes in youth born preterm: A systematic review. *Child Neuropsychology*, 27(3), 351-370. (PMID: 33342364; PMC7969400).

3. Longitudinal Modeling to Understand Neurodevelopmental Processes: Longitudinal and growth curve modeling are central to my research on infants and neurodevelopmental disorders. These analytic methods are crucial within the developmental cascades framework^a, which underpins my theoretical approach to understanding the emergence of autism spectrum disorder. I use both linear mixed modeling and structural equation modeling approaches in my work, employing analytic techniques such as non-linear modeling (e.g, splines)^b and latent growth curve models^c, respectively.

- a. **Bradshaw, J., Schwichtenberg, A. J., & Iverson, J. M. (2022).** Capturing the complexity of autism: Applying a developmental cascades framework. *Child Development Perspectives*, 16(1), 18-26. (PMC9673985).
- b. **Bradshaw, J., Klin, A., Evans, L., Klaiman, C., Saulnier, C., & McCracken, C. (2020).** Development of attention from birth to 5 months in infants at risk for autism spectrum disorder. *Development and psychopathology*, 32(2), 491-501. (PMID: 31012398; PMC6812597).
- c. **Bradshaw, J., Shi, D., Federico, A., Klaiman, C., & Saulnier, C. (2023).** The pull-to-sit task: examining infant postural development in autism spectrum disorder. *The Journal of pediatrics*, 253, 225-231. (PMC11009487).

4. Early Identification of ASD in Infants and Toddlers: My contributions in early identification and of ASD have thus far focused on evaluating early developmental profiles of infants between 9-12 months who are later diagnosed with ASD as well as early behavioral trajectories that distinguish typically developing infants from those at elevated genetic likelihood for ASD.

- a. **Bradshaw, J., McCracken, C., Lewis, M., Bran, N., Delehanty, A., Day, T., Klaiman, C., Saulnier, C., Klin, A., Wetherby, A. (2021).** Early Social Communication Development in Infants with Autism Spectrum Disorder. *Child Development*, 92(6), 2224-2234. (PMC: 8935345)
- b. **Foster, M., Federico, A., Klaiman, C., & Bradshaw, J. (2022).** Early Sleep Differences in Young Infants with Autism Spectrum Disorder. *Journal of Developmental & Behavioral Pediatrics*, 44(8), e519-e526. (PMID: 37556592)
- c. **Pileggi, M., Brane, N., Bradshaw, J., Delehanty, A., Day, T., McCracken, C., Stapel-Wax, J., Wetherby, A. (2021).** Early Observation of Red Flags in 12-month-old Infants Later Diagnosed with Autism Spectrum Disorder. *American Journal of Speech-Language Pathology*. (PMID: 33989505; PMC8702865).
- d. **Bradshaw, J., Klin, A., Evans, L., Klaiman, C., Saulnier, C., McCracken, C. (2019).** Development of Attention from Birth to 5 Months in Infants at Risk for Autism Spectrum Disorder. *Development and Psychopathology*, 32(2) 491-501. (PMID: 31012398; PMC6812597).

5. Early Intervention for Infants and Toddlers: In order to translate biomarkers to actionable intervention strategies, a large part of my research has also focused on developing and evaluating early interventions for infants and toddlers.^{a-b} I have investigated novel methods of outcome measurement^c and published one of the first studies to focus behavioral intervention for 6-12-month-old infants.^d This work documented preliminary effectiveness of increasing social engagement in the first year of life.

- a. **Bradshaw, J.**, Wolfe, K., Hock, R., & Scopano, L. (2022). Advances in Supporting Parents in Interventions for Autism Spectrum Disorder. *Pediatric Clinics*. (PMID: 35934491).
- b. **Bradshaw, J.**, Trumbull, A., Stapel-Wax, J., Gillespie, S., George, N., Saulnier, C., Klaiman, C., Woods, J., Call, N., Klin, A., Wetherby, A. (2020). Factors Associated with Enrollment into a Clinical Trial of Caregiver-Implemented Intervention for Infants at Risk for Autism Spectrum Disorder. *Autism*. (PMID: 32594763)
- c. **Bradshaw, J.**, Shic, F., Holden, A., Horowitz, E., Barret, A., German, T., Vernon, T. (2019). The Use of Eye Tracking as a Biomarker of Treatment Outcome in a Pilot Randomized Clinical Trial for Young Children with Autism. *Autism Research*, 12(5), 779-793. (PMID: 30891960).
- d. Koegel, L., Singh, A., Koegel, R., Hollingsworth, J., & **Bradshaw, J.** (2013). Assessing and Improving Early Social Engagement in Infants. *Journal of Positive Behavior Intervention*, 16(2), 69-80. (PMID: 25313271).

Complete list of published works on MyBiobibliography:

<https://www.ncbi.nlm.nih.gov/myncbi/jessica.bradshaw.1/bibliography/public/>

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Kayla C. Everhart, PhD, RN

eRA COMMONS USER NAME (credential, e.g., agency login): EVERHAKC

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
University of North Carolina at Wilmington	BSN	12/2010	Nursing
University of North Carolina at Charlotte	MSN	12/2017	Nursing Education
University of South Carolina at Columbia	PhD	05/2021	PhD Nursing

A. Personal Statement

For the proposed R01 study I am well-suited for the role of Co-Investigator. I have ongoing research in the Prisma Richlands NICU and have worked in this unit as a nurse. I will be responsible for working with Dr. Rao and the Prisma Health research nurse to consent mothers of participants, be a liaison with the Prisma Health IRB, and provide oversight and assistance between the USC Project Coordinator and the Prisma Research Nurse. My research is grounded in over 10 years' experience as a clinical neonatal nurse in Level III NICUs which has allowed me to care for critically ill infants of varying gestations. In addition to this proposed R01, I am currently funded by The Gerber Foundation as Principal Investigator to examine commercial blood warmers to decrease hypothermia during blood transfusions in very preterm infants at Prisma Health Richlands NICU. The target enrollment number for this randomized control trial is 140 very preterm infants. Enrollment began September 2022 and will end Fall 2024. With my vast early experience participating in and leading research as a nurse scientist and with consistent mentorship from a nationally known expert in neonatal research (Dr. Robin Dail), I am well prepared to serve in the role of Co-Investigator. This study is promising for identifying infants who are at heightened developmental risk prior to NICU discharge is essential to guide very early intervention efforts and offer the best chance at improving childhood outcomes.

Ongoing and recently completed projects that I would like to highlight include:

Ongoing Research Support

The Gerber Foundation

Everhart (PI)

01/01/2022-12/31/2024

Evaluation of a blood warming device for packed red blood cell transfusions to decrease hypothermia in very preterm infants

NIH/NINR: 1R01NR017872

Dail (PI); Everhart (Co-I)

09/26/2018-06/30/2024

Central & peripheral body temperature in VLBW preterm infants during the neonatal period: Relationship to neonatal infection and necrotizing enterocolitis

US DHHS PDG B-5 Subaward Chappell and Davis

Everhart (Consultant)

Supporting community care transitions for infants discharged from high-level newborn care (Childcare Ready for All: CRAwl) and children with stable medical needs seeking childcare (Baby Coach)

Citation:

- Bradshaw, J., O'Reilly, C., Everhart, K. C., Dixon, E., Vinyard, A., Tavakoli, A. Iskersky, V. Dail, R. B. Neonatal autonomic regulation as a predictor of autism symptoms in very preterm infants. *Journal of Perinatology*, 2024.

B. Positions and Honors

Academic Appointments:

2023-present The University of South Carolina, Columbia, SC Assistant Professor-Tenure Track

2021-2023 The University of South Carolina, Columbia, SC, Research Assistant Professor

2019-Midlands Technical College, Columbia, SC, Clinical Instructor

2018-2021 The University of South Carolina, Columbia, SC Research Coordinator-The PATH Study

2017-The University of South Carolina, Columbia, SC, Clinical Instructor

Hospital Positions:

2019-2021 Prisma Health- Site Research Nurse- The PATH Study

2018-2021 The University of South Carolina, Columbia, SC

Research Coordinator-The PATH Study

2017-Prisma Health, Columbia, SC Staff Nurse

Neonatal Intensive Care Unit

2010-2017 New Hanover Regional Medical Center, Wilmington, NC, Staff Nurse

Adult Medical Surgical Oncology/Neonatal Intensive Care Unit

Honors:

2022 University of South Carolina College of Nursing New Scholar Award

2017 Induction into Sigma Theta Tau

2017 Induction to Phi Kappa Phi

C. Contributions to Science

1. Examining the relationship of blood transfusions and acute morbidity in preterm infants. Preterm infants, those born less than 37 weeks gestational age, often suffer major morbidity and mortality during hospitalization in the neonatal intensive care unit due to their prematurity; one of the major morbidities is Necrotizing Enterocolitis (NEC), a devastating gastrointestinal disease. The exact etiology of NEC is unknown; however, we hypothesize thermal instability and/or stress related to blood transfusions are etiological factors in the cascade of events leading to occurrence of NEC in extremely preterm infants.

My PhD dissertation included 1) a scoping literature review examining the relationship of packed red blood cell transfusions and Necrotizing Enterocolitis in preterms, 2) the study team's evaluation, selection, and adaptation of a conceptual framework, 3) a national survey of nursing practices around giving blood transfusions to preterm infants, and 4) an exploratory multiple case study using mixed methods and a within case design to examine the relationship of blood transfusions and acute morbidity in preterm infants. **Central findings:** 1) Some researchers postulate blood transfusions are linked to increased incidence of NEC, while others disagree and still others feel the association is due to feeding or not feeding during the blood transfusion. We identified a gap in the literature highlighting the need for further research around the link between blood transfusions and the occurrence of NEC in extremely preterm infants. 2) A holistic look at extremely preterm infant morbidity is needed to examine all possible pathways contributing to the cascade of events leading to NEC. 3) There is great practice variation across the nation in the procedures around delivering blood transfusions to preterm infants. 4) Preterm infants may experience acute hypothermia during

and after blood transfusions as well as physiological instability seen in heart rate variation and oxygenation. The more premature infants are affected more than mature infants. **Influence on science and health:** Future studies will aim to provide the best evidence so that clinicians can standardize practice, improve consistency, and decrease complications related to blood transfusions in very preterm infants. It is evident that transfusing preterm infants with life-saving blood is not without consequence to their physiologic stability. Studies are needed to investigate warming of blood transfusions and exploration of the link between hypothermia during transfusions and the occurrence of NEC.

- **Everhart, KC.**, et al. "Examining etiological pathways to necrotizing enterocolitis in very preterm infants." *Journal of Neonatal Nursing* 27.2 (2021): 77-81. <https://doi.org/10.1016/j.jnn.2020.07.003>
- **Everhart, K, Iskersky, V,** Donevant, S, Wirth, M, Dail, RB. (2021). Exploring contributing factors to the pathogenesis of necrotizing enterocolitis in very preterm infants: An adaptation of Neuman's Systems Theory. *Nursing Science Quarterly*.
- **Everhart K,** Donevant S, Iskersky V, Wirth M, & Dail RB. (2022). Examining standard practices used for packed red blood cell transfusions for preterm infants: A National Survey. *Reports and Reviews in Neonatology*.
- **Everhart, KC** Donevant SB, Iskersky V, Wirth M, & Dail RB. (2022). Preterm infant stability during packed red blood cell transfusions: A case comparison. *Nursing Research*.

2. Using thermal gradients to predict infection in premature infants. My role on this study is a Co-I, after transitioning from the role of research coordinator. This study is examining infants' longitudinal central and peripheral temperature differences in relationship to the onset of infection in 330 preterm infants at five NICUs in North and South Carolina. Each site has a site PI and a research nurse. Infant quantitative data are collected prospectively with temperature dataloggers and HeRO monitors for their first 28 days of life.

- Dail, R. B., Everhart, K., Hardin, J., Chang, W., Kuehn, D., Iskersky, V. Fisher, K. & Murphy, H. J. (2020). Predicting Infection in Very Preterm Infants: A Study Protocol. *Nursing Research*. doi: 10.1097/NNR.0000000000000483. [Epub ahead of print] PubMed PMID: 33060417.
- Dail, R.B.; Everhart, K.C.; Iskersky, V.; Chang, W.; Fisher, K.; Warren, K.; Steflik, H.J.; Hardin, J. W. Prenatal and postnatal disparities in very-preterm infants in a study of infections between 2018–2023 in southeastern US. In Press 2024, *Tropical Medicine, and Infectious Disease*. PMID: PMC11054784.

3. Evaluation of a blood warming device for packed red blood cell transfusions to decrease hypothermia in very preterm infants. A randomized controlled trial to evaluate the use of a commercial blood warmer to decrease hypothermia in very preterm infants (<less than 32 weeks) gestational age who receive blood transfusions within their first month of life. This RCT is funded by The Gerber Foundation and enrollment began in September 2022. These study results will determine if blood should be warmed while infusing to prevent the incidence of hypothermia during and after blood transfusions.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. DO NOT EXCEED FIVE PAGES.

NAME: O'Reilly, Christian

eRA COMMONS USER NAME (credential, e.g., agency login): coreilly

POSITION TITLE: Assistant Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	END DATE MM/YYYY	FIELD OF STUDY
Ecole Polytechnique Fédérale de Lausanne, Lausanne, Vaud, Switzerland	Postdoctoral Fellow	11/2018	Computational neuroscience
McGill University, Montreal, Quebec, Canada	Postdoctoral Fellow	07/2015	Brain imaging and autism
Université de Montréal, Montreal, Quebec, Canada	Postdoctoral Fellow	05/2014	Biomedical sciences - sleep EEG
Polytechnique Montréal, Montreal, Quebec, Canada	PHD	05/2012	Biomedical engineering
Polytechnique Montréal, Montreal, Quebec, Canada	MENG	05/2011	Biomedical engineering
Ecole Polytechnique, Montréal, Quebec, Canada	BENG	12/2007	Electrical Engineering

A. Personal Statement

Throughout the years, I gathered diversified experience in analyzing and modeling biological systems. I worked with various types of physiological recordings, including human movement kinematics, electroencephalogram, polysomnogram (including ECG), magnetoencephalogram, patch-clamp recordings, and implanted electrodes (local field potentials). Through this work, I have mastered the techniques required to process the biosignals involved in this project. I also worked on diagnostic applications (e.g., for brain stroke, ADHD) based on pattern recognition and machine learning techniques. I gained further expertise in designing automatic classifiers (e.g., for sleep spindles) and biomarkers (e.g., for early diagnosis of ASD using EEG and ECG recordings). Aside from developing significant expertise in signal processing, statistical analysis, and machine learning in the biomedical field, I also specialized in ASD. Further, apart from my professional experience, I can count on a well-established network of local (e.g., at the Artificial Intelligence Institute of South Carolina) and international collaborators. For example, my active involvement Quebec 1000 Families (Q1K) project – a world-class cohort on ASD and related neurodevelopmental disorders led by renowned Canadian institutions like McGill's Azrieli Centre for Autism Research – allows me to draw support from world-renown experts in ASD. Therefore, I am well-positioned to support the PIs to successfully carry on the objectives of this proposal.

Further, I already have established a fruitful collaboration with both PIs. For example, in collaboration with J. Bradshaw, we studied how ECG can be used to characterize atypical autonomic control in autism. In the context of this work, we published a paper on modeling ECG signals. One of my Ph.D. students (D. Tilwani) also published a biomarker study on using atypical autonomic activation, as characterized through ECG, to predict the likelihood of ASD in infants. Another of my Ph.D. (V. Srivastava) students is currently working on implementing in Python and validating in ASD the novel indices of sympathetic and parasympathetic activation proposed by Valenza et al. (J Appl Physiol; 2018). In collaboration with R. Dail, we also conducted analyses on temperature gradients and their usage as a biomarker for neonatal sepsis. For these analyses, I mentored three K12 students and one undergraduate student. Our work on these data prepared us well to extend these analyses to the ASD population, as we recently did in a publication on the use of neonatal autonomic regulation to predict ASD in very preterm infants.

1. Bradshaw J, O'Reilly C, Everhart KC, Dixon E, Vinyard A, Tavakoli A, Dail RB. Neonatal Autonomic Regulation as a Predictor of Autism Spectrum Disorder in Very Preterm Infants. Journal of Perinatology. 2024 Mar 29. doi: 10.1038/s41372-024-01942-2. Online ahead of print.

2. O'Reilly C, Huberty S, van Noordt S, Desjardins J, Wright N, Scoriah J, Webb SJ, Elsabbagh M. EEG functional connectivity in infants at elevated familial likelihood for autism spectrum disorder. *Mol Autism*. 2023 Oct 7;14(1):37. PubMed Central PMCID: PMC10559476.
3. Tilwani D, Bradshaw J, Sheth A, O'Reilly C. ECG Recordings as Predictors of Very Early Autism Likelihood: A Machine Learning Approach. *Bioengineering (Basel)*. 2023 Jul 11;10(7) PubMed Central PMCID: PMC10376813.
4. O'Reilly C, Oruganti SDR, Tilwani D, Bradshaw J. Model-Driven Analysis of ECG Using Reinforcement Learning. *Bioengineering (Basel)*. 2023 Jun 7;10(6) PubMed Central PMCID: PMC10295052.

B. Positions, Scientific Appointments and Honors

Positions and Scientific Appointments

2023 -	Editorial board member for Scientific Reports, Nature Portfolio, Belin
2021 -	Assistant Professor, University of South Carolina, Columbia, South Carolina
2020 - 2021	Research Associate, McGill University, Montreal Neurological Institute, Montreal, Canada
2015 - 2015	Visiting Scholar, Wellcome Center for Human Neuroimaging, UCL, London, UK
2013 - 2016	Guest editor, <i>Frontiers in Human Neuroscience</i> , Frontiers, Lausanne, Switzerland
2011 -	Senior Member, Institute of Electrical and Electronics Engineers, New York, New York

Honors

2013 - 2015	Postdoctoral fellowship, Natural Sciences and Engineering Research Council of Canada
2010 - 2012	Doctoral fellowship, Natural Sciences and Engineering Research Council of Canada
2005 - 2006	Bourses UPIR - Unité de participation et d'initiation à la recherche, Polytechnique Montreal
2013	Postdoctoral fellowship, Fonds de recherche Nature et technologies du Québec
2012	Recommended for best thesis award, Polytechnique Montreal
2011	Best student paper - Conference of the International Graphonomics Society, Maryland University
2011	Best paper, Association des étudiants des cycles supérieurs de Polytechnique
2010	Doctoral fellowship, Fonds de recherche Nature et technologies du Québec
2006	Undergraduate Student Research Awards, Natural Sciences and Engineering Research Council of Canada

C. Contribution to Science

1. EEG functional connectivity in ASD: Following the publication of a very well-cited systematic review paper on EEG and magnetoencephalography functional connectivity in autism, I studied how atypical neural connectivity develops in infants. To improve EEG analyses for neurodevelopmental cohorts, I prepared and publicly released a set of 13 realistic head models for infants between two weeks and 24 months of age. Researchers can now use these templates to perform source reconstruction in infants with common neuroimaging software. Further, we published a paper using these templates to study in detail the functional connectivity between EEG cortical sources in infants at elevated familial likelihood for autism and how it correlated with Autism Diagnostic Observation Schedule (ADOS) severity scores.
 - a. O'Reilly C, Huberty S, van Noordt S, Desjardins J, Wright N, Scoriah J, Webb SJ, Elsabbagh M. EEG functional connectivity in infants at elevated familial likelihood for autism spectrum disorder. *Mol Autism*. 2023 Oct 7;14(1):37. PubMed Central PMCID: PMC10559476.
 - b. O'Reilly C, Elsabbagh M. Intracranial recordings reveal ubiquitous in-phase and in-antiphase functional connectivity between homotopic brain regions in humans. *J Neurosci Res*. 2021 Mar;99(3):887-897. PubMed PMID: 33190333.
 - c. O'Reilly C, Larson E, Richards JE, Elsabbagh M. Structural templates for imaging EEG cortical sources in infants. *Neuroimage*. 2021 Feb 15;227:117682. PubMed Central PMCID: PMC7901726.

- d. O'Reilly C, Lewis JD, Elsabbagh M. Is functional brain connectivity atypical in autism? A systematic review of EEG and MEG studies. *PLoS One*. 2017;12(5):e0175870. PubMed Central PMCID: PMC5414938.
2. Zero-lag synchrony in the brain: In a series of papers, I used intracranial recordings and scalp electroencephalography (EEG) to study the existence of zero-lag synchrony in the brain. I showed that most functional connectivity between homotopic brain regions has no lag. Besides highlighting a generally neglected property of brain communication potentially involved in important mechanisms like inter-hemispheric inhibition, these results also question commonly accepted assumptions when estimating functional connectivity using EEG and magnetoencephalography (MEG) recordings, i.e., that zero-lag synchrony can safely be discarded to eliminate the confounding effect of volume conduction. This systematic investigation of zero-lag connectivity also shows how researchers can use surrogate approaches to control for volume conduction when computing functional connectivity without having to discard zero-lag neural synchrony
 - a. O'Reilly C, Elsabbagh M. Intracranial recordings reveal ubiquitous in-phase and in-antiphase functional connectivity between homotopic brain regions in humans. *J Neurosci Res*. 2021 Mar;99(3):887-897. PubMed PMID: 33190333.
 - b. C. O'Reilly, M. Elsabbagh. Computing Realistic Surrogate EEG for the Study of Functional Connectivity. 2021 IEEE EMBS International Conference on Biomedical and Health Informatics (BHI). 2021; :1-4.
 - c. O'Reilly C, Lewis J, Theilmann R, Elsabbagh M, Townsend J. In-phase and in-antiphase connectivity in EEG. *bioRxiv*. 2021 June. DOI: 10.1101/2021.05.19.444800
3. Large-scale biophysically-detailed modeling of the thalamocortical loop: For three years I led the effort to model the thalamocortical loop at the Blue Brain Project. This is a fundamental big-science project that is likely to have a transformative impact on neuroscience in the long run. In the context of this project, we shared with the community a new approach to harvest information manually or automatically from the neuroscientific literature and include it in neuronal models in a systematic and fully traceable way. We also released new highly biophysically-detailed models of thalamocortical cells and we recently published a thorough review of the somatosensory thalamocortical loop in rodents, which I am confident will become a highly respected reference on this topic.
 - a. Iavarone E, Simko J, Shi Y, Bertschy M, García-Amado M, Litvak P, Kaufmann AK, O'Reilly C, Amsalem O, Abdellah M, Chevtchenko G, Coste B, Courcol JD, Ecker A, Favreau C, Fleury AC, Van Geit W, Gevaert M, Guerrero NR, Herttuainen J, Ivaska G, Kerrien S, King JG, Kumbhar P, Lurie P, Magkanaris I, Muddapu VR, Nair J, Pereira FL, Perin R, Petitjean F, Ranjan R, Reimann M, Soltuzu L, Sy MF, Tuncel MA, Ulbrich A, Wolf M, Clascá F, Markram H, Hill SL. Thalamic control of sensory processing and spindles in a biophysical somatosensory thalamoreticular circuit model of wakefulness and sleep. *Cell Rep*. 2023 Mar 28;42(3):112200. PubMed Central PMCID: PMC10066598.
 - b. O'Reilly C, Iavarone E, Yi J, Hill SL. Rodent somatosensory thalamocortical circuitry: Neurons, synapses, and connectivity. *Neurosci Biobehav Rev*. 2021 Jul;126:213-235. PubMed PMID: 33766672.
 - c. Iavarone E, Yi J, Shi Y, Zandt BJ, O'Reilly C, Van Geit W, Rössert C, Markram H, Hill SL. Experimentally-constrained biophysical models of tonic and burst firing modes in thalamocortical neurons. *PLoS Comput Biol*. 2019 May;15(5):e1006753. PubMed Central PMCID: PMC6541309.
 - d. O'Reilly C, Iavarone E, Hill SL. A Framework for Collaborative Curation of Neuroscientific Literature. *Front Neuroinform*. 2017;11:27. PubMed Central PMCID: PMC5395614.
4. Study of sleep spindles correlates: During my first postdoctoral project, I studied correlates of sleep spindles with memory consolidation and clinical conditions like rapid eye movement sleep behavior disorder (RBD). We provided further insights into the role of sleep spindles by showing that their density is associated with improvement in the Tower of Hanoi test (a problem-solving, executive function task sensitive to frontal lobe functioning) and that RBD is associated with lower fast and higher slow sleep spindle densities. Further, as a guest editor, I led a special issue on sleep spindles. To allow reproducible outcomes across this special issue and in future research, we also released an open-access database comprising 200 high-quality full-night polysomnographic recordings including a full 10-20 electrode grid and

a rich set of other biosignals as well as multi-expert scoring for sleep stages, sleep spindles, and K-complexes. The paper presenting this database has since then been consistently well-cited and the database has been used by a wide array of research teams across the world. I also published within this special issue a paper systematically benchmarking four different automatic sleep spindle detectors. In this study, I highlighted the large bias introduced by testing detectors on different datasets and expert scorings, i.e., the effect of the dataset and the expert scoring was typically larger than the difference in performance between different detection approaches. These results further supported the importance of providing a common open-access database and expert scorings to benchmark contributions from various research laboratories using the same gold standard. Further, in a two-part paper series, I developed and tested a novel method to assess how sleep spindles propagate across the scalp. This approach revealed a systematic pattern across subjects, with slower propagation from posterior to anterior and from central to lateral regions. These patterns are potentially indicating a systematic topology in the thalamocortical generation of sleep spindles which might provide key insights into rhythmogenesis in healthy thalamocortical interactions as well as in diseased conditions like epilepsy.

- a. O'Reilly C, Godin I, Montplaisir J, Nielsen T. REM sleep behaviour disorder is associated with lower fast and higher slow sleep spindle densities. *J Sleep Res.* 2015 Dec;24(6):593-601. PubMed PMID: 26041532.
 - b. O'Reilly C, Nielsen T. Automatic sleep spindle detection: benchmarking with fine temporal resolution using open science tools. *Front Hum Neurosci.* 2015;9:353. PubMed Central PMCID: PMC4478395.
 - c. O'Reilly C, Gosselin N, Carrier J, Nielsen T. Montreal Archive of Sleep Studies: an open-access resource for instrument benchmarking and exploratory research. *J Sleep Res.* 2014 Dec;23(6):628-635. PubMed PMID: 24909981.
 - d. O'Reilly C, Nielsen T. Assessing EEG sleep spindle propagation. Part 2: experimental characterization. *J Neurosci Methods.* 2014 Jan 15;221:215-27. PubMed PMID: 23999173.
5. Human movement modeling: As a Ph.D. student, I used a lognormal decomposition to model the kinematics of human movements. In a highly cited paper, I explained how complex movements like those involved in handwriting signatures can be decomposed into lognormal components to efficiently and accurately model fine motor control. I then used this approach to study correlates of the neuromuscular system with physiological processes like aging and clinical conditions like the main risk factor for brain stroke. Further, I demonstrated how such an approach can be used to develop diagnostic tools based on movement kinematics.
- a. O'Reilly C, Plamondon R, Lebrun LH. Linking brain stroke risk factors to human movement features for the development of preventive tools. *Front Aging Neurosci.* 2014;6:150. PubMed Central PMCID: PMC4085479.
 - b. Plamondon R, O'Reilly C, Rémi C, Duval T. The lognormal handwriter: learning, performing, and declining. *Front Psychol.* 2013;4:945. PubMed Central PMCID: PMC3867641.
 - c. O'Reilly C, Plamondon R. Impact of the principal stroke risk factors on human movements. *Hum Mov Sci.* 2011 Aug;30(4):792-806. PubMed PMID: 20888057.
 - d. O'Reilly C, Plamondon R. Development of a Sigma-Lognormal representation for on-line signatures. *Pattern Recognition.* 42(12):3324-3337.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: Rao, Raviprasad G.

eRA COMMONS USER NAME (credential, e.g., agency login): RAVIPRASADRAO

POSITION TITLE: Director of Neonatology, Medical Director for Neonatal services for Prisma Health, Midlands

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Government Medical College Vadodara, Gujarat, India	MBBS	05/1988	Medicine
Shree Sayaji General Hospital Vadodara, Gujarat, India	MD	12/1992	Pediatric Residency
University of Illinois at Chicago Chicago, Illinois	MD	06/1996	Pediatric Residency
University of Illinois at Chicago Chicago, Illinois	MD	10/2004	Fellowship in Neonatal-Prenatal Medicine

A. Personal Statement

I have the expertise, training and experience to serve as a site PI for the proposed research study. I have nearly 20 years of experience in the field of neonatal perinatal medicine and have been in a variety of leadership roles for the past 14 years. I have served as the Unit director at Lexington Medical center since 2010 and have been the Quality improvement director for the Midlands Pediatrix Medical Group since 2012. Most recently in February 2024, I was appointed to the position of Medical Director for Neonatal services for the Pediatrix Medical Group as my colleague, Dr. Victor Iskersky retired at Prisma Midlands NICU. My responsibilities include directing a 65 bed Level III C NICU at Prisma Health Richland which serves as the Regional Perinatal center for 16 counties in the Midlands region. I oversee a busy Neonatal Practice in the Midlands which includes the 65 bed Level III C unit, a 30 bed Level III NICU, a 20 bed Level II NICU and a 16 bed community perinatal center.

Over the past 3 years, we have been one of the highest enrolling NICU sites for Dr. Dail's study (NIH R01NR017872) "Central and peripheral body temperature in VLBW preterm infants during the neonatal period : Relationship to neonatal infection and Necrotizing enterocolitis." We collaborated with Drs. Dail and Bradshaw to conduct a pilot study with a subset of 20 very preterm infants from Dr. Dail's study to analyze autonomic nervous dysfunction with neurodevelopmental outcomes and signs of autism spectrum disorder. These data offer support for the methods of the

proposed study and highlight the innovation in studying ANS dysfunction to identify infants at risk for ASD. We admit around 220 VLBW infants annually and we are confident that our NICU can support the enrollment for this study over the period proposed. I am very enthusiastic about this proposed collaboration.

Citations:

Dail, R. B., Everhart, K., Hardin, J., Chang, W., Kuehn, D., Iskersky, V. Fisher, K. & Murphy, H. J. (2021). Predicting Infection in Very Preterm Infants: A Study Protocol. *Nursing Research*. 70 (2): 142-149. doi: 10.1097/NNR.0000000000000483. PMCID: PMC8044260; PMID: 33630537

Dail, R.B.; Everhart, K.C.; Iskersky, V.; Chang, W.; Fisher, K.; Warren, K.; Steflrik, H.J.; Hardin, J. W. Prenatal and postnatal disparities in very-preterm infants in a study of infections between 2018–2023 in southeastern US., *Tropical Medicine, and Infectious Disease*. 2024 Mar 28;9(4):70. PMID: 38668531 PMCID: PMC11054784 DOI: 10.3390/tropicalmed9040070

Bradshaw, J., O'Reilly, C., Everhart, K. C., Dixon, E., Vinyard, A., Tavakoli, A. Iskersky, V. Dail, R. B. Neonatal autonomic regulation as a predictor of autism symptoms in very preterm infants. *Journal of Perinatology*, 2024 Mar 29. PMID: 38553604. doi: 10.1038/s41372-024-01942-2. Online ahead of print.

B. Positions, Scientific Appointments, and Honors

Positions and Scientific Appointments

Feb 2024 – Present	Director of Neonatology, Prisma Health – Midlands , Columbia, SC
Jan 2024 – Present	Senior Medical Director's Committee , Prisma Health Children's Hospital- Midlands, Columbia, SC
Jan 2024- Present	Operations Committee, Prisma Health Children's Hospital- Midlands, Columbia, SC
Jan 2024- Present	Patient Care and Safety council, Prisma Health Richland Hospital, Columbia, SC
Jan 2024 – Present	Perinatal morbidity and mortality committee, Prisma Health Richland, Columbia, SC
2019	Vice President of the SC chapter of the American Physicians of Indian Origin
July 2014-June 2015	Chair, Department of Pediatrics, Lexington Medical center, Columbia, SC
Jan 2012 – Present	Quality improvement director , Prisma Health NICU – Columbia, SC
Jan 2010- Present	Unit Director, Lexington Medical Center neonatal and newborn services, Columbia , SC
Jan 2010- Present	Chair, Maternal and neonatal morbidity and mortality review committee, Lexington Medical Center, Columbia, SC
Jan 2010 to Present	Clinical Associate Professor of Pediatrics, University of South Carolina , Columbia, SC

Board Certifications and Clinical Licensure

2005- Present Board certified in Neonatal Perinatal Medicine with continued recertification

2004 – Present Active physician license in South Carolina

1997- Present Board certified In General Pediatrics with continued recertification
1995 Educational Commission for Foreign Medical Graduates certification

Professional Memberships

American Academy of Pediatrics
SC chapter of American Academy of Pediatrics
Society for Neonatal Perinatal Medicine
SC chapter of American Physicians of Indian Origin

Honors

2006 Recognized as best faculty by the Department of Pediatrics Residency Program
1998 Voted the best resident by the Faculty at the Pediatric Residency Program at the University of Illinois at Chicago Hospitals
1997/1998 Voted the best resident of the year by the Pediatric Residency Program at the University of Illinois at Chicago Hospitals
1992 Awarded the Gold Medal for the Best Candidate in the Certifying exam for the Pediatric Residency at the Shree Sayaji General Hospital Pediatrics residency program.
1988 Awarded the Pfizer Gold medal for meritorious performance in the final year of Medical School at the Government Medical College, Vadodara India.
1983 Awarded for the highest score in Mathematics in the State of Gujarat, India for the High School Graduation Board exams

C. Contributions to Science

1. Opioid use in neonates is ever increasing with the need for post operative pain control and for pain control with multiple procedures to which the high risk neonates in the NICU are exposed. Opioid exposure in the extremely premature neonate has been studied extensively including long term neurodevelopmental and neurobehavioral effects with prolonged opioid exposure. We looked at the neurobehavioral effects, opioid receptor binding and CDK5 signaling on immediate postnatal exposure of morphine using a rat model. We also looked at the long term learning and spatial memory functions in the same opioid exposed newborn rat pups. We also looked at cocaine exposure in the newborn pups in the immediate postnatal period and with long term spatial memory and learning. The same was done using a Morris water maze. Prenatal Morphine and cocaine exposed rat pups demonstrated mild delays in spatial memory and learning compared to non-exposed pups.
 - a) Abstract for Society for Pediatric Research (2004) and Poster Presentation, Rao R .G. & Bhat, R. "Neurobehavioral effects, opioid receptor Binding and CDK5 signaling , Effect of immediate postnatal Morphine (M) on the developing rat."
 - b) Abstract for Society for Pediatric Research (2003) and Poster Presentation, Rao, R.G. & Bhat, R. "Long term Learning and spatial memory functions and Opiate receptor binding in the developing rat: Effect of Prenatal cocaine and/or morphine."
 - c) Applied as a Grant to the NIH (2004) with Dr. Bhat, R. as the PI and Rao, R.G. as the co-investigator, "Differential effects of chronic Prenatal cocaine and/or Morphine exposure on Cyclin dependent kinase (CDK 5) in the brain of the offspring using a rat model."
2. Infants with meconium aspiration are at a very high risk for immediate postnatal morbidities including respiratory and infectious etiologies. Some of the infants with MAS end up on extensive and prolonged respiratory support including the need for Extra corporeal membrane oxygenation. This ultimately effects their length of stay and the need for support at the time of discharge. Moreover, the inflammatory process involved with Meconium aspiration might generate cytokines

and radicals which might lead to long term morbidities including neuro-developmental impairments. We looked at the Neurodevelopmental outcomes at 6, 12, 18, and 24 months using the High risk follow up clinic in infants who were discharged from the NICU after having sustained Meconium aspiration syndrome at birth.

- a) Abstract and poster presentation for the Society of Pediatric Research. (2002). Rao, R.G., "Neurodevelopmental outcomes of infants with Meconium Aspiration syndrome."
3. Phenobarbital kinetics could potentially vary significantly in the pediatric population on Continuous Arterio- Venous Hemofiltration dialysis. We looked at a single case of an 18 year old who was on prolonged CAVH and had a h/o seizures.
 - a) Poster Presentation in the Society of Pediatric Research. (2002). Rao, R.G. & John, E., "Phenobarbital kinetics in an adolescent on Continuous Arterio-Venous hemofiltration."
4. Abdominal wall defects are a relatively common neonatal congenital defect which requires involvement of a multidisciplinary team for immediate and ongoing care. We looked at the long term neurodevelopmental outcomes of the infants who were born with abdominal wall defects and discharged from the NICU.
 - a) Poster presentation in the Society of Pediatric Research (2004). "Long term developmental follow up of infants with Abdominal wall defects."
5. Worked on my thesis, "Cerebral Malaria and complicated Presentations" as part of my Pediatric Residency Program at the Shree Sayaji General Hospital in Vadodara, India. We presented the same at the local chapter of the Indian Academy of Pediatrics.
 - a) Presented a paper "Clinical aspects of Neonatal Hypoglycemia" in the State Chapter of the Indian Academy of Pediatrics.
 - b) Presented and worked on a project "Rheumatic fever – epidemiology and clinical presentations" as part of the Public Health policy project in the Final year of Medical School at the Government Medical College, Vadodara, India.

BIOGRAPHICAL SKETCH

Provide the following information for the Senior/key personnel and other significant contributors.
Follow this format for each person. **DO NOT EXCEED FIVE PAGES.**

NAME: John E. Richards

eRA COMMONS USER NAME (credential, e.g., agency login): jrichards

POSITION TITLE: Carolina Distinguished Professor Emeritus, Research Professor

EDUCATION/TRAINING *(Begin with baccalaureate or other initial professional education, such as nursing, include postdoctoral training and residency training if applicable. Add/delete rows as necessary.)*

INSTITUTION AND LOCATION	DEGREE (if applicable)	Completion Date MM/YYYY	FIELD OF STUDY
Biola College, La Mirada, CA California	B.A.	May, 1976	Psychology, Humanities
State University, Fullerton, CA	M.A	Dec, 1978	Psychology
University of California, Los Angeles, CA	Ph.D.	May, 1982	Psychology

A. Personal Statement

The present proposal will study autonomic nervous system development in preterm infants with respect to their neurodevelopmental outcome. I have expertise in the analysis of HR data, including measures of parasympathetic vagal activity on HR. I have used HR measures in preterm and full term infants, and infants with neurodevelopmental disorders (e.g., ASD, ASD-siblings, FXS). I have expertise in the study of infant attention, using a wide variety of measures (EEG, ERP, HR, MRI, fMR). I have a well-known model for using infant heart rate as an index of sustained attention.

Ongoing and recently completed projects that I would like to highlight include:

R01 HD19894-31-35
Richards (PI)
8/01/17 – 8/31/2023
Development of sustained attention in infants

R01 MH090194-01
Roberts, Jane E. (PI); role CO-I
5/01/2016 – 4/30/2021

Citations:

a. Mallin, B.A., & Richards, J.E. (2012). Peripheral stimulus localization by infants of moving stimuli on complex backgrounds. *Infancy*, 17, 692-714.

b. Xie, W., & Richards, J. E. (2016). Effects of interstimulus intervals on behavioral, heart rate, and event-related potential indices of infant engagement and sustained attention. *Psychophysiology*, doi:10.1111/psyp.12670. PMID: PMC4949138.

c. Tonnsen, B., Richards, J.E., & Roberts, J.E. (2018). Heart Rate-Defined Sustained Attention in Infants at Risk for Autism. *Journal of Neurodevelopmental Disorders*. 10(1), 7. PMID: PMC5811956.

d. Bradshaw, J., Fu, X., & Richards, J.E. (2024). Infant sustained attention differs by context and social

content in the first 2 years of life. *Developmental Science*, doi: 10.1111/desc.13500

B. Positions and Honors

Positions and Employment

Carolina Distinguished Professor, Emeritus, January 2021 to present

Carolina Distinguished Professor, August 2003 to 2020.

Research Professor, January, 2021 to present.

Professor, August, 1992 to 2020

Associate Professor, August, 1987, to July, 1992.

Assistant Professor, August, 1982, to July, 1987.

Other Experience and Professional Memberships

Fellow, Society for Psychophysiological Research, 2022 -

External Advisory Committee, Baby Siblings Research Consortium, Autism Speaks, 2015-2017.

Executive Committee, International Society on Infant Studies, 2012-2018.

Fellow, American Association for Advancement in Science, 2010-...

Fellow, American Psychological Society, January 2008-...

MERIT Award, NICHD, National Institutes of Health, March, 2007 to February, 2018

Fellow, Division 3, American Psychological Association. January, 2003-...

Carolina Distinguished Professor, University of South Carolina, August, 2003-2020

Fellow, Division 7, American Psychological Association. January, 2000 - ...

Associate Editor: *Infancy*, 1998-2004

Associate Editor: *Biological Psychology*, 1996-2001

Research Scientist Development Award, NIMH, May 1992 to April, 1997.

Associate Editor: *Psychophysiology*, 1991-1993

Russell Award for Research in Humanities and Social Sciences, University of South Carolina. 1991.

New Investigator Research Award, NICHD, August, 1985 to July, 1988.

Member, Cognitive Neurosciences Society; International Society of Infant Studies; Society for Psychophysiological Research; Society for Research in Child Development

C. Contributions to Science

1. I have a long-standing and continuing expertise in the development of sustained attention in young infants. Using psychophysiological measures and innovative behavioral manipulations, I have shown that fixation in young infants towards interesting visual stimuli can be distinguished into stimulus orienting, sustained attention, and attention termination. These attention phases show different developmental pathways; i.e., early onset of stimulus orienting, and extended development of sustained attention throughout the first year of postnatal life.

This distinction has been applied across a wide range of stimuli and testing situations, such as to simple stimuli, interesting video stimuli, and television and movie programs.

- a. Mallin, B.A., & Richards, J.E. (2012). Peripheral stimulus localization by infants of moving stimuli on complex backgrounds. *Infancy*, 17, 692-714. PMID: PMC3963520.
- b. Xie, W., & Richards, J. E. (2016). Effects of interstimulus intervals on behavioral, heart rate, and event-related potential indices of infant engagement and sustained attention. *Psychophysiology*, doi:10.1111/psyp.12670. PMID: PMC4949138.
- c. Tonnsen, B., Richards, J.E., & Roberts, J.E. (2018). Heart Rate-Defined Sustained Attention in Infants at Risk for Autism. *Journal of Neurodevelopmental Disorders*, 10(1), 7. PMID: PMC5811956.
- d. Bradshaw, J., Fu, X., & Richards, J.E. (2024). Infant sustained attention differs by context and social content in the first 2 years of life. *Developmental Science*, doi: 10.1111/desc.13500

2. I have integrated my study of attention with studies of responses to stimuli using EEG and ERP. The EEG is a direct measure of brain activity. I have shown that a commonly studied ERP component in infant participants, the Nc, is primarily related to attention, rather than short-term recognition memory or other long-term recognition processes. To study the brain sources of the ERP during attention, I developed new methods for using cortical source analysis of ERP with infant participants. This includes the use of structural MRI with individual infants who then participate in psychophysiological experiments. From the structural MRIs a realistic model of the head is

made, and then realistic finite-element-method meshes can be used to do source analysis. We have shown the cortical sources for the Nc and other components, and are currently expanding this work to include infants' responses to faces. One strategy of mine is to do experiments with adult participants to sharpen the cortical source tools, and then to apply these to infant participants; so, I have several papers using cortical source analysis of ERP with adults.

- a. Gao, C, Conte, S., Richards, J.E., Xie, W., & Hanayik, T. (2019). The neural sources of N170: Understanding timing of activation in face-selective areas. *Psychophysiology*. 56 DOI:10.1111/psyp.13336. PMCID: PMC6508977.
- b. Conte, S., Richards, J.E., Guy, M.W., Xie, W., & Roberts, J.E. (2020). Face-sensitive brain responses in the first year of life. *Neuroimage*, <https://doi.org/10.1016/j.neuroimage.2020.116602>. PMCID: PMC7085434.
- c. Conte, S., & Richards, J.E. (2022). Cortical source analysis of event-related potentials: A developmental approach. *Developmental Cognitive Neuroscience*. 2022 Apr: 54:101092. Doi:10.1016/j.dcn.2022.101092. Epub 2022 Feb 25. PMCID: PMC8885610.
- d. Conte, S., Richards, J.E., Fox, N.A., Valadez, E.A., McSweeney, M., Tan, E., Pine, D.S., Winkler, A.M., Liuzzi, L., Cardinale, E.M, White, L.K., & Buzzell, G.A. (2023). Multimodal study of the neural sources of error monitoring in adolescents and adults. *Psychophysiology*, <https://doi.org/10.1111/psyp.14336>. PMCID: PMC10524909 (available on 2024-10-01).

3. I have examined the development of attention with measures of brain activity in infants and children with neurodevelopmental disorders, including children with FXS, infants at risk for ASD outcomes, and children with ASD diagnoses. These studies include measures of heart rate the EEG/ERP, and the brain sources of face processing.

- a. Tonnsen, B., Richards, J.E., & Roberts, J.E. (2018). Heart Rate-Defined Sustained Attention in Infants at Risk for Autism. *Journal of Neurodevelopmental Disorders*, 10:7, doi: 10.1186/s11689-018-9224-2. PMCID: PMC5811956.
- b. Guy, M. W., Richards, J. E., Hogan, A. L., & Roberts, J. E. (2021). Neural Correlates of Infant Face Processing and Later Emerging Autism Symptoms in Fragile X Syndrome [Original Research]. *Frontiers in Psychiatry*, 12(2019). <https://doi.org/10.3389/fpsyt.2021.716642>
- c. Guy, M.W., Richards, J.E., & Roberts, J.E. (2022). Head model selection and cortical source analysis of the N290 ERP component in infants at high risk for autism. *Brain Sciences*. 12(9), 1129. PMCID: PMC9497227.
- d. Richards, J.E., Guy, M.W., Hogan, A.L., & Roberts, J.E. (2023). Neural correlates of face processing among preschoolers with fragile X syndrome, autism spectrum disorder, autism siblings, and typical development. *Autism Research*. <https://doi.org/10.1002/aur.3045>.

4. Many neuroimaging methods use a standard MRI template to do media segmentation (WM, GM, CSF) and stereotaxic atlas identification (e.g., macroanatomical identification with Talairach maps). However, the brain composition of infants is significantly different than that of adults, so I began a set of studies to develop neuroanatomical MRI templates for infants. As a result of this impetus, I now have a "Neurodevelopmental MRI Database" that provides average MRI templates, head media priors, and stereotaxic atlases in small age increments across the lifespan (e.g., 1.5 increments in first year; 3 months in second year, yearly through age 5; every six months from age 5 to 20-24; 5 year increments to age 89 years This database is publicly available and is has been accessed by over 150 labs both in the US and internationally.

- a. Sanchez, C.E., Richards, J.E., & Almli, C.R. (2011). Neurodevelopmental MRI brain templates for children from 2 weeks to 4 years of age. *Developmental Psychobiology*. Doi:10. 1-02/dev20579. PMCID: PMC3184192.
- b. Richards, J.E., & Xie, W. (2015). Brains for all the ages: Structural neurodevelopment in infants and children from a life-span perspective. In J. Benson (ed), *Advances in Child Development and Behavior* (Vol 7).
- c. Fillmore, P.T., Richards, J.E., Phillips-Meek, M.C., Cryer, A., & Stevens, M. (2015) Stereotaxic MRI brain atlases for infants from 3 to 12 months. *Developmental Cognitive Neuroscience*. DOI:10.1159/000438749. PMCID:

PMC4644510.

- d. Richards, J.E., Sanchez, C., Phillips-Meek, M., & Xie, W. (2015) A database of age-appropriate average MRI templates. Neuroimage, doi:10.1016/j.neuroimage.2015.04.055. PMCID: PMC4630162.

5. I have used the “Neurodevelopmental MRI Database” to develop tools to be used for neuroimaging with infants and children. I have structural MRIs from over 4000 participants, including nearly 200 infants from age of 3 to 12 months. I have used these MRIs and the segmenting from the database to develop new tools that can be used with infant neuroimaging. For example, I developed techniques that determine scalp positions on the infants head and the underlying cortical areas at those scalp locations. This may be used to identify the neural sources that are likely generating the optical measurements used in “NIRS” (near-infrared spectroscopy).

Additionally, I have created average electrode positions for adults, adolescents, children, and infants. I have used these with collaborators to study infant attention and cognition with neuroimaging tools.

- a. Richards, J.E., Boswell, C., Stevens, M., & Vendemia, J.M.C. (2015). Evaluating methods for constructing average high-density electrode positions. Brain Topography, 28, 70-86 doi:10.1007/s10548-014-0400-8 PMCID: PMC4293342.
- b. Emberson, L.E., Cannon, G., Palmeri, H., Richards, J.E., & Aslin, R.N(2016). <http://dx.doi.org/10.1016/j.dcn.2016.11.002>. Using fNIRS to Examine Occipital and Temporal Responses to Stimulus Repetition in Young Infants: Evidence of Selective Frontal Cortex Involvement. Developmental Cognitive Neuroscience. PMCID: PMC5253300.
- c. Fu, X., & Richards, J.E. (2022) Age-related changes in diffuse optical tomography sensitivity profiles from childhood to adulthood, Journal of Biomedical Optics, 27(8), 083004 (2022), doi: 10.1117/1.JBO.27.8.083004. PMCID: PMC9270691.
- d. Conte, S., Zimmerman, D., & Richards, J.E. (2024). White matter trajectories over the life span. PLOS One. 19(5), e0301520. PMCID: PMC11101104.

Full bibliography:

<http://www.ncbi.nlm.nih.gov/myncbi/browse/collection/40569197/?sort=date&direction=ascending>

PHS 398 Cover Page Supplement**1. Vertebrate Animals Section**

Are vertebrate animals euthanized? ☐ Yes ☒ No

If "Yes" to euthanasia

Is the method consistent with American Veterinary Medical Association (AVMA) guidelines?

☐ Yes ☐ No

If "No" to AVMA guidelines, describe method and provide scientific justification

.....

2. *Program Income Section

*Is program income anticipated during the periods for which the grant support is requested?

☐ Yes ☒ No

If you checked "yes" above (indicating that program income is anticipated), then use the format below to reflect the amount and source(s). Otherwise, leave this section blank.

*Budget Period	*Anticipated Amount (\$)	*Source(s)
----------------	--------------------------	------------

3. Human Embryonic Stem Cells Section

*Does the proposed project involve human embryonic stem cells? ☐ Yes ☒ No

If the proposed project involves human embryonic stem cells, list below the registration number of the specific cell line(s) from the following list: http://grants.nih.gov/stem_cells/registry/current.htm. Or, if a specific stem cell line cannot be referenced at this time, check the box indicating that one from the registry will be used:

☐ Specific stem cell line cannot be referenced at this time. One from the registry will be used.

Cell Line(s) (Example: 0004):

4. Human Fetal Tissue Section

*Does the proposed project involve human fetal tissue obtained from elective abortions? ☐ Yes ☒ No

If "yes" then provide the HFT Compliance Assurance

If "yes" then provide the HFT Sample IRB Consent Form

5. Inventions and Patents Section (Renewal applications)

*Inventions and Patents: ☐ Yes ☒ No

If the answer is "Yes" then please answer the following:

*Previously Reported: ☐ Yes ☐ No

6. Change of Investigator/Change of Institution Section

☐ Change of Project Director/Principal Investigator

Name of former Project Director/Principal Investigator

Prefix:

*First Name:

Middle Name:

*Last Name:

Suffix:

☐ Change of Grantee Institution

*Name of former institution:

PHS 398 Research Plan

Introduction	
1. Introduction to Application (for Resubmission and Revision applications)	Dail_Introduction_Final_June_28_2024.pdf
Research Plan Section	
2. Specific Aims	Dail_Specific_Aims.pdf
3. Research Strategy*	Dail_Research_Strategy_Final.pdf
4. Progress Report Publication List	
Other Research Plan Section	
5. Vertebrate Animals	
6. Select Agent Research	
7. Multiple PD/PI Leadership Plan	Dail_Multiple_PD_Plan_Final.pdf
8. Consortium/Contractual Arrangements	Dail_Consortium_Contractual.pdf
9. Letters of Support	Dail_Letters_of_Support.pdf
10. Resource Sharing Plan(s)	Dail_Resource_Sharing_Plan_Final.pdf
11. Other Plan(s)	DATA_MANAGEMENT_AND_SHARING_PLAN.pdf
12. Authentication of Key Biological and/or Chemical Resources	
Appendix	
13. Appendix	

Introduction

We are pleased the reviewers found our conceptual framework and proposed methods innovative and of high significance. Our project was seen as feasible with novel analytic approaches, novel theoretical ideas, strong pilot data, and led by stellar MPIs with a strong investigator team and ample resources to perform the work. Our team is very grateful for the issues/weaknesses the reviewers raised, as we have substantially improved our proposal in addressing the following concerns.

We have clarified that our goal for **this** proposal, at **this stage** in our research is **not to outperform more scalable/translatable predictors**, **not** to assess feasibility of uptake in standard of care, and **not** to suggest potential interventions. These are all **later-stage** research programs. There are currently no established, scalable, neonatal predictors (e.g., birth weight, cumulative risk score) that are specific to ASD/NDI outcomes. **Our goal here is to evaluate ANS as an etiological pathway to ASD in VPTs** and ultimately, following study completion, to develop methods for scaling up ANS dysfunction as a viable biomarker for VPTs. Regarding potential interventions, a highly predictive biomarker can, at the very least, lead to psychoeducational and parent coaching interventions that help parents prepare for their family and child's future. **We acknowledge that the development of ASD-specific features in VPT infants has been documented**; we now emphasize our focus on characterizing both ASD and NDI from as early as 12 months, along with medical and maternal factors.

Recruitment/Feasibility: Due to the concern that the research nurse would not have enough time to **conduct study procedures**, we have increased the research nurse effort from 20% to **50%**, providing ample time to consent and conduct study activities, and is now the same as effort for our last neonatal study with the same study activities. **There was a concern over study activities**, and additional detail has been added to point out this is standard of care biosignal monitoring and was used successfully in our last 5-year study in the same NICU. We have added detail about study activities to take place in the NICU. **Enrollment concerns:** We have added detail through supporting documents that this NICU admits ~220 VPTs annually, which is more than enough to recruit 12 infants per month (144 annually), to support our intention to enroll 260 infants in 2 years. **Retention:** We have added consultation with our existing community advisory board, created by MPI Bradshaw and leveraged in her ongoing studies to support retention. **Family history of ASD** will be assessed during parent interview and with ASD screeners for any biological siblings. In addition, we have added exclusion criteria of VPTs with known **genetic/chromosomal abnormalities**, per reviewer suggestion.

Consideration of Confounds, Social Inequities/Social Drivers of Health, Maternal/Infant Factors: Due to reviewer concern, we have more explicitly added consideration for **social drivers of health** (geographic location, income, maternal education), **maternal/early caregiver factors** (psychopathology, parenting stress), and **postnatal infant factors** (early intervention, medical events). We clarify why NICU data is not uploaded from EMR to REDCap directly. The proposed **morbidity index** was created from a VPT group who were largely comparable to our proposed participants (i.e., expected to survive to NICU discharge). However, all neonatal morbidities and severity will be recorded and could be used for Aim 2 **in case the morbidity score shows limited variability**.

Analyses: Due to reviewer concern over lack of statistical expertise, we have added detail around **statistical expertise** of MPI Bradshaw (latent growth curve analysis, linear mixed effects modeling for longitudinal data, modeling nonlinear developmental change in infancy) and Co-I O'Reilly (advanced statistical analyses). **A reviewer noted limited and underspecified analyses**, which prompted us to give details analyzing cross-model temporal associations. A classical supervised ML clustering approach will maximize our chances of obtaining stable clusters that group trajectories predictive of specific outcomes, thus supporting the use of early trajectories to predict diagnostic outcomes. We have added **more detail on QA/QC and machine learning methods**: use of the Silhouette Coefficient, t-SNE visualization, and cluster stability for QA/QC of clustering analysis; analysis of redundancy between features (hierarchically clustering of correlation matrices for Aims 1-2; Principal Component Analysis for Aim 3); pruning of highly correlated features or reduction of redundancy of feature set, as necessary to address issues with **multicollinearity**. **To provide additional sensitivity** (Aim 1), additional features will be extracted (1st, 50th, and 99th percentiles, inter-quartile range) for the measured values across NICU stay. **The training vs. testing sets ratio** for subject-level nested-cross-validation is as follows: train=70%; test=15%; validation=15%. **Sex will be evaluated as a moderator for Aim 2 analyses**. **Due to concern over missing data**, we have added details on ECG outlier/artifact detection as well as missing data and imputation. Finally, we will **share all autonomic and phenotypic data** (see *Resource Sharing Plan*).

Autonomic nervous system (ANS) development begins in the prenatal period and is critical for a successful transition to extrauterine life. ANS supports vital functions, like cardiac activity and thermal regulation, as well as higher-level cortical processes from birth through adulthood. ANS function can be impaired or altered by preterm birth and/or preterm medical morbidities¹ and ANS dysfunction has been associated with several neurodevelopmental disorders, including **autism spectrum disorder (ASD)**.² Interestingly, the rate of ASD in infants born preterm (PT) is more than triple that of the general population,³ with recent reports of a 10-fold increase for very PT infants (VPT; born <32 weeks gestation)⁴ and a 20-fold increase for extremely PT infants (<28 weeks).⁵ Yet, **the etiological link between PT birth and the development of ASD remains unknown.**

Significant ANS maturation occurs in the last prenatal trimester, and PT birth before 32 weeks significantly disrupts ANS development and the overall functioning of the PT infant.⁶ ANS dysfunction may therefore represent an underlying mechanism contributing to the higher incidence of ASD among children born VPT. Early identification and intervention for ASD is critical for improving developmental outcomes and supporting overall child and family well-being. Thus, identifying the **specificity and time course of ANS dysfunction** in VPT infants in relation to **emerging ASD symptoms** and **neurodevelopmental impairment (NDI)** could significantly improve developmental outcomes for this vulnerable population and contribute to novel mechanistic frameworks of ASD.

ANS function in VPTs can be determined through standard physiological vital signs, e.g., heart rate and body temperature, already recorded in the neonatal intensive care unit (NICU). Heart rate and thermal gradients (central and peripheral body temperatures) show an infant's ability to control blood flow to the brain and the rest of the body, through parasympathetic and sympathetic activity. Our preliminary studies show that **continuous, multimodal** measures of ANS function – **heart rate indices (HRIs)** and **thermal gradients** – in the **NICU** are strong predictors of ASD over a year later in VPT infants ($r=.81$) and represent a promising avenue for early risk assessment.⁷ Identification of infants who are at heightened developmental risk prior to NICU discharge can guide very early intervention efforts and improve childhood outcomes.⁸ This **prospective, longitudinal, study of 200 VPT** infants builds on our preliminary work and fills critical research gaps in three distinct ways, as follows.

Aim 1: Examine preterm ANS dysfunction as a specific predictor of ASD symptoms at outcome. In ASD, ANS dysfunction is pervasive over the lifespan, and ANS measures have been suggested as candidate biomarkers for ASD.^{9,10} In VPT infants, HRIs of ANS dysfunction have been documented as predictors of brain injury^{11,12} and NDI.¹²⁻¹⁵ Similarly, recent research suggests an association between preterm HRIs and ASD.^{7,16} But the predictive specificity of **multimodal** measures of PT ANS dysfunction for **ASD** versus broader NDI outcomes has never been examined. This study is critical to determining the mechanistic links between VPT birth, PT ANS dysfunction, and ASD. We will characterize ANS function using complementary, continuous measures – HRIs and thermal gradients – from birth to NICU discharge. A machine learning approach will be used to examine associations among these PT ANS features to ASD (symptom severity) and NDI (motor/cognitive/language impairments)¹⁵ at outcome (36 months). Our work suggests ANS dysfunction exists in VPT infants¹⁷ and is associated with ASD symptoms.^{7,18,19} We hypothesize that PT ANS dysfunction will be associated with 3-year outcomes, with stronger (i.e., more specific) associations to ASD than NDI.

Aim 2: Evaluate preterm morbidities and sex as potential moderators in the association between PT ANS dysfunction and symptoms of ASD and NDI at outcome. We will use medical records to extract PT-related morbidities that may moderate the association between ANS dysfunction and ASD/NDI symptoms at outcome (36 months). Established significant morbidities related to PT ANS dysfunction include:²⁰ severe brain injury, infection²¹ or necrotizing enterocolitis (NEC), bronchopulmonary dysplasia (BPD) or need for mechanical ventilation, retinopathy of prematurity (ROP),^{21,22} and/or patent ductus arteriosus (PDA). PT morbidities and male sex have been associated with both ASD and NDI outcomes,^{16,23-25} and we hypothesize that PT morbidities and male sex will strengthen the association between ANS dysfunction and ASD/NDI.

Aim 3: Characterize and differentiate the emergence of ASD symptoms and NDI in VPT infants. VPT infants and infants with ASD show significant overlap in clinical presentation in the first 3 years of life,²⁶ and recent findings suggest that VPTs with ASD show worsening broad developmental impairment from 12-24 months.²⁷ Moreover, our data in full-term infants with ASD point to the first year of life as a critical developmental window, with ASD-specific symptoms emerging as early as 9 months.^{28,29} These early-emerging developmental complexities call for rigorous, repeated quantification of ASD symptomology and NDI in VPT infants using highly sensitive and specific measures across the first 3 years of life. Such an approach can characterize phenotypic profiles of VPT infants and elucidate complex, heterogeneous developmental pathways that may lead to ASD. VPT infants experience a myriad of developmental vulnerabilities; the use of both developmentally broad and ASD-specific measures at each time point (months 12, 24, 36) is essential to support differential characterization of ASD-specific and broad NDI symptoms across the first 3 years.

RESEARCH STRATEGY

SIGNIFICANCE

Early Identification Catalyzes Early Intervention, *Especially for Vulnerable Infants*

Autism spectrum disorder (ASD) is a neurodevelopmental disorder that affects ~2.3% of the population and carries a lifelong financial cost of \$1.4-2.4 million per individual, thus representing a pressing public health challenge.^{30,31} Early intervention plays a pivotal role in the lives of children and families affected by ASD – improving functional outcomes for the individual, enhancing the quality of life for the family, and reducing the lifelong financial cost.^{32,33} Unfortunately, with an ASD diagnosis and intervention onset occurring after age 4-6 years,^{34,35} we are missing out on a critical period for early intervention that could capitalize on peak neuroplasticity in the first 2 years of life.^{36,37} Later identification of ASD occurs disproportionately in more vulnerable populations,^{30,31,38} such as those with medical comorbidities, neurodevelopmental impairments (NDIs),³² as well as those from marginalized racial/ethnic groups and lower socioeconomic status (SES), all of which are highly prevalent in infants born preterm (PT).

Across the world, 15 million PT infants are born each year.³⁹ Prematurity is the leading cause of short and long-term morbidities^{40,41} and PT infants are at a significantly increased likelihood for ASD and NDI. The risk for NDI outcomes increases exponentially with lower gestational age (GA)¹⁵ and the occurrence of neonatal morbidity in the Neonatal Intensive Care Unit (NICU).^{21,22,42} Socioeconomic disparities play a substantial role in PT birth outcomes, with evidence for multiplicative effects of SES and GA on NDI outcomes.^{26,43,44} The burden of PT birth equates to at least \$26 billion in the United States (US), which includes medical, educational, and lost productivity expenses.⁴⁵ The American Academy of Pediatrics emphasizes the importance of early intervention and coordinated efforts for follow-up care to maximize childhood health and development outcomes.⁴⁶

The rate of ASD in PT infants is more than triple that of the general population,³ with recent reports of a 10-fold increase for very PT infants (VPT; born <32 weeks GA)⁴ and a 20-fold increase for extremely PT infants (<28 weeks GA).⁵ Thus, PT infants are at an elevated likelihood for ASD while also experiencing medical, developmental, and socioeconomic risk factors that put them at a very high risk of later identification of ASD and even later intervention. **This is a critical barrier in the field** – earlier identification of ASD within the first year of life of PT infants can help address this health disparity by *catalyzing earlier intervention*, leading to greater *improvements in long-term outcomes*. Evidence that forms the **scientific basis of this study** includes: 1) the importance of autonomic nervous system (ANS) in environmental adaptation, social interaction, and learning, 2) the presence of ANS dysfunction at birth in VPT infants and throughout the lifespan of individuals with ASD, 3) the exacerbation of ANS dysfunction by PT morbidities, and 4) the elevated prevalence of ASD and NDI phenotypes among VPT infants. This strong scientific foundation and the absence of clear mechanistic explanations for distinct developmental pathways present three gaps:

- (1) **Multimodal measurement** has not yet been leveraged to analyze the specific and combined contributions of ANS subcomponents (i.e., sympathetic and parasympathetic) to PT autonomic dysfunction and developmental outcomes.
- (2) **Prematurity-associated morbidities** have a compounding effect on ANS dysfunction, but their impact on the association between ANS and developmental outcomes is not well understood.
- (3) **Prospective, comprehensive characterization of trajectories** of ASD-specific features and neurodevelopmental impairment throughout infancy in VPTs is lacking.

These gaps **call for further research** on ANS dysfunction as a mechanism for ASD in VPT infants.

ANS Dysfunction: A Mechanistic Hypothesis for Increased Incidence of ASD in PT Children

Existing evidence suggests that ANS dysfunction is experienced by both PT infants and children with ASD. In this study, we will evaluate **ANS dysfunction** as a novel mechanism for the increased incidence of ASD in VPT infants by surveilling NICU-based indicators of ANS functioning and comprehensively phenotyping the emergence of ASD symptoms and NDI across the first 3 years of life. Determining the association between PT ANS dysfunction and ASD/NDI outcomes by age 3 years will significantly improve scientific knowledge of ASD pathogenesis. Ultimately, it will pave the way for translatable neonatal indicators of risk that will lead directly to tailored developmental follow-up and **reduce disparities in timely diagnosis and intervention** for this vulnerable population. This study is well aligned to address PAR-21-201 which calls for studies designed to elucidate the etiology, epidemiology, and diagnosis in relation to ASD.

Significant ANS maturation occurs in the second half of pregnancy, and PT birth (born <37w GA) significantly disrupts ANS development and the overall functioning of the PT newborn.^{47,48} For PTs, reduced ANS activity fails to normalize at term age equivalent,⁴⁹ and may alter long-term temporal programming of ANS maturation.⁵⁰ Following birth, ANS plays a vital role in the adaptive competence of the newborn through balanced adaptive responses of two major subcomponents: the sympathetic nervous system (“fight or flight”; SNS) and the

parasympathetic nervous system (“rest and digest”; PNS). In response to ever-changing environmental contexts, these systems enact physiological changes that regulate basic functions, such as heart rate and body temperature. In early prenatal development, physiological functions are driven primarily by sympathetic influence. Higher cortical processes begin to integrate autonomic control in the late prenatal and early postnatal period leading to increasing influence of the PNS. As ANS matures, parasympathetic tone serves as the foundation for several higher-level behaviors that support calm states of alertness and sustained attention, ultimately providing countless opportunities for social and language learning. In this way, ANS is considered a “developmental resource” in the neonatal and early infant period that mediates infant-environment interactions.⁵¹

In the case of PT birth, a drastic transition to the extrauterine environment occurs amidst ongoing ANS maturation and is a clash that is compounded by exposure to medical stressors. As a result, autonomic maturation is impaired throughout the PT period, initiating a developmental cascade that may disrupt the physiological adaptation to the environment and integration of autonomic control into later-developing, higher-level cortical processes important for social learning and cognitive development. These early disruptions may be at the root of the emergence of ASD symptoms (social-communication delays, social interaction impairments, repetitive behaviors, atypical sensory reactivity). Emerging evidence for this developmental pathway from our group links early infant ANS/attention regulation in full-term infants to later social-communication and ASD.^{18,52,53}

1. Multimodal Measurement of PT ANS Dysfunction

Current studies of PT ANS dysfunction are largely limited to unimodal measurement of ANS function, most often heart rate indices (HRIs), such as heart rate variability, derived from electrocardiographic (ECG) recordings⁵⁴. The current study overcomes this limitation by leveraging multimodal measurement of ANS function that includes recordings of **peripheral and central body temperature** and **ECG**.

Temperature-Derived Measures of ANS Dysfunction. Body temperature is regulated in part by vasomotor tone, or constriction and dilation of blood vessels, which may become altered with autonomic instability.⁵⁵ In an infant with a healthy, stable ANS, the temperature is higher for the central body than the periphery, with a Central (abdominal) - Peripheral (foot) Temperature difference (CPTd) of 1-2° C.⁵⁶ Thus, CPTd is a measure of the **thermal gradient**, or blood perfusion across the infants’ body. The SNS plays a critical role in preventing abnormal thermal gradients through peripheral vasoconstriction.⁵⁷ Under normal conditions, the vasomotor center in the brain continuously transmit signals to the sympathetic vasoconstrictor nerve fibers throughout the body.⁵⁸ This continual firing is called vasoconstrictor tone and is responsible for maintaining vasomotor tone, which allows constriction of peripheral vessels to conserve heat. Peripheral temperature correlates with peripheral blood flow.⁵⁹ As blood flow to the extremities increases, higher peripheral temperatures occur. Our research has found that assessing vasomotor tone through surveillance of the CPTd offers a novel approach to evaluating ANS dysfunction.^{60,61} Longitudinal abnormal thermal gradients (CPTd <0° or CPTd ≥2°C) are a sign of microcirculatory dysfunction and can be used as a sign of ANS dysfunction (likely driven by abnormal SNS activation), which may be an indicator of neurodevelopmental outcome and ASD.⁷ As in our previous work,^{61,62} this study will use continuous measurement of PT abnormal thermal gradients as an index of ANS dysfunction.

ECG-Derived Measures of ANS Dysfunction. ECG recordings of heart activity yield several **HRIs** that can characterize ANS function, including beat-to-beat heart rate, transient heart rate accelerations/decelerations (i.e., tachycardias and bradycardias), and heart rate variability (HRV), and heart rate variability at the frequency of respiration (i.e., respiratory sinus arrhythmia; RSA). Each HRI represents unique and combined effects of sympathetic and parasympathetic activity, where sympathetic activity generally increases heart rate and parasympathetic activity reduces heart rate. The magnitude of HRV and RSA measures increase over the first years of life, and higher magnitudes indicate a more adaptive, balanced ANS system.⁶³ This process is disrupted in PT infants, resulting in an increase in RSA magnitude that is 2-3 times slower than that of term infants by 6 months age.⁶⁴ Time and frequency domain analyses of HRV show that ANS function is adversely affected by PT-related morbidities such as mechanical ventilation, infections, and patent ductus arteriosus (PDA).²⁰ This detrimental influence of PT birth on RSA and HRV has been found to persist into childhood and even adulthood.^{65,66} Existing research, including our own, link abnormal HRIs (e.g., transient heart rate accelerations) in the PR period with the later emergence of ASD, with atypical RSA emerging in later infancy (age 2-3 years) for those with ASD, suggesting a potential decline in parasympathetic regulatory mechanisms.^{7,67} While RSA could be affected by external influences on respiration (e.g., ventilatory rate, pressure support, and continuous positive airway pressure), previous studies have documented success in measuring RSA in PT infants on mechanical ventilation and we will include respiratory support and severity of lung disease as covariates in all analyses of ANS dysfunction based on RSA.^{68,69} HRIs can be obtained through standard cardiopulmonary monitoring systems and, for this study, will be obtained via HeRO bedside monitors,⁷⁰ as we have done

previously.^{7,62} Continuous heart activity recording offers a novel way to evaluate ANS function, including both sympathetic and parasympathetic activation and balance, in VPT infants in the NICU.

The multiplicative effects of ANS dysfunction on several biological functions make mechanistic interpretations of unimodal data insufficient. For example, abnormal thermal gradients may be caused by vasoconstriction resulting from sympathetic overactivation or by reduced vasodilation caused by parasympathetic under activation. Imbalance of the SNS and PNS has been suggested as an ASD-specific biomarker,¹⁰ but a debate remains on what precisely causes this imbalance (e.g., overactive sympathetic or underactive parasympathetic). Multimodal assessment (temperature, ECG) that includes **multiple** biological indicators will allow for a clear, specific description of ANS function and a nuanced understanding of adaptive and maladaptive sympathetic and parasympathetic responses. This will help parse contributions of SNS and PNS in the overall evaluation of ANS dysfunction and test their relative predictive contributions to ASD and NDI.

2. Prematurity-Associated Morbidities, ANS Dysfunction, and Neurodevelopmental Outcome

Infants born PT experience multiple morbidities (see Table 1) and higher major morbidity rates are associated with worse neurodevelopmental outcomes.⁷¹

Preterm Morbidities and ANS Dysfunction. The risk for morbidities increases with earlier GA^{72,73} and recent research suggests that PT morbidities may mediate the adverse effect of GA on PT ANS dysfunction.^{20,71,74} All PT morbidities are associated with periods of ANS dysfunction due to the morbidity-related stress on infants’ already fragile homeostasis – many directly attacking the brain, heart, or lungs. Some researchers posit that ANS dysfunction underlies the need for oxygen in PTs and, in turn, this increased oxygen exposure can cause hyperoxia and induce oxygen radical disease, which links ANS dysfunction to BPD, ROP and NEC morbidities in VPT infants.^{72,75} Thus, early-life ANS dysfunction may both cause and exacerbate PT morbidities.

Table 1. Preterm Infants <32 weeks GA Morbidities and Incidence

Morbidity		Definition	Incidence
	Severe Brain Hemorrhage	Grade III-IV, Periventricular Leukomalacia	6-19%
	Sepsis	Late onset blood infection > 7 days of age	9-50%
	Necrotizing Enterocolitis (NEC)	Infection of the GI tract, requiring medical and/or surgical treatment	5-15%
	Bronchopulmonary Dysplasia (BPD)	Chronic Lung Disease, requiring oxygen use at 36 weeks postnatal age	32-44%
	Patent Ductus Arteriosus (PDA)	The Ductus Arteriosus fails to close post birth and requires medical or surgical treatment for closure if hemodynamically significant	5-60%
	Retinopathy of Prematurity (ROP)	Disorder of the developing retina of preterm infants that can lead to blindness	3-42%

Preterm Morbidities and Developmental Outcome. As ANS plays a large regulatory role in mitigating stressors of extrauterine life in the VPT infant, the combined effects of ANS dysfunction and PT morbidities can greatly decrease optimal health outcomes. The risk for ASD and NDI increases with decreasing GA and limited research has shown an association between PT morbidities and ASD.⁷⁶ For example, the occurrence of 2-4 PT-related morbidities has been associated with positive ASD screening (M-CHAT-R/F)²⁴ and PT sepsis has been associated with a later ASD diagnosis.⁷⁷ Some medical treatments for morbidities (e.g., steroids for BPD) have also been associated with ASD.⁷⁸ This already limited research largely comes from retrospective analysis of medical records and has not addressed how PT morbidities *moderate* associations between ANS dysfunction and *dimensional* ASD and NDI outcomes. Aim 2 will evaluate these PT morbidities as potential moderators of the association between ANS dysfunction in the PT period and ASD/NDI outcomes at 36 months.

Social Drivers of Health (SDOH). It is also critical to consider the role of **SDOH** in child development and mental health research.⁷⁹ Demographic factors and SDOH, such as race/ethnicity, maternal education level, insurance, and housing stability, have major effects on access to maternal prenatal and postnatal healthcare as well as infant and child healthcare, developmental supports, and medical and mental health diagnoses.^{80,81} These barriers to care can heavily influence preterm morbidities and neurodevelopmental outcomes, as our work has suggested.⁸²⁻⁸⁶ Likewise, early infant experience, including maternal stress, postnatal hospitalizations, and early interventions, can influence outcomes. As such, demographic variables, SDOH,^{87,88} maternal stress,⁸⁹ and infant context (intervention, hospitalizations) will be assessed and accounted for in analytic models.

3. Comprehensive Characterization of Emerging ASD-Specific Symptoms and NDI

Rates of both ASD and NDI in VPT infants are significantly greater than that of the general population,³ with many overlapping symptoms common to VPTs both with and without ASD including: motor dysfunction, language deficits, social impairment, atypical sensory processing, and executive function challenges.²⁶ This is further complicated by health disparities wherein PT birth disproportionately effects minoritized and lower SES families and, in turn, lower SES adversely effects developmental outcomes for individuals with and without ASD,^{90,91} as our work has also shown.⁸³ These commonalities complicate differential diagnosis of ASD and NDI. Yet another

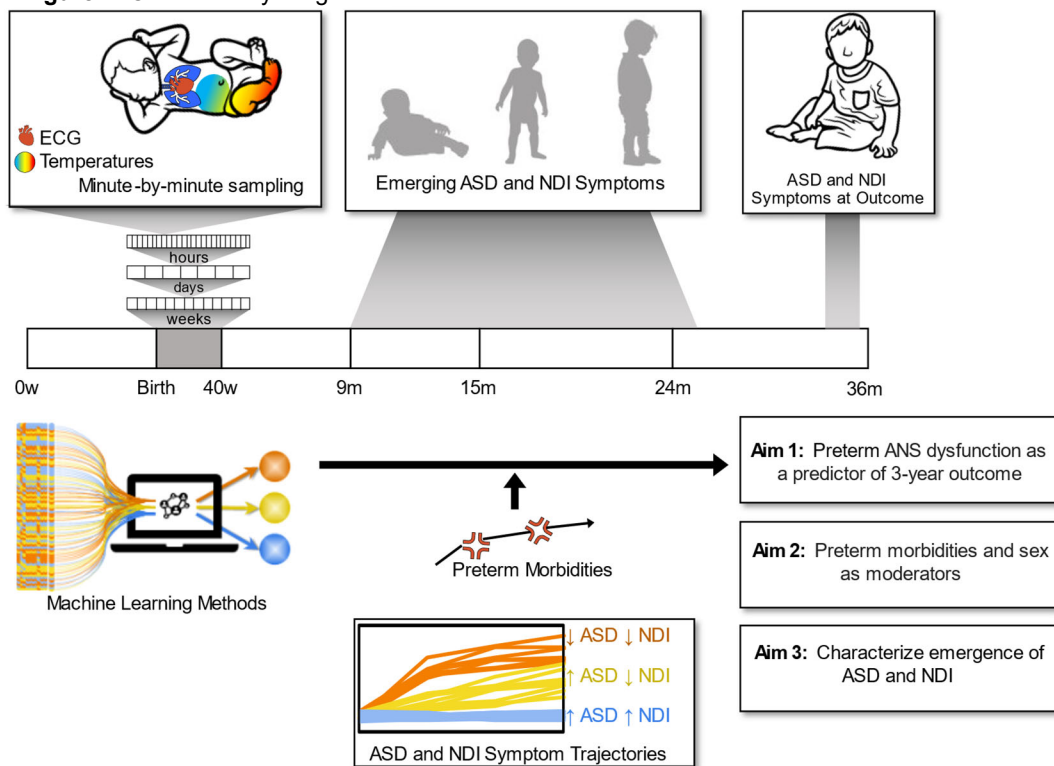
complication is characterizing development and change during a critical period of rapid brain development and skill acquisition. Our preliminary data, along with existing research,^{27,92} suggest differential trajectories of development across domains for VPT infants where strengths and weaknesses are not stable. Moreover, direct observation of ASD-specific symptoms in the first year of life using validated assessment tools²⁹ have yet to be applied to a cohort of VPT infants. Understanding individual differences and shifts in phenotypic profiles of both ASD-specific characteristics and NDI (cognitive, motor, language impairments) from as early as 12 months of age, over the first 3 years of life, is critical for informed and specific diagnostic and intervention.

INNOVATION

This proposal seeks to shift current research paradigms related to PT birth and ASD by prospectively studying a large cohort of VPT infants for ANS dysfunction during their entire NICU hospitalization and monitoring them for the emergence of ASD/NDI symptoms over their first 3 years of life. Through this investigation, we aim to identify abnormal thermal gradients and HRIs that can be used as a biomarker in infancy to forecast atypical neurodevelopmental outcomes and, specifically, ASD. Study results will inform the novel mechanistic hypothesis that ANS dysfunction is an underlying link between PT birth and ASD outcome. Although our extensive experimental data collection protocol is not meant for direct translation into a revised standard of care, it will inform the relevance of considering ANS monitoring, potentially via contactless, more scalable strategies.

Continuous Measurement of ANS Dysfunction in the NICU. PT infants are born with an exceptionally immature autonomic regulatory system that may not fully develop until well after birth.^{65,93-95} We propose to measure ANS dysfunction through continuous recording of biological measures (temperature, ECG) from birth to discharge. Sampling for an average of 9 weeks per infant (average NICU stay = 9 weeks) will yield an average

Figure 1. Overall Study Diagram



of 90,720 minute-by-minute temperature samples and 1512 hours of continuous recordings of ECG *per infant*. This innovative approach will generate an unprecedented volume of longitudinal, multimodal indicators of ANS.

Machine Learning Approach. A machine learning (ML) approach is necessary for the high volume of biological signals that will be collected and processed throughout this study. Our analytic approach leverages ML and artificial intelligence (AI) techniques that support nonlinearity and investigation of multiple ANS measures, while also retaining interpretability, clinical significance, and clinical relevance, of ML algorithm results. Specifically, we will use support vector machine regression, a supervised ML tool for nonlinear regression, in addition to permutation feature importance techniques developed in the context of explainable AI, to identify the most meaningful ANS features in our prediction models. We will also use DBSCAN, an unsupervised ML algorithm to identify clinically meaningful longitudinal phenotypic profiles. These innovative methods represent powerful tools that provide this study with enormous potential for understanding the impact of PT birth on the emergence of ASD symptoms.

APPROACH

Preliminary Data

Thermal Gradients as a Measure of ANS Dysfunction. Our series of studies have shown that ANS dysfunction in VPT infants is observable via abnormal thermal gradients, defined by a condition of warmer, better perfused extremities with simultaneous colder and less perfused core body parts intermittently over their first month of life.^{60,61,96-98} CPTd is a measure of microcirculatory dysfunction that can capture this phenomenon.⁹⁹ We, and others, have shown that abnormal thermal gradients, defined as a CPTd $< 0^{\circ}$ or CPTd $> 2^{\circ}$ C, is correlated with the occurrence of infection in VPT infants.^{61,99,100} The state of circulatory dysfunction and centralization of blood flow in relationship to the presence of bacteria, will cause the CPTd to become increased with researchers measuring CPTd values on average of 2° C, but as high as 9° C^{99,101} in PT infants with sepsis. In severely ill infants, with decreased GA and immature ANS, the CPTd can be quite small ($0-1^{\circ}$ C) or negative, indicating warmer periphery than core (see **Fig. 2**).^{101,102} We used Grant dataloggers and thermistors in our study (R01NR017872) to measure CPTd to predict infection,⁶² a preliminary analysis showed that when an infant had CPTd $>2^{\circ}$ C for more than $> 50\%$ of the time, it had an 86% higher odds of infection compared to infants without temperature differentials $>2^{\circ}$ C.¹⁰³ These data demonstrate that continuous CPTd, derived from readily measurable dual temperature vital signs, can be continuously recorded in VPT infants over their NICU hospitalization and effectively utilized as one of two multimodal measures of ANS function.

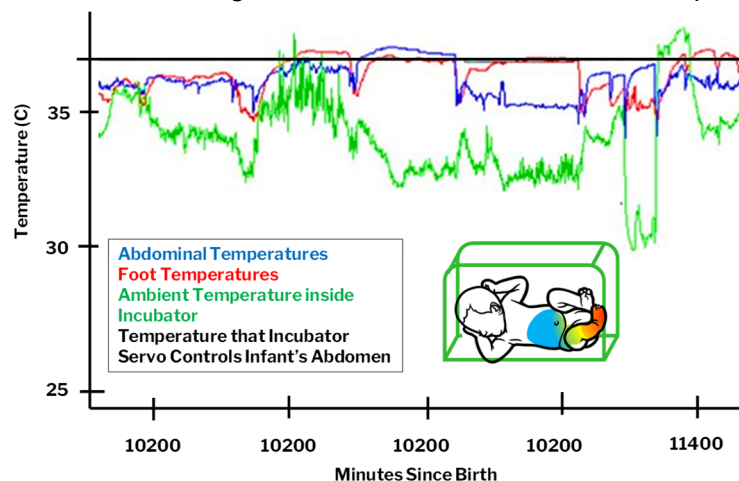


Figure 2. VPT infant body temperatures with intermittent hypothermia, negative central-peripheral temperature difference and normal thermal gradient over 12 hours.

Heart Rate Indices (HRI): A Gold Standard

Measure of ANS Dysfunction in the NICU. Heart rate is regulated by the ANS, with sympathetic activation causing accelerations and parasympathetic activation causing decelerations.¹ ANS dysfunction can cause arrhythmias, vascular irritability, and hypo and hypertension.¹ HRIs, including beat-to-beat heart rate, HRV, mean, standard deviation, and skewness of IBI can be recorded through bedside cardio-pulmonary monitors as we and others have done.^{16,20,104} For example, we have used simultaneous, continuous measurement of temperature and HRIs in the NICU to find the body temperature where infants were most thermally stable.¹⁰⁴ Researchers have used HRV through continuous ECG recordings to measure autonomic maturation in VPT infants hospitalized in a NICU.¹⁰⁵ We recently completed enrollment of 352 infants (R01NR017872) employing HRIs⁷⁰ as an index of ANS dysfunction and predictor of infection.⁶² In an interim analysis of 55 VPT infants, we found that more abnormal HRIs were associated with significantly increased odds of infection on that day.¹⁰³

ANS Dysfunction Measured with Thermal Gradients and HRIs: Associations with Later ASD Symptoms. Preliminary findings⁷ from a funded pilot study (PIs: Dail, Bradshaw) significantly contribute to the scientific basis for the proposed work. In this prospective, longitudinal trial of N=20 VPT infants (a subsample of participants enrolled in R01017872, PI: Dail), continuous measures of minute-by-minute thermal gradients and hour-by-hour abnormal heart rate indices, measured using Heart Rate Characteristic (HRC) scores derived from HeRO Monitors,⁷⁰ were collected for 28 days after birth ($>40,000$ samples/infant). Following NICU discharge, comprehensive neurodevelopmental follow-up consisting of standardized measures of cognition, language, and motor skills were collected at adjusted ages 6, 9, and 12 months for N=12 infants. At 12 months, assessments of social communication and early ASD symptoms were administered. Across all infants in the first 28 days of life, ANS dysfunction was indicated by an average HRC score >1 and abnormal negative thermal gradients for about 20% of the time (6 of 28 days). Neonatal abnormal HRC scores were strongly associated with ASD symptoms at 12 months ($r=0.81$, $p<.01$; **Fig. 3**), as was birth GA, birth weight (BW), and abnormal negative thermal gradients. A regression model including GA, BW, and HRC scores revealed a significant, unique predictive effect of abnormal HRI (HRC score) on ASD outcomes (model $r=0.86$, $p=.017$; HRC Score: $r=.72$, $p=.029$).⁷ **This strong predictive model stands as compelling preliminary data for Aims 1-2.** This R01 is critical in delineating precise HRIs (e.g., tachycardia, bradycardia, HRV, etc.) and other indices (thermal

gradients) that are associated with ASD outcomes. Each index may be sensitive to different underlying neural processes and parasympathetic-sympathetic dynamics. We will therefore go beyond the HRC scores to evaluate how different indices differentially map to ASD versus global ANS dysfunction.¹⁰⁶⁻¹⁰⁹

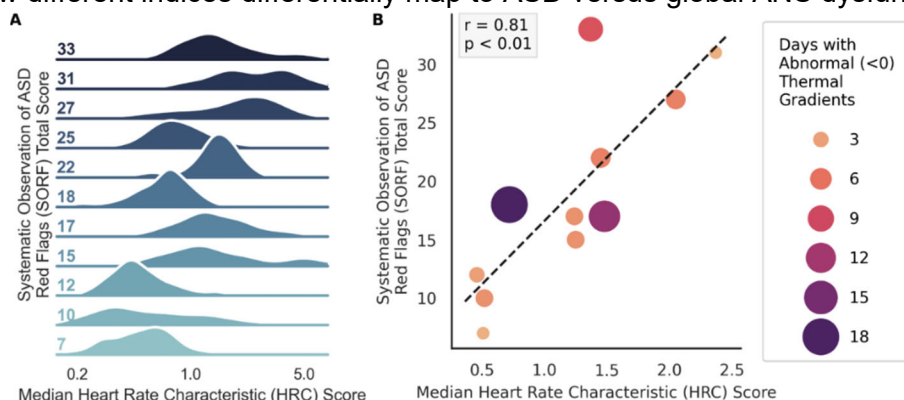


Figure 3. Association between neonatal abnormal heart rate characteristics (HRCs) and 1-year ASD symptoms. Higher scores indicate more heart rate abnormalities and more ASD symptoms. A) Ridgeline plot displaying the distribution of neonatal abnormal HRC scores across participant ASD symptoms (SORF total composite score) at 1 year. B) Scatterplot and associated regression between neonatal abnormal HRC scores across 1-year ASD symptoms (SORF total composite score).

Precise timing and dynamics of multiple ANS measures. HRC scores used in our preliminary data are coarse, hourly estimates of a single composite score. In this study, we will derive **more temporally resolved HRIs** from ECG recordings. Collecting temperatures continuously, with high temporal resolution, will allow for sophisticated analyses of the complex interplay of neonatal autonomic regulation across the preterm period.

Disentangling ASD from NDI in VPT Infants Across the First Three Years. Our preliminary data suggest that significant neurodevelopmental impairments did not emerge in VPTs until 12 months of age when over 50% of the sample exhibited social-communication delays and exceeded the clinical cutoff for ASD risk (Fig 4).⁷ At 12 months, infants were characterized by ASD and NDI as follows: ASD features (high = 33%; moderate = 25%; little-to-none = 42%); NDI manifested by cognitive and/or motor delays (NDI = 67%; no NDI = 33%). 25% of the sample had no evidence of ASD or NDI. Of interest, 50% of the infants with high ASD features evidenced no motor or cognitive NDI. These preliminary descriptive statistics illustrate the **significant variability of developmental outcomes** for VPTs using direct clinical observation and ASD-specific measures. Minimal change in developmental skills was observed prior to 12 months, but significant developmental divergence was observed at 12 months. These **preliminary data** suggest **feasibility** for Aim 3: 1) participant retention and **completion of longitudinal follow-up measures** with VPT infants at evidence-informed time points (based on preliminary data); 2) obtention of **sufficient variability in developmental trajectories and ASD/NDI outcomes** with significant, distinct ASD and NDI symptoms already emerging at 12 months; and 3) use of measures with **sufficient sensitivity** to uniquely quantify ASD and NDI symptoms in infancy. Our data (Fig. 4) suggest significant variability and complexity in development and outcomes and Aim 3 is crucial for unveiling developmental nuances and advancing etiological frameworks of ASD in VPTs.

Research Team

This study is a natural collaboration between **University of South Carolina's (USC) MPI Robin B. (Knobel) Dail**, Health Sciences Endowed Professor, College of Nursing (CON) and MPI Jessica Bradshaw, Associate Professor of Psychology, with the mutual interest of ANS dysfunction in infancy. The current project is an extension of the MPIs' individual research programs and will capitalize on existing infrastructure, resources, and ongoing/completed awards (Dail: R01NR017872;^{62,82} Bradshaw: R01MH1364619, K23MH120476, foundation).

Neonatal Expertise: MPI Robin Dail is a nurse scientist with expertise in morbidity and mortality in PT infants. She has collaborated with **Prisma Richland NICU** neonatologists, on studies of ANS in VPTs over the last 6 years. **Co-I Ravi Rao**, Neonatologist and Medical Director will oversee a 50% research coordinator hired for the proposed study (see Budget Justification). **Co-I Kayla Everhart** is a CON neonatal researcher and has worked with MPI Dail since 2020. She is currently running a funded RCT in Prisma Richland's NICU with MPI Dail. Drs. Dail, Everhart, Rao, and research nurses/coordinators are all trained in collecting thermal gradient and

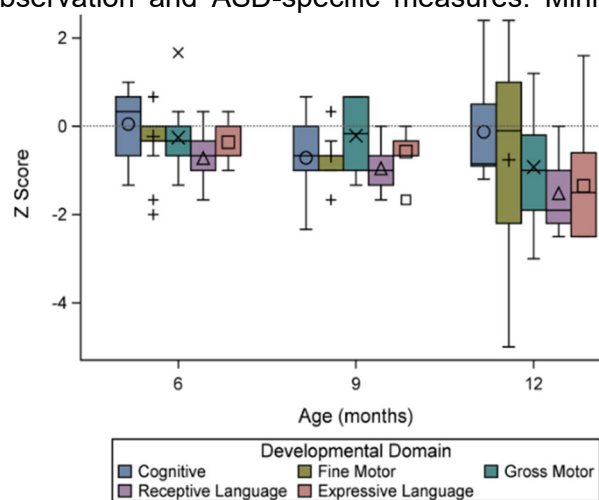


Figure 4. Bayley-4 subscale scores over time represented in z-scores where 0 is the average, -1 is one standard deviation (SD) below the mean, -2 is two SDs below the mean, etc.

HRI data from Prisma NICU patients and maintaining a REDCap database with demographic and morbidity data. Dail's Data Manager will oversee data transfer/storage.

ASD, Neurodevelopment, and Longitudinal Data Expertise: MPI Jessica Bradshaw (Psychology) has expertise ASD/NDI in infants/toddlers and in prospective, longitudinal designs and longitudinal modeling.^{28,85,110,111} Bradshaw maintains a lab staffed with 1 postdoc, 5 research coordinators, 1 psychologist, 2 recruitment coordinators, 5 graduate and 20 undergraduate students. Staff/students are trained in proposed data collection and processing procedures, 1 of whom will be dedicated to this project. 3 additional RAs will be hired specifically for this project (see *Budget Justification*) and Bradshaw's clinical staff (clinical psychology postdoc and social worker) will be available for RA training, case consultation, and family feedback when needed.

Physiological Data Science and Statistical Expertise: Co-I Christian O'Reilly (College of Engineering & Computing) is an electrical and biomedical engineer and neuroscientist with expertise in scientific computing, advanced statistical analyses, biosignal processing, and computational neuroscience with applications to neurodevelopment. He has established collaboration with the other investigators on this proposal (R01MH132925; Institutional grants).^{7,108,112,113} Co-I O'Reilly will supervise 1 graduate student to support the analyses proposed in this project. **Co-I John Richards** (Psychology) is a renowned developmental cognitive neuroscientist with methods expertise in heart rate indices, EEG, and MRI as well as content expertise in nervous system development in PT infants.^{17,114,115} Richards and Bradshaw have collaborated closely on studies of infant attention and physiology for the past 5 years (Co-I, R01MH132925; mentor K23MH120476) and has collaborated with Co-I O'Reilly on cortical source analyses in infancy.¹¹³

Study Design

This prospective, longitudinal study will enroll and follow 260 VPT infants. Minute-by-minute, continuous ANS monitoring will be provided during their entire NICU stay (admission duration range: 5-16 weeks). Following NICU discharge, comprehensive assessments will be administered at adjusted ages 12, 24, and 36 months.

Setting and Equipment. Participant enrollment will occur at Prisma Richland NICU, which is 2 miles away from USC campus where all study investigators are located. Standard-of-care electronic medical records (EMR), SpaceLabs cardiopulmonary monitors, and HeRO bedside monitors (equipped to store continuous R-R intervals, see *Letter of Support*), will be used for data acquisition. Central and peripheral temperatures will be recorded and saved using Grant Squirrel Dataloggers,⁶² (n=26 available for use). This setting and available equipment allows for enrollment of 12 infants per month, with measures collected for an average of 3 months per infant. Prisma NICU capacity (70 beds) and admission rate (~250/yr) far exceeds this planned enrollment rate, ensuring study feasibility. **A NICU Research Nurse (RN) will dedicate 50% effort to this project** and will: obtain consent from mothers within 24 hours of the infant's birth, educate clinical staff on study protocols throughout the study, place and monitor infant instrumentation and skin condition (described below), monitor data acquisition, and download/enter/save all collected data (weekly for EMR data, 2x/month for physiological data). Upon NICU discharge, the RN will be responsible for ensuring all medical and demographic data from EMR is entered into a REDCap and all physiological data is transferred to USC Project Manager. All data will be housed at USC on a secure server. Developmental assessments (adjusted age 12, 24 and 36 months) will be completed in the participants' homes by MPI Bradshaw's research staff (team of 2 per visit). In-home data collection has resulted in adequate retention, particularly for our underrepresented populations, but families may come to the lab if they prefer. Research Specialists will have clinical experience and training in assessments to be administered. MPI Bradshaw's lab also has 5 clinical psychology graduate students, 4 of whom are supported by training grants (NIH T32, F31, NSF GRFP), and who can contribute to clinical data collection and training as needed. Our team has conducted in-home data collection for >5 years and has designed custom hardware and software solutions. Our in-home data collection kits include 5 A/V recording systems, assessment kits (2 Bayley-4s, 3 CSBSs, 3 ADOSs), and 3 Surface Pros for completing REDCap surveys. Due to different home sizes and resources, we bring a variety of infant highchairs, adult chairs, and tables to ensure a standardized setup in any context. Data will be entered into REDCap and videos stored on secure servers.

Participants. Over the first two years of the study, we will enroll a total of 260 VPT infants born at Prisma Richland NICU between **24 and 31 6/7 weeks GA and at least 500 grams birthweight**. Infants born less than 24 weeks GA are fragile and frequently unstable. We consistently have used these enrollment GAs and weights for our studies with little attrition due to instability or mortality.^{61,62,116} Exclusion criteria include: major anatomical or cardiac defect known at birth, anatomical brain damage or known in utero infant asphyxia, and known genetic or chromosomal abnormality that is known to affect heart function or neurodevelopment (e.g., Down Syndrome, Fragile X). To account for attrition, we are over-enrolling by 30%. Infants may be withdrawn from the study by parents or attending physician during NICU hospitalization, or a very few may die due to complications of prematurity. Based on our pilot data and previous studies, enrolling 260 VPT infants in the NICU should yield at

least 200 retained infants who will complete the final 3-year assessment visit and enable sufficiently powered analyses for our aims. We expect to enroll $n=12/\text{month}$. Prisma NICU admits 200-250 VPTs annually (~50% males; 56% Black/African American, 35% White, 9% Hispanic). We will monitor enrollment demographics to ensure our sample is representative of NICU hospital admission and our analyses will account for sex as a biological variable. Given the NICU capacity and admissions rate, our enrollment success rate (90% for NICU families), and our conservative plan to over-enroll by 30%, we will easily reach our target enrollment within 2yrs.

Procedures

All study procedures are outlined in PHS Human Subjects and Clinical Trials Information form. Consent will be obtained from parents before their infant is born or after birth up to 24 hours of age. Data will be collected on the primary variables of abdominal and foot skin temperature, and ECG for each infant included in the study until NICU discharge, transfer, or death. Demographic data, maternal/prenatal history, family history of ASD, and PT-related morbidity data will be collected from the EMR and follow-up questionnaires to be used as covariates (race, maternal education, GA) and moderators (sex, morbidities) in analyses. Family history of ASD will be assessed via parent interview and ASD screeners for any biological siblings. Developmental assessments will begin at 12 months, following a virtual touch point at 6 months with parents. This virtual touch point will take place over the phone and will include a brief medical history since NICU discharge, re-explanation of the study and study procedures, demographic update (contact information, maternal education, income), and visit scheduling. Follow-up data will be collected in participant homes at adjusted ages 12, 24 and 36 months.

Primary Variables/Measures

Neonatal ANS Function: Thermal Gradients. Abdominal (central) and foot (peripheral) temperatures will be measured continuously with a thermistor (disposable skin sensor, 499B, Cincinnati Sub Zero, Cincinnati, OH) covered by Mepitac Silicone adhesive tape.⁶² These thermistors and tape are used for temperature monitoring in standard care. Thermistors record temperatures every minute using portable, high-precision dataloggers (Squirrel SQ data logger, Grant Instruments), resulting in a CPTd value every minute, as done in our previous studies.^{61,62} Abnormal thermal gradients are either negative (CPTd < 0° C; foot temperature warmer than abdomen) or high (CpTd > 2° C; abdominal temperature much warmer than foot). Per each 24-hour period, days with < 80% of valid temperature data will be dropped. Our previous studies suggest this method results in < 10% data loss. **Heart Rate Indices: Pre-Processing.** HRIs will be derived from continuous ECG recordings from standard of care cardiopulmonary and HeRO monitors within the NICU (see *Letter of Support*). Continuous, beat-by-beat R-R intervals will be recorded from birth to discharge via the FDA-approved device HeRO Solo from Medical Predictive Science Corporation (MPSC). This same system was used for preliminary data reported here.⁷ MPSC utilizes an FDA-approved, automated QRS detector to remove noise/artifacts and to detect R-waves; it has been validated against hand-annotated ECG waveforms with precision >99% and sensitivity >90%.^{117,118} Further validation is evident in its use in the largest randomized controlled trial conducted among VPTs, where displaying HRIs based on R-R intervals generated by the MPSC QRS detector to clinicians resulted in a 20% reduction in all-cause mortality and a 40% reduction in mortality after infection.^{119,120} In addition, our team has developed custom pre-processing pipelines that will further validate the MPSC-generated R-R intervals, as we have done in previous work.^{108,110,121} Specifically, isolated missed beats will be detected using standard non-parametric outlier analysis and corrected through interpolation. Segments with >5 contiguous missing/edited beats will be rejected from analysis. Windows of 1h with >10% discarded data will be inspected and dropped as needed. This procedure will be validated and adjusted using a large dataset of manually cleaned ECG available from previous and ongoing projects run in Bradshaw's lab. **HRIs.** The following HRIs will be extracted: HRV, RSA, NN sample entropy, mean NN, NN standard deviation, and NN skewness.⁶⁷ To support comparison, we will also include four ECG-derived features that we used in a previous study,¹⁰⁸ namely, the root mean square of the difference between successive NN (RMSSD), the ratio of the two main axis of the ellipse fitting the scatter plot of NN vs preceding NN (i.e., Poincaré plot; SD1SD2), the coefficient of variation (std/mean) of NN (CVNN), and the HRV triangular index (HTI) calculated as the number of NN values in the modal bin of the normalized 8-ms-bin NN histogram. Continuous wavelet transforms (CWT) will be used to estimate RSA (see Table 2b below). We will complement our analysis of the relationship between ASD and ANS by refining our assessment of sympathetic and parasympathetic branches of this system. Toward this goal, we will investigate the relationship between ASD and the Sympathetic and Parasympathetic Activity Indices (SAI and PAI),¹⁰⁶ the Cardiac Sympathetic Index (CSI), and the Cardiovagal Index (CVI).^{107,108} This extensive set of indices will be used specifically for ML prediction of ASD and NDI (Aim 1).

PT-associated Morbidities (birth to NICU discharge). Analyses for Aim 2 will use an established Morbidity Score previously adopted to study associations between ANS development and PT-related morbidities in the NICU.²⁰ This summative morbidity score quantifies the number of morbidities experienced at any time and

includes the following factors: incidence of culture positive infections, NEC diagnosis, presence of PDA requiring medical or surgical intervention, severe brain injury (ultrasound readings of Grade III or IV intraventricular hemorrhage and/or periventricular leukomalacia), diagnosis of BPD or chronic lung disease, need for mechanical ventilation, and ROP at $\geq$ stage I (Table 1).²⁰ Morbidities will be extracted from the EMR weekly.

Social Drivers of Health (SDOH) and Early Infant Experience. Given the significant effect of SDOH, and the prenatal and postnatal environment on infant/maternal care and the developing brain,^{122,123} the following measures will be accounted for in analyses. At consent (birth) and outcome (3 years), mothers will complete the 15-item, evidence-based, patient-centered tool PRAPARE (Protocol for Responding to and Assessing Patients' Assets, Risks, and Experiences).^{87,88} PRAPARE measures SDOH across four domains: 1) personal characteristics (e.g., race, ethnicity), 2) family and home, 3) money and resources, and 4) social-emotional health, and provides an overall Risk Tally Score that represents the number of social drivers of health risk (range: 0-22). PRAPARE is a recommended for assessing maternal unmet social needs as potential barriers to healthy prenatal and postnatal outcomes.¹²⁴ In addition, the following four measures (variables) will be collected on an ongoing basis to evaluate infant caregiving context and early life experience¹²⁵: maternal perceived stress (any moderate to high score > 20),^{126,127} depression (any score > 12)^{128,129}, infant health (number of days of infant post-discharge hospitalization), and infant developmental support (months of early intervention participation). Note that clinically significant/elevated levels of maternal stress/depression, as defined by the instruments, will be flagged immediately and caregiver will be contacted by the lab's clinical psychologist and/or MPI Bradshaw (licensed clinical psychologist) who will provide resources/referrals as needed (same protocol as ongoing R01).

ASD and NDI Outcome (36 Months): ASD Symptoms. The ADOS-2 will be administered at age 36 months adjusted and Calibrated Severity Scores (CSS)¹³⁰ for Social Affect (SA CSS) and Restricted and Repetitive Behavior (RRB CSS) will be used as dimensional measures of ASD outcomes in Aim 1-2. ADOSs will be administered by clinicians trained to research reliability and supervised by MPI Bradshaw and lab psychologist, with co-scoring to consensus occurring for at least 20% of the sample. CSS ranges from 1-10 and represents an excellent measure of ASD symptoms, relatively independent from IQ and language abilities,^{130,131} that has been shown by our group and others to have sufficient variability for biomarker prediction.^{132,133} For Aim 3, the Total CSS score will be used to compare derived clusters. **Neurodevelopmental Impairment.** Consensus is lacking on the definition of severe NDI outcome for PT populations.¹³⁴ But when assessing NDI, studies tend to use normative thresholds to define impairment (e.g., 1.5 or 2 standard deviations below the mean) for the following areas: motor, cognitive, language.^{15,134} However, this dichotomous approach fails to capture subtle challenges and nuanced profiles in VPT infants.¹³⁵ This study will use a dimensional measure of neurodevelopmental impairment as the NDI outcome for Aims 1-2. In line with other studies of neurodevelopmental outcome in VPTs,¹³⁶ the Bayley Scales of Infant and Toddler Development, 4th Edition (Bayley-4)¹³⁷ will be administered at adjusted age 36 months, and the Cognitive, Motor, and Language standard scores (mean=100, SD=15) will be used as continuous measures of impairment. These scales explain a sizable proportion of later cognitive and academic outcome variability and are invaluable for characterizing early development and guiding intervention efforts.¹³⁸

Emerging ASD and NDI (12-24 Months): ASD Symptomatology. Emerging ASD symptom profiles will use a direct measure of very early ASD symptoms, as has been used in our previous studies,^{7,18,28,29,52,85} at 12 and 24 months. Specifically, we will use the Systematic Observation Rating of Flags for ASD (SORF), an observational measure of ASD red flags based on the DSM-5 framework that will be scored from the Communication and Symbolic Behavior Scales-Behavior Sample (CSBS).¹³⁹ The SORF has been found to distinguish children with ASD from children with NDI (but no ASD) and typically developing children, and our study team has documented the sensitivity and specificity of this measure for differentiating 12-month-old infants later diagnosed with and without ASD.¹⁴⁰ In infants 16-24 months of age, the SORF has large effects and good discrimination (area under the curve=.84-.87).¹⁴⁰ For Aim 3, the SORF Social Communication Domain score and the SORF RRB Domain score will be used to characterize emerging ASD symptoms. **Neurodevelopmental Impairment.** NDI will be measured dimensionally using standard scores from all 5 subscales of the Bayley-4: Cognitive, Fine Motor, Gross Motor, Receptive Communication, and Expressive Language. Scores will be adjusted for infants' GA. These subscales will be used in cluster analyses described in Aim 3.

Data Management

A HIPAA-compliant, USC-based, REDCap database will be created to collect all data in this study. The NICU RN will review and enter neonatal demographic and clinical data. Direct import from EMR into REDCap is not feasible as an RN is required to harmonize EMR data across individual patients to be suitable for analysis. Per our recent 5-site R01, this protocol involves clinical reading and interpretation of multiple tests results with varying response options. Co-I Everhart will train the RN to reliability in this process and will oversee all data

harmonization procedures, as she has done for over 350 VPT infants in our multisite R01. The RN will download and store temperature data from dataloggers and ECG data from HeRO monitors every two weeks until the infant is discharged from the NICU. All data will be accessible over an encrypted and password-protected connection. The Data Manager at USC will clean temperature data and transfer data into SAS (9.4) datasets. Co-I O'Reilly's team will pre-process ECG data and compute HRIs (see above). Clinical data collected after NICU discharge will be stored on our secure server (videos) and entered in REDCap (assessment scores) following each study visit. Double data entry will be used to ensure data validity.

Data Analysis

Aim 1: Examine PT ANS dysfunction as a specific predictor of ASD symptoms at outcome. The analysis for Aim 1 will leverage ML by using a **support vector regression** from Scikit-Learn¹⁴¹ to implement non-linear regression models that predict symptoms of ASD (Social and RRB CSS from ADOS-2) and NDI (Cognitive, Motor, Language standard scores from Bayley-4) at outcome from a set of features describing the preterm ANS and its dysfunction. **Table 2a** lists the measures that will be considered for this analysis. For each measure, four primary features will be extracted: 1st, 50th, and 99th percentiles, and the inter-quartile range for the measured values across the NICU stay. The slope of the relationship between the median value of these measures (averaged per 24-h period) and GA will also be considered. We will also investigate cross-modal temporal associations by calculating the correlation

Table 2a. ANS Measure Descriptions for Aim 1 Machine Learning

Measures	Modality	Interpretation
% time CPTd < 0° C	Temperature	Peripheral body warmer than core body, indicative of ANS dysfunction or immaturity. ^{59,123}
% time CPTd > 2° C	Temperature	High core to peripheral temperature indicative of Sympathetic activity, seen with sepsis and stressor activity in PT infants. ^{89,124}
SAI	Heart Rate	Index of sympathetic activity
PAI	Heart Rate	Index of parasympathetic activity
CSI	Heart Rate	Index of sympathetic activity
CVI	Heart Rate	Index of parasympathetic activity
HRV	Heart Rate	Measured as the natural logarithm of RMSSD. Decreased values indicate ANS dysfunction and are characteristic among individuals with ASD. ¹²⁵
Mean NN	Heart Rate	Increased with overactive sympathetic and decreased parasympathetic or vagal tone. ¹⁶ Highly predictive features for likelihood of ASD according to previous study. ⁹⁹
NN standard deviation	Heart Rate	Spread of the NN distribution. Index both sympathetic and parasympathetic activity. ¹²⁶
NN skewness	Heart Rate	Transient HR decelerations (bradycardia) result in negative skewness (parasympathetic). Transient HR accelerations (tachycardia) result in positive skewness (sympathetic).
NN sample entropy	Heart Rate	Decreased values are thought to be associated with a decoupling of the ANS from external inputs (i.e., non-responsiveness, non-adaptability). ^{127,128}
RMSSD	Heart Rate	Parasympathetic activity; vagally mediated variation in heart rate. ¹²⁹
SD1SD2	Heart Rate	Reflect the balance between short-term vagal activity and sympathetic modulation. ¹²⁹
CVNN	Heart Rate	STD(NN) captures both low and high-frequency components of the heart-rate signals. High-frequencies variations are due to RSA (parasympathetic). The normalization of STD(NN) by MEAN(NN) makes CVNN more stable when comparing individuals or conditions with different baselines. ¹²⁹
HTI	Heart rate	Depressed HTI is a robust index of sympathovagal imbalance. ¹³⁰
RSA	Respiration & Heart Rate	Periodic slowing and speeding up of heart rate with the respiratory cycle. Higher RSA indexes more vagal flexibility, which promotes balanced parasympathetic and sympathetic influences to the heart.

Table 2b. RSA Calculation

RSA Calculation Method: Continuous Wavelet Transform (CWT)	
Step	Relevant References
1. Transform R-R (beat-by-beat) series into a sample-rate (10Hz) time series, then use "cwt" function of MATLAB. The output from cwt is all the wavelet-transformed IBI series , that is the IBI time series transformed by the wavelet filterbank.	Parameters used: "analytic Morlet", "amor", and 48 "voices per octave" (# of wavelets). CWT Analysis and RSA: Grossmann & Morlet, 1984; Cnockaert et al., 2008; Wachowiak et al., 2016
2. Use inverse CWT ("icwt") with frequency bands set to the min and max of respiration, determined by age-appropriate norms. The output of icwt is the RSA time series for the respiration frequency chosen for this age.	Respiration rate in infants: Gagliardi & Rusconi, 1997; O'Leary et al., 2015; Rusconi et al., 1994; Tveiten et al., 2016
3. To be consistent with established RSA calculation methods (PB-MPF), use log(var(icwt time series)) to calculate RSA .	Porges-Bohrer Moving Polynomial Filter: Porges, 1982; Porges & Bohrer, 1990
Advantages. Compared to other filters, the CWT approach results in higher resolution for frequencies, better drop-off, and is more modifiable.	

across time between all pairs of median measures (elevated per 1h-hour period) and Fisher Z-Transforming these correlations. We will add these correlations to the feature set if their intra/interparticipant variance ratio (estimated with bootstrapping) is statistically significant compared to chance-level assessed on a surrogate dataset obtained by randomizing the temporal order of the median measure time series. The correlation matrix between these features will be hierarchically clustered using the Seaborn *clustermap* function, and highly correlated features will be pruned to remove multicollinearity. Linearity of this relationship will be tested, and a power transform used if necessary. We will further evaluate the association of these features with demographic covariates (race, sex, maternal education, GA) and, if statistically significant, the effect of these covariates will be regressed out of the features.

Social Drivers of Health (SDOH) and Early Infant Experience. Further, we will test for significant associations between these features and relevant SDOH (PRAPARE Risk Tally Score), and early infant experience (maternal stress, maternal depression, infant health, infant intervention) variables. If these associations are statistically significant, the effect of these covariates on the features will be regressed out. We recognize the complex, interconnected, and multifaceted nature of demographics, SDOH, and early infant experience, all of which cannot be fully and meaningfully captured in a single score, while also acknowledging the possibly high number of influential variables and the collinearity therein, which raises significant analytic challenges.¹⁴² Thus, our analytic plan for Aims 1-2 will account for these summative variables, while also allowing us to further probe individual items in subsequent analyses.

We will then use permutation feature importance techniques developed in the context of explainable artificial intelligence and implemented in Scikit-Learn¹⁴¹ to identify which features contributed the most to this prediction. Since most of our features can be interpreted in a mechanistic way (e.g., see Table 2a), this analysis will allow us to confirm and refine our hypotheses on the role of ANS dysfunction in the emergence of ASD and NDI. All ML analyses will be conducted using nested subject-level-cross-validation (train: 70%; test: 15%; validation: 15%) to allow reliable and reproducible model fitting and hyperparameter tuning. The effect size for ANS as a predictor of outcomes at 3 years of age will be evaluated as model's R^2 values (i.e., coefficients of determination) and their statistical significance will be assessed by looking at whether their bootstrapped 95% confidence intervals overlap with 0. Since model assessment is cross-validated, negative R^2 values and confidence intervals overlapping with 0 are expected in the absence of a true relationship.

Aim 2: Evaluate preterm morbidities and sex as potential moderators in the association between PT ANS dysfunction and symptoms of ASD and NDI at outcome. While Aim 1 (above) leverages machine learning, the statistical analysis for this Aim 2 relies on a classical moderation modeling where the association between PT ANS dysfunction (independent variables; listed in the next paragraph) and symptoms of ASD and NDI at outcome (dependent variables) will be tested for **moderation by sex and a previously validated preterm morbidity index**,²⁰ as defined in the PT Morbidities section. The linearity of this relationship will be tested, and a power transform will be used if necessary. The correlation matrix between independent features will be analyzed and highly correlated features will be discarded to avoid multicollinearity issues. If a significant moderation effect is found for the morbidity index, we will conduct post hoc moderation analyses to determine which of its constituting factors (severe brain injury,¹⁴³ infection, or NEC¹⁴⁵, BPD^{75,146} or need for mechanical ventilation,¹⁴⁷ ROP,¹⁴⁹ and/or PDA) contribute to the moderation. Python (StatsModels Mediation class), which supports both mediation and moderation analyses, will be used.

We will use the same dependent variables as those used in Aim 1 (i.e., ADOS-2 CSS and Bayley-4 Scales). Our analyses will align with our preliminary investigations⁷ and will characterize ANS function during PT periods using 1) the proportion of days during which negative thermal gradients are present > 30% of the time, 2) the proportion of days during which CPTd > 2° C are present > 10% of the time, 3) IBI skewness (based on its key role in HRI predictors used in our previous work⁵³ and the work of others⁶⁷), and 4) a small set (≤ 3) of features that will be shown to have the highest predictive power in Aim 1. By using a combination of ANS measures that are determined based on preliminary data and that are highly predictive according to the analyses of Aim 1, we ensure that Aim 2 does not significantly depend on Aim 1. We will correct for multiple independent tests using the Bonferroni method. We will first perform these analyses using average values (e.g., rates and frequencies of ANS measures; no repeated measures). Then, we will add the effect of GA by considering 24-h averages for the independent variables and replace the fixed effect linear moderation by a mixed-effect one, including both the intercept and slope of the effect of GA. For this second analysis, we will test if GA needs to be transformed (e.g., power transform) prior to its use for linear modeling and evaluate the effects of potential covariates, including SDOH and early infant experience, as listed above.

Aim 3: Characterize and differentiate the emergence of ASD symptoms and NDI in VPT infants. We will characterize the development of VPT infants using direct measures of ASD symptoms and NDI administered at ages 12 and 24 months. Given the developmental heterogeneity observed in both infants with ASD and VPT infants, we will use both SORF subdomains (Social Communication and RRB Domain scores) to characterize ASD symptoms and all subtests of the Bayley-4 to characterize NDI (Cognitive, Fine Motor, Gross Motor, Receptive Communication, Expressive Communication). We will use a supervised clustering approach to identify subgroups of developmental trajectories that differ in terms of the emergence of ASD symptoms and NDI and that are predictive of the outcome at 36 months. To better capture the trajectories of autonomic ASD characteristics, and considering that we have two time points, we will consider the mean value across time points and the slope between time points for the two ASD and five NDI subscales, for a total of $2 \times 7 = 14$ features. The supervised approach will compute SHAP (Shapely Additive exPlanation) values associated with the features

used for an XGBoost regression predicting the ADOS CSS Total score at 36 months (5-fold cross validation).¹⁵⁰ The SHAP values will then be used as input to the unsupervised The DBSCAN algorithm for clustering.^{151,152} This approach will be implemented using Scikit -Learn¹⁴¹ and the SHAP Python package.¹⁵³ Although the nonlinear XGBOOST and DBSCAN algorithms are not vulnerable to multicollinearity issues, we will use a Principal Component Analysis to evaluate the proportion of variance captured by every component and reduce redundancy in the feature set, if necessary. The Silhouette Coefficient¹⁵⁴ will be computed to assess the quality of the clustering, the t-distributed Stochastic Neighbor Embedding approach¹⁵⁵ will be used to visualize the separability of the clusters, and the stability of the clustering will be assessed through bootstrapping.^{156,157}

By using a supervised clustering approach based on the outcome, we maximize our chances of obtaining stable clusters that group trajectories predictive of different outcomes. These clusters will, therefore, support the possibility of using early symptom trajectories to predict ASD diagnosis. Clusters will be visualized as seven subplots (one per ASD and NDI subscales) with the time points along the x-axis and the cluster mean values in y-axis with bootstrapped 95% confidence intervals as obtained from Python Seaborn *lineplot* function. We will test whether extracted clusters form statistically significant subgroups by testing differences in centroids using a Hotelling T-squared test. We will validate the clinical significance of extracted clusters by assessing whether subgroups differ based on 36-month ASD outcome (ADOS CSS) using t-tests, adjusted for multiple comparisons.

Sample Size and Statistical Power Considerations. **Aim 1** uses a ML-based regression approach to establish the statistical and clinical significance of the relationship between PT ANS abnormality and ASD/NDI outcomes. For this aim, the R^2 values for regression model fitting can be seen as the squared values of Pearson's coefficient of correlation which can be used for power analysis. With $N=200$, we can expect to detect a correlation of at least 0.20 with a 20% false negative rate (i.e., $\beta=0.2$) and a 5% false positive rate (i.e., $\alpha=0.05$). This coefficient of correlation corresponds to a 0.04 R^2 value, which can be interpreted as the percentage of variance explained. Once corrected for 7 independent tests (i.e., 7 subscales, Bonferroni correction), this R^2 increases to about 0.0625. Therefore, we consider this sample size sufficient, as effects that cannot explain reliably more than 6.25% of the total variance in the final diagnosis are probably of low clinical significance. For **Aim 2**, we used G*Power¹⁵⁸ to determine the minimal sample size required to have a 0.80 power (i.e., $\beta=0.2$) to test for moderation (two moderators: medical morbidity index and sex) in a model including six predictor main effects and their centered interaction with moderators, and using an $\alpha=0.01$ (i.e., 0.05 Bonferroni-corrected for testing 5 dependent variables). For a $f^2=0.25$ (considered a medium effect size), we will need $N=130$, which demonstrates that we have a large enough sample to run our main analysis and expect to obtain significant results if a clinically relevant effect exists. Our larger sample size ($N=200$) affords us extra maneuvering margin in case of smaller effects and for integrating co-variables (race, GA, and maternal education). For **Aim 3**, we will be able to detect significant differences in multivariate means between pairs of clusters each grouping 50 participants with $\beta=0.2$, $\alpha=0.05$, and 14 response variables if the effect size has at least a Mahalanobis distance (a multivariate generalization of the Cohen's d) of 0.92, implying that we will be able to establish that these clusters are different only for relatively large effect sizes. If significant effects cannot be detected with our sample, we will increase the statistical power of this test by reducing the number of clusters (hence increasing the number of participants per cluster) and reducing the number of response variables (i.e., feature selection). Similarly, we will be able to detect effects having a Cohen's d of at least 0.57 for differences in mean ASD symptoms (Total CSS) between pairs of clusters with $N=50$ each (power=0.8; $\alpha=0.05$; two-tailed).

Potential Challenges and Alternative Solutions. Many of the potential problems and alternative solutions have been uncovered in our previous studies, and effective solutions have always been implemented. As in the past, all potential issues will be discussed with study team members and immediately addressed as needed and as outlined in the MPI plan. All-investigator meetings will occur biannually to review progress, identify challenges, and generate solutions. **Enrollment:** Due to the ebb and flow of Prisma Richland NICU enrollment, we have adopted conservative enrollment quotas with a minimum of 12 infants per month (~50% of their annual VPT admissions), which allows us to over-enroll by 30% to account for attrition and loss to follow-up. For further backup, we have another NICU in the Prisma system in Greenville, SC that we can use as a second site for enrollment. Thus, we are confident we will meet our enrollment quotas for this study. **Follow-up Assessments.** We have built in a 6-month virtual visit (and incentive) to retain infants from discharge to the first follow-up ~1 year later. All follow-up assessments will be completed in the families' homes at their convenience to increase research accessibility. This strategy has greatly increased participation in our current studies, but in-lab visits are still available if desired. We will also access our Community Advisory Board as needed to overcome any barriers. All neonatal morbidities and severity will be recorded and could be used for Aim 2 in case the morbidity score shows limited variability. Later morbidities that may impede diagnostic determination (e.g., severe motor delays) will be evaluated in all-investigator meetings to determine course of action for inclusion in analyses.

PHS Human Subjects and Clinical Trials Information

OMB Number: 0925-0001

Expiration Date: 01/31/2026

Use of Human Specimens and/or Data

Does any of the proposed research in the application involve human specimens and/or data *

☒ Yes

☐ No

Provide an explanation for any use of human specimens and/or data not considered to be human subjects research.

Are Human Subjects Involved

☒ Yes

☐ No

Is the Project Exempt from Federal regulations?

☐ Yes

☒ No

Exemption Number

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

☐ 8

Other Requested Information

Human Subject Studies

Study#	Study Title	Clinical Trial?
<u>1</u>	Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants	No

Section 1 - Basic Information (Study 1)

1.1. Study Title *

Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants

1.2. Is this study exempt from Federal Regulations *

☐ Yes

☒ No

1.3. Exemption Number

☐ 1

☐ 2

☐ 3

☐ 4

☐ 5

☐ 6

☐ 7

☐ 8

1.4. Clinical Trial Questionnaire *

1.4.a. Does the study involve human participants?

☒ Yes

☐ No

1.4.b. Are the participants prospectively assigned to an intervention?

☐ Yes

☒ No

1.4.c. Is the study designed to evaluate the effect of the intervention on the participants?

☐ Yes

☒ No

1.4.d. Is the effect that will be evaluated a health-related biomedical or behavioral outcome?

☐ Yes

☒ No

1.5. Provide the ClinicalTrials.gov Identifier (e.g. NCT87654321) for this trial, if applicable

Section 2 - Study Population Characteristics (Study 1)

2.1. Conditions or Focus of Study

- Autonomic Nervous System
- Premature Infants
- Autism Spectrum Disorder
- Heart Rate
- Thermal Gradients

2.2. Eligibility Criteria

Premature infants born at Prisma Health Richlands NICU, between 24-31 6/7 weeks gestational age.

2.3. Age Limits	Min Age: 12 Hours	Max Age: 3 Years
2.3.a. Inclusion of Individuals Across the Lifespan	Dail_Inclusion_Across_Lifespan.pdf	
2.4. Inclusion of Women and Minorities	Dail_Inclusion_of_Women_and_Minorities_Final.pdf	
2.5. Recruitment and Retention Plan	Dail_Recruitment_and_Retention_Final.pdf	
2.6. Recruitment Status	Not yet recruiting	
2.7. Study Timeline	Dail_Timeline_Final.pdf	
2.8. Enrollment of First Participant	07/01/2025	Anticipated

Inclusion of Individuals Across the Lifespan

This study aims to examine preterm infants during their NICU hospitalization for autonomic nervous system (ANS) dysfunction and evaluate associations with emerging neurodevelopmental impairment and autism spectrum disorder (ASD) symptoms assessed at ages 12, 24, and 36 months. To meet this aim, 260 preterm infants, born at less than 32 weeks gestational age will be enrolled upon birth and admission to the NICU at Prisma Richland's NICU, in Columbia, South Carolina. After NICU discharge these preterm infants will be followed in their home by research staff from age 12-36 months. Intensive data collection of ANS function in the first months of life followed by repeated assessments of ASD symptomology and neurodevelopmental impairment (NDI) in very preterm infants using highly sensitive and specific measures across the first 3 years of life has enormous potential for understanding the impact of VPT birth on neurodevelopment.

The MPIs have decades of experience conducting research with infants and children, especially preterm infants and those children at an elevated likelihood for ASD. Thus, MPIs have finely-tuned data acquisition and assessment procedures in place to keep infants, children, and their families happy, comfortable, and engaged in the proposed research measures. Further, all research staff who will be interacting with preterm infants in the NICU will have experience as NICU nurses and are quite experienced with placing thermistors for temperature monitoring, as this is routinely done as part of standard nursing care. The NICU Research Nurse will be supervised by site PI/Co-Investigator Dr. Ravi Rao. All research staff who will be interacting with participating families and administering study protocols for follow up assessments will have a background in child development, complete knowledge of the research protocol, and will be trained and closely supervised by Dr. Bradshaw (MPI).

Assessment procedures will be adapted or expanded upon as needed for the proposed study. As such, data collection procedures will be family-friendly and developmentally appropriate with survey questionnaire and interview completion, as well as direct assessment, scheduled to be both efficient and flexible based on each family's need. The length of direct assessment generally increases as the child ages, from 1-1.5 hours at age 12 months to 2-3 hours at age 24 and 36 months. These later timepoints can be completed in two study visits. Each visit will allow ample time for breaks, naps, feeding, etc., to ensure participant comfort and reduce participant fatigue. Data will be primarily collected at each family's home for the family's convenience. All assessments will be scheduled to be efficient but flexible per family preference by offering accommodability in scheduling (e.g., evenings and weekends).

Inclusion of Women and Minorities

INCLUSION OF WOMEN

A total of 260 very preterm infants will be enrolled in this study, who are preterm infants born at less than 32 weeks gestational age. Target ratios for gender enrollment are 50% female for infants. We will analyze gender as a biological variable in Aim 2. Additionally, while not the primary participant in this project, the biological mothers of each infant will participate; thus 260 women will participate.

INCLUSION OF MINORITIES

Our targeted infant enrollment will reflect the ethnic and racial demographics for the Prisma Richland's NICU, in Columbia, SC. Total expected sample should be approximately 64% African American infants, 34% White infants, and 2% Hispanic, with a goal of 50% males and 50% females, as projected from enrollment demographics in our last study for the same inclusion criteria (NIH/NINR: R01017872, 2018-2024). In a pilot study with a 20-infant subset of this R01 to conduct follow up with preterm infant until 12 months of age, consented infants for follow up are 74% African American and 26% White. We will stringently aim to recruit a participant sample across groups that is representative of the local population in the Prisma Richland's NICU, and as such, we aim for no more than 40% of our sample to be White. However, we recognize that barriers exist with under-represented minorities participating in research and we have built in procedures to mitigate some of these barriers. For example, all follow-up assessments will be done in the participant's homes, if they agree, to mitigate any travel or inconvenience for the participants' families associated with lab assessments. We have found, in our pilot work and previous studies, that bringing follow-up equipment and personnel to the parents' homes has greatly reduced attrition and improved satisfaction with follow-up. We will continue to work with under-represented minority participants to improve our procedures, increase convenience, and lessen any research burden to enable increased research recruitment and participation. To this end, we will also employ participation of the USC Carolina Autism and Neurodevelopment (CAN) Research Center's Community Advisory Board (CAB). Our CAB is comprised of a racial/ethnically diverse group of community parents, providers, and state officials who have lived and professional experience with high-needs families and who have a vested interest in helping research succeed. Of note, 50% of our CAB are racial/ethnic minorities who are traditionally underrepresented in research. At least 2x/year and as-needed, we will report on our inclusion of minorities and CAB will provide feedback on our progress, protocols, and procedures in recruiting and retaining this population.

Recruitment & Retention Plan

This proposed, longitudinal study aims to enroll 260 very preterm (VPT) infants delivered at the Prisma Richlands Hospital, Columbia, South Carolina, and admitted to the NICU. Infants will be discharged home to their families after they reach near term gestational age (34-37 weeks) and are stable with the heart rate, respirations, feeding and growing. Our conservative attrition estimates, based on pilot data that evidence 10-20% for PTs, allows for 23% attrition, therefore we anticipate that approximately 200 children will be assessed at all follow-up time points at ages 12, 24, and 36 months. Prisma Richlands Hospital NICU admits around 220 very preterm infants annually, and Dr. Rao (site PI) is confident that we will easily meet enrollment goals.

We will enroll infants up to 24 hours in age, and we have enjoyed a consent rate of VPT infants for a similar study since 2018 of 90%. In this study, which just completed enrollment of 145 VPT infants at Prisma Richland's NICU, infants were placed on the same instrumentation. Less than 1% of parents removed their infants from study procedures before NICU discharge. While infants are in the NICU on study instrumentation, the dedicated Research Nurse rounds at least 2-3 times per week and talks to parents of study infants to elicit any concerns with the study. The Research Nurse also speaks to the bedside nurse for the same purpose and to ensure that instrumentation is working properly with no issues. The average VPT infant may have a NICU hospitalization anywhere from 3-10 weeks duration.

Once a discharge plan is initiated (typically ~3 days prior to discharge date), a warm handoff from the Research Nurse to the Psychology Research Specialists who will be conducting in-home follow-up is initiated. This hand-off would ideally take place in-person but may take place virtually through Zoom or phone with the Research Nurse at the bedside with the parents. The study team will connect to the family by reaching out via phone and email when the infant reaches 6 months of age to collect demographic information, provide a study incentive for survey completion, and schedule the 12-month in-home study visit. From our pilot study, we know that up to 20% of families may not follow through and keep their first in person follow up appointment; therefore, we have planned for attrition to ensure that we complete follow up assessments on 200 VPT infants to meet the aims of this study. Our analytic procedures are designed to handle missing data (Aim 3) and rely most heavily on NICU measures and 36-month outcome measures (Aims 1-2) and so participants will continue to be contacted for study visits up to 36 months of age or until they request to withdraw from the study. Written communication (emails, texts, letters) to schedule study visits will include easily accessible methods for withdrawing from the study for those participants who have not responded to scheduling requests. Participating families are likely at an elevated risk of parental stress associated with caring for a preterm infant with the associated medical morbidities; thus, we will implement strategies that will not only ensure the collection of high-quality data but enhance participants' willingness to participate and remain in the study. As the recruitment, retention, and assessment of our participants will require a great deal of effort and expertise, we will train a number of project staff to address these challenges. Additionally, we will facilitate the development of a personal relationship with our recruited families by assigning two research staff to conduct each assessment. To enhance continuity, one Research Specialist will be designated to scheduling all participants and we aim to have at least one Research Specialist who has previously assessed the family to be assigned to all future study visits for that family. This continuity will facilitate the personal relationships that are core to the successful retention of participants over the course of longitudinal studies.

The study will also capitalize on the MPI's previous and existing protocols that support participant retention. As such, data collection procedures will be "family-friendly" with survey questionnaires, interview completion, and direct assessment scheduled to be both efficient and flexible based on each family's need. Data collection will occur in-home, which we hope will remove the potential burden associated with travel and assessment in an unfamiliar space. Furthermore, Dr. Bradshaw's lab has established several free resources to support parents participating in research. These include a mindfulness-based stress reduction group and a respite care program that support families by: (1) providing an opportunity for parents to meet, connect, and share resources with each other, (2) delivering free respite care services at least one Saturday a month, and (3) creating a space for children to play and socialize with their peers and trained research staff.

In addition, to support retention efforts, we have the support of the USC Carolina Autism and Neurodevelopment (CAN) Research Center's **Community Advisory Board (CAB)**. This CAB is comprised of

a racial/ethnically diverse group of community parents, providers, and state officials who have lived and professional experience working with high-needs families and who have a vested interest in helping research succeed. As detailed in CAN's *letter of support*, the CAB is supportive of this study and our study will be on CAB's agenda at least 2x/year to discuss challenges with recruitment and retention, and propose solutions. In addition, CAB members will be available for consultation as needed. Retention will be a highly prioritized discussion topic in these meetings as we garner input for how to accurately and ethically convey the value of this study to families, discuss ways to modify our procedures and protocols to be more family-friendly and culturally relevant, and strategize solutions to potential barriers to participation for these families.

Dr. Bradshaw's lab staffs a Clinical Research and Outreach Coordinator (Elizabeth Dixon, LCSW) who will monitor recruitment and retention rates closely, as she has done the past year for our ongoing NIH-funded longitudinal study. Examples of efforts to bolster retention include (1) soliciting continued feedback on the informed consent process and research protocols through participation satisfaction surveys and (2) promoting community engagement by organizing information sessions, workshops, and outreach events for families and providers within the autism community. Given the collective success of the MPIs and investigative team in recruiting preterm infants, and our established recruitment/retention infrastructure at Prisma Richlands NICU, we do not anticipate issues meeting our recruitment goals. Further, Dail and Bradshaw have over 20 years and 10 years, respectively, of experience conducting highly similar research protocols with great success.

STUDY TIMELINE

Progress and benchmarks for this project will be assessed weekly by MPIs Dail & Bradshaw. In general activities will occur as follows: enrollment Y1-2, data collection and processing Y1-5, data analysis Y2-5, and publications Y1, 3-5. Specifically relevant to the work described in this study, we anticipate the following timeline:

- **Research Staff Training.** We anticipate having four research staff for MPI Dail (1) and MPI Bradshaw (3) already-trained research coordinators ready to begin the project upon its start date. In the event that this is not the case, staff will be hired immediately with training ensuing right away. MPI Dail has used the NICU study methods for the last 5 years in an ongoing study at the proposed study site, Prisma Richland NICU, and has existing operating procedures, training manuals and nursing education. We anticipate the research nurse for the enrollment site to be trained and ready to begin project enrollment, as MPI Dail, Co-Is Rao (site PI) and Everhart, have existing studies in the Prisma Richland NICU with multiple research nurses participating in their studies. MPI Bradshaw has used the neurodevelopmental assessments in her ongoing studies and maintains current experimental protocols and training manuals. These documents will be updated to align with the proposed study methods. A third research coordinator will be hired by MPI Bradshaw in Year 2 to assist with data collection. This individual will begin training immediately. While training is likely an ongoing effort, the primary effort will take place in the first 1-3 years. MPI Dail will oversee training for enrollment and data collection in the Prisma Richland NICU in Y1- 2 and MPI Bradshaw will oversee training for data collection for follow-up participant assessments.
- **Participant Enrollment.** We anticipate beginning infant enrollment within the first 2 months of the start of the study, once NICU nurses and physicians have been given education as to the study methods. The research nurse and Dr. Rao will consent parents for their infant to participate, and the research nurse will enroll and collect data on each infant. Infants will remain on data collection until they are discharged from the NICU, which can range from 3-12 weeks, depending on their gestational age at birth. We will also over-enroll by 60 infants for a total of 260. All infants should be enrolled by the end of year 2.
- **Data Collection.** NICU data collection will occur throughout Years 1-2, until each infant is discharged or transferred from the NICU or dies. Following the 6-month remote touch point that will occur between the participants and Bradshaw's research team, neurodevelopmental assessments will occur at ages 12, 24, and 36 months in the infant's home over Y2-Y5. These assessments will be completed on 200 children (we anticipate loss to follow-up of up to 60 infants) by Y5.
- **Data Processing** will be an ongoing process with reliability standards established for data entry and verification as one of the first steps. The study team includes a data manager who will process all NICU data per established methods used in MPI Dail's current R01. The REDCap database will be developed prior to the study start, with entry points for NICU morbidity data and all neurodevelopmental follow-up data. Co-I O'Reilly will hire and train a graduate research assistant in Y1 to process all ECG data obtained in the NICU and assist with analysis of data from Y2-5.
- **Data Analysis** will take place Y2-5 and primarily be led by Co-I O'Reilly, with MPI Bradshaw and Dail's assistance. Co-I Richards will also be consulted throughout Y2-5 concerning heart rate data interpretation.
- **Manuscript preparation and submission** will begin with a protocol paper in Y1, and publications of early preliminary data in Y3 of the award.

Activities	Year 1	Year 2	Year 3	Year 4	Year 5
Enrollment and Staff Training	→				
Data Collection and Processing	→	→	→	→	→
Data Analyses		→	→	→	→
Publications	→	→	→	→	→

2.9. Inclusion Enrollment Reports

IER ID#	Enrollment Location Type	Enrollment Location
<u>Study 1, IER 1</u>	Domestic	

Inclusion Enrollment Report 1

1. Inclusion Enrollment Report Title* : Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants
2. Using an Existing Dataset or Resource* : ☐ Yes ☒ No
3. Enrollment Location Type* : ☒ Domestic ☐ Foreign
4. Enrollment Country(ies): USA: UNITED STATES
5. Enrollment Location(s):
6. Comments: Inclusion is based on demographics of our ending R01 study that enrolled infants using same criteria at same enrollment site.

Planned

Racial Categories	Ethnic Categories				Total
	Not Hispanic or Latino		Hispanic or Latino		
	Female	Male	Female	Male	
American Indian/ Alaska Native	0	0	0	0	0
Asian	2	2	0	0	4
Native Hawaiian or Other Pacific Islander	0	0	0	0	0
Black or African American	83	83	0	0	166
White	37	37	8	8	90
More than One Race	0	0	0	0	0
Total	122	122	8	8	260

Cumulative (Actual)

Racial Categories	Ethnic Categories									Total
	Not Hispanic or Latino			Hispanic or Latino			Unknown/Not Reported Ethnicity			
	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	Female	Male	Unknown/ Not Reported	
American Indian/ Alaska Native	0	0	0	0	0	0	0	0	0	0
Asian	0	0	0	0	0	0	0	0	0	0
Native Hawaiian or Other Pacific Islander	0	0	0	0	0	0	0	0	0	0
Black or African American	0	0	0	0	0	0	0	0	0	0
White	0	0	0	0	0	0	0	0	0	0
More than One Race	0	0	0	0	0	0	0	0	0	0
Unknown or Not Reported	0	0	0	0	0	0	0	0	0	0
Total	0	0	0	0	0	0	0	0	0	0

Section 3 - Protection and Monitoring Plans (Study 1)

3.1. Protection of Human Subjects

Dail_3.1_Protection_Human_Subjects.pdf

3.2. Is this a multi-site study that will use the same protocol to conduct non-exempt human subjects research at more than one domestic site?

☐ Yes ☒ No ☐ N/A

Single IRB plan attachment

3.3. Data and Safety Monitoring Plan

3.4. Will a Data and Safety Monitoring Board be appointed for this study?

☐ Yes ☒ No

3.5. Overall structure of the study team

3.1 Protection of Human Subjects

1. Risks to Human Subjects

A. Human Subjects Involvement, Characteristics, and Design

The proposed study is a 5-year prospective, longitudinal study focused on evaluating autonomic nervous system (ANS) dysfunction in the neonatal period and associations with neurodevelopmental outcomes, specifically autism spectrum disorder (ASD) symptoms and other neurodevelopmental impairments (NDI), by 3 years of age in 260 very preterm infants born at Prisma Richland Neonatal Intensive Care Unit (NICU) in Columbia, South Carolina. Very preterm (VPT) infants are those born before reaching 32 weeks gestational age (GA) or 31 6/7 days. In this study, we will include VPT infants born after reaching 24 weeks GA, because infants born between 22-24 weeks GA are many times too unstable and too fragile to include in study measures. This study will have a minimal birthweight requirement of 500 grams, because those preterms born less than 500 grams have skin too fragile and are often too unstable to enroll.

Specifically, the study aims to evaluate complementary measures of preterm ANS dysfunction (heart rate and thermal gradients) as predictors of ASD and NDI, and as an ASD-specific mechanism. Further, this study aims to examine whether and how sex and neonatal morbidities (e.g., severe brain injury, infection, bronchopulmonary dysplasia, retinopathy of prematurity, patent ductus arteriosus) moderate the association between ANS dysfunction and ASD/NDI outcomes. Because it is critical to consider the role of social drivers of health (SDOH), we will evaluate demographic variables, SDOH, maternal stress, and early infant experience (e.g., intervention history) in the analytical models. We will also comprehensively document emerging ASD and NDI features across the first 3 years of life for VPTs who are at increased likelihood for ASD and NDI.

As previously detailed in the Research Strategy, parental consent will be sought while mother is a patient in the Prisma Richland Labor & Delivery unit in non-active labor for her infant/s to participate in the study. Research nurses can also obtain consent for infant participation from the mother until the infant/s is 24 hours of age and after being admitted to the NICU. The first five days of life for VPTs is a critical period during which the infant's ANS system is undergoing a significant transition and medical stabilization is reached. In order to capture this critical period of time, infants will be enrolled in the first 24 hours after birth. This recruitment/enrollment strategy has been used in our previous studies at 6 different hospitals, including in our last study at Prisma Richland NICU with great success and a > 90% consent rate. We will over enroll by 30% (N=260 infants) to ensure 3-year longitudinal follow-up for the target sample size of 200 infants. This is based on a very conservative attrition/loss to follow-up estimate evidenced by our attrition rate of <10% in the majority of our existing infant longitudinal studies, and rate of up to 20% in our current pilot study of VPTs, that follow infants from birth into early childhood.

Inclusion Criteria are: Born at Prisma Richland NICU between 24-31 6/7 weeks GA (determined by obstetrical dating in electronic medical record), enrolled within 24 hours of birth, and have a birthweight of 500 grams or more. **Exclusion Criteria** will include: major anatomical or cardiac defect known at birth, anatomical brain damage or known in utero infant asphyxia, or known genetic or chromosomal abnormality that is known to affect heart function or neurodevelopment (e.g., Down Syndrome, Fragile X). Multiple births (e.g., twins) will not be excluded, but multiple birth will be considered in analyses.

B. Study Procedures, Materials, and Potential Risks

Procedures, materials, and potential risks can be separated by those occurring while the infant participants are in the NICU and those that occur after discharge.

NICU Participation Procedures & Materials: Once the VPT infant is enrolled in the study and assigned a de-identified study number, the infant will be assigned a study ID and record in REDCap in a dedicated study database. The Research Nurse, dedicated to the study at Prisma Richland NICU, will place a thermistor on the infant's abdomen and a separate thermistor on the sole of the infant's foot. These two thermistors will be attached to a Grant Squirrel Datalogger. The logger will record temperatures every minute until removed. T A standard of care SpaceLabs cardio-pulmonary monitor will be connected to a HeRO Solo monitor from Medical

Predictive Science Corporation (MPSC) and attached to the infant with standard ECG leads, per routine medical management in this NICU. Temperature and ECG data will be downloaded by the Research Nurse approximately every two weeks and stored locally and on a backup hard drive until the infant is discharged home, transferred to another unit/hospital or dies. Upon discharge, transfer, or death, data files will be sent via Prisma hospital's secure file transfer system to the Research Coordinator at USC or placed directly on our secure USC server via VPN. Our server is backed up weekly.

The Research Nurse will review the infant's electronic medical record (EMR) weekly during the NICU hospitalization to enter infant medical morbidity data into the REDCap database. In our experience, morbidity and other clinical information that is entered into the EMR system by medical staff is not always consistent, clear, or in a format that is appropriate for data analysis. Our current studies suggest that direct upload from the EMR into our REDCap database is not a feasible method of data acquisition and can compromise data accuracy and validity. Our procedure, in which the Research Nurse evaluates clinical/morbidity data, makes interpretations, and enters medical morbidity data into the database, is essential for data harmonization. Doing this weekly also allows us to ask the medical staff any clarifying questions for data purposes. Once the infant is ready to leave the NICU due to discharge home orders, transfer to another unit or hospital, or death, all instruments will be removed on the day of departure and reason for NICU discharge will be recorded. In cases of death, data will be kept in the REDCap database as deaths during the study is reported to the NIH on the RPPR. Neonatal death is rare; but may occur due to severe morbid conditions such as sepsis, lung disease, or intracranial hemorrhage with extreme prematurity due to risk of extreme prematurity. For all infants that are discharge or transferred from this NICU, a warm handoff between the Research Nurse and follow-up Research Specialists (who will conduct post-discharge in-home follow-up) will occur virtually or in-person as soon as a discharge plan is initiated. All contact information for follow-up, including back-up contact information, will be confirmed at this time.

Potential Risks: VPT infants are intermittently unstable, and have high risks for infection, cardiac or respiratory compromise, or even death due to being born prematurely. This study causes no potential for risk above normal risks experienced by premature infants, and there is no more than minimal risk anticipated in this study. The thermistors placed on infants for measurement of body temperature are the same as are used in standard of care and routinely used in this NICU for temperature measurement. As with standard of care, infants are examined for any skin breakdown due to attaching thermistors and probes, and any breakdown is noted in the EMR. Placement of thermistors or ECG leads are rotated every 8-24 hours depending on NICU protocol at the time and depending on skin integrity. No additional testing or monitoring is being done above standard of care for the study participants while in the NICU. Although there is a potential risk for breach of confidentiality with the research data, the anticipated risk is minimal, and every effort will be made to safeguard participants' confidentiality. We have used these same procedures with over 350 very preterm infants in 5 NICU sites without incident.

Neurodevelopmental Follow-Up Visits Procedures and Materials: Mothers will be contacted when the infant reaches 6 months corrected age to collect demographic information and post-discharge medical information via REDCap survey or over the phone. Caregivers will receive \$40 incentive payment for time and effort to complete this virtual survey and have their infant continue in the study. Study procedures will be reiterated, and the first in-home visit (corrected age 12 months) will be scheduled. Subsequent study visits will occur at infant ages 12, 24, and 36 months and caregivers will receive \$50 incentive payment for time and effort for these visits. Several sources of research materials will be collected at study visits that will occur age 12, 24, and 36 months, including (1) developmental testing of the child; (2) behavioral ratings from examiners; (3) questionnaires from caregivers; (4) video recordings. Study visits are video recorded for offline behavioral coding and reliability assessments. Only core research staff on the study's Institutional Review Board (IRB) who have completed ethics and compliance training will have access to identifiable information. The collection of identifiable information is necessary to maintain contact with participants throughout this longitudinal study and to set up repeated assessments.

Potential Risks: There is no more than minimal risk anticipated in this study. Although there is a potential risk for breach of confidentiality with the research data, the anticipated risk is minimal, and every effort will be made to safeguard participants' confidentiality. Additionally, there is minimal risk anticipated for the collection of

behavioral data, although participants may experience mild fatigue or fussiness. The length of direct child assessment generally increases as the child ages, from 1-1.5 hours (12-month time point) to 2-3 hours at the 24- and 36-month timepoint. Longer assessments can be completed across two study visits. All study visits will be scheduled at each family's convenience (e.g., weekends, evenings) and frequent breaks (e.g., naps, feedings) will be allotted as needed. VPT infants are at an elevated risk of neurodevelopmental impairment (NDI) and ASD, but there is the chance that families will be unsuspecting of the occurrence of NDI or ASD. The MPI (Bradshaw) is a licensed clinical psychologist and will be alerted on a weekly basis to infants with possible NDIs, defined by scoring 2 standard deviations below the mean in one developmental domain, 1.5 standard deviation below the mean in 2 domains, and/or clinical impressions. After review of testing results, the family will be contacted by Dr. Bradshaw to describe the results and gather more information and/or refer to our 0-3 state agency (BabyNet) for further evaluation and possibly intervention. Regardless of developmental testing results, all families who report developmental concerns on questionnaires or during any study visits will be contacted to determine if a referral is warranted. In addition, we collect measures of maternal perceived stress and depression. These measures are reviewed immediately upon completion by study staff and clinically significant/elevated levels of maternal stress/depression, as defined by the instruments, will be flagged immediately and parents will be contacted by the lab's clinical psychologist and/or MPI Bradshaw (licensed clinical psychologist) who will provide resources/referrals as needed (same protocol as ongoing R01). The PI will be alerted immediately (including during the study visit if the questionnaire is completed during the visit) if the parent endorses the critical item that reflects potential danger/harm to self or others and appropriate safety precautions and referrals will take place when necessary. The PI has extensive experience with families in need of acute care. This referral process is currently being used in Bradshaw's active R01 and will be supported by her team of clinical psychologists/trainees (project psychologist/postdoc, advanced doctoral students).

2. Adequacy of Protection Against Risks

A. Informed Consent and Assent

In accordance with policy 45 CFR 46.408, informed consent will be obtained from at least one parent of all infants prior to data collection. Trained research staff will review the consent form with parents, allowing time for the parents to review the consent and ask any questions. Parents who choose to proceed will sign a paper copy of the consent form, which will be retained in a secure office space in a locked cabinet. Parents will also receive a paper copy of the consent form for their records. A copy of the consent will be placed in the infant's EMR at Prisma Richlands NICU, consistent with IRB-approved study procedures of our current grants.

B. Protections Against Risk

The proposed research protocol will be approved by the IRB at Prisma Richland Hospital and the University of South Carolina prior to the attainment of data, and approval will be obtained for any subsequent modifications. Participant confidentiality will be safeguarded by assigning all subjects a unique identification number, not linked with any identifiable information, which will be associated with all raw data collected. Identifiable information will be kept separate from all raw data during transport and will be stored separately. All personal, identifying information will be stored in a locked filing cabinet in a locked office or on a secure server in a password-protected database. Only core research staff on the study's IRB who have completed research ethics and compliance training will have access to identifiable information.

Data collection for all NICU days overseen by the Co-I Dr. Ravi Rao. A NICU nurse, employed as a Research Nurse will work with Dr. Rao (Site PI), Dr. Everhart (Co-Investigator) and MPI Dail to ensure valid and reliable data collection. The Research Nurse, overseen by the Site PI, will work with each study infant's bedside nurse to minimize any risk related to data collection. In the event of skin breakdown, the bedside nurse will notify the research nurse and Site PI, who will in turn notify MPI Dail. As the Site PI is the Medical Director of the Prisma Richland's NICU, he will prescribe any treatment needed for altered skin integrity and/or remove the infant from study procedures if necessary. The Site PI or MPI Dail will correspond with the study infant's parents for notification of skin breakdown if treatment or removal from study procedures is necessary. Our team has used these procedures over the last 20 years and no infants have needed to be removed from the study due to study procedures.

Data collection for the 12, 24, and 36-month timepoints will be overseen by MPI Bradshaw. Research Specialists, who have at least a bachelor's degree in psychology, child development, or a related field, will work with the PI and project psychologist to be trained on data collection with sensitive populations. Data collection will occur in the family's home to maximize participant comfort, thus minimizing the potential of stress from travel and anxiety over unfamiliar surroundings. Furthermore, to mitigate the potential for fatigue and fussiness during data collection, all study visits will be scheduled to allot for frequent breaks during the assessment. Visits will also be scheduled to occur at each family's convenience to accommodate individual schedules, including work, nap times, and intervention services.

C. Vulnerable Subjects

As noted above, the rationale for including infants between birth at 32 weeks GA or less and 36 months of age is to assess ANS dysfunction in relationship to neurodevelopment as a mechanism and predictor for early symptoms of ASD and NDI. MPIs have over 20 years (Dail) conducting research with VPT infants and over 15 years of experience (Bradshaw) conducting research with infants and young children with ASD. The proposed study follows highly similar procedures to the MPIs' past and current studies on development, attention, and physiology, which have occurred without incident. Given the success of previous study procedures and feedback, we anticipate no more than minimal risk for this study given the study team's experience with children.

3. Potential Benefits of the Proposed Research to Research Participants and Others

There are no direct benefits from this study for participants and their families. The primary anticipated benefit is close monitoring of infant development by licensed clinical psychologists. Parents in the MPI's (Bradshaw) current, ongoing studies have expressed deep appreciation for this monitoring and feedback. Parents will receive diagnostic information and a report if their child develops ASD over the course of the study and parents may view prospective surveillance and access to a diagnostic report as beneficial. We expect this study to advance research on the etiology of ASD and our hope is that families will find it satisfying to contribute to increasing the scientific knowledge in this domain and the potential for advancement in early intervention options. The MPIs make valiant efforts to ensure recent findings are available to the families that participate in their studies through their website, newsletter, and Facebook page, providing both links to publications and family-friendly summaries. Although participants are not expected to personally benefit from the results of this study, the projected potential benefits to the autism community outweigh the minimal risks associated with the study.

4. Importance of the Knowledge to be Gained

Increased knowledge about the autonomic dysfunction in VPT infants and how it relates to the emergence of ASD symptoms will increase knowledge of the etiology of ASD and help lead to targeted interventions and support for families that can optimize outcomes. The results from the proposed study are expected to yield high-impact results that promote a more comprehensive understanding of the role of autonomic function in the daily lives of children with ASD.

Section 4 - Protocol Synopsis (Study 1)**4.1. Study Design****4.1.a. Detailed Description****4.1.b. Primary Purpose****4.1.c. Interventions**

Type	Name	Description
------	------	-------------

4.1.d. Study Phase

Is this an NIH-defined Phase III Clinical Trial? ☐ Yes ☐ No

4.1.e. Intervention Model

4.1.f. Masking ☐ Yes ☐ No

☐ Participant ☐ Care Provider ☐ Investigator ☐ Outcomes Assessor

4.1.g. Allocation**4.2. Outcome Measures**

Type	Name	Time Frame	Brief Description
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4.3. Statistical Design and Power**4.4. Subject Participation Duration**

4.5. Will the study use an FDA-regulated intervention? ☐ Yes ☐ No

4.5.a. If yes, describe the availability of Investigational Product (IP) and Investigational New Drug (IND)/ Investigational Device Exemption (IDE) status

4.6. Is this an applicable clinical trial under FDAAA? ☐ Yes ☐ No

4.7. Dissemination Plan

Delayed Onset Studies

Delayed Onset Study#	Study Title	Anticipated Clinical Trial?	Justification
The form does not have any delayed onset studies			

Multiple PD/PI Leadership Plan

MPIs Drs. Dail and Bradshaw will provide oversight of the entire research project and will be responsible for the implementation of the Study Methods to accomplish Study Aims. Drs. Dail and Bradshaw come together with nearly 4 years of active research collaboration and complementary expertise, with Dr. Dail ensuring all methods, data collection, and data storage are completed for enrollment and data collection from the NICU site and Dr. Bradshaw ensuring all methods, data collection, and data storage are completed for 6-month check-in and all 12-36-month follow up assessments.

Dr. Dail is the Health Sciences Endowed Professor at the University of South Carolina, College of Nursing (CON). Dr. Dail is a Registered Nurse, with over 30 years of clinical practice experience in a variety of NICUs. She has conducted research with preterm infants for over 20 years and specifically in the proposed study site NICU for the last 6 years. Dr. Dail will oversee the Project Coordinator at USC, CON who will work from the Dail Research Lab to maintain study IRB regulatory procedures and documentation, develop standard operating procedures for enrollment and data collection in the NICU, develop the study REDCap database which will be maintained on a USC server, maintenance of instrumentation (Grant Dataloggers and HeRO monitor) and supplies, training of the NICU site Research Nurse, arranging study team meetings, and NIH reports. Dr. Dail will also oversee study recruitment, which will occur in the NICU, working with Dr. Rao and the Prisma (enrollment site) Research Nurse for enrollment activities. Dr. Dail will oversee the Data Manager, Christopher Bush, who will clean and maintain data for all enrolled infants until their NICU discharge.

Dr. Bradshaw is an Associate Professor in the Department of Psychology with the College of Arts and Sciences at USC. Dr. Bradshaw is a licensed clinical psychologist with expertise in infants and children with neurodevelopmental disorders and ASD. She has years of research and clinical experience in all developmental measures to be used in this study and is a research reliable ADOS examiner. Dr. Bradshaw will primarily be responsible for training and oversight of Psychology Research Specialists who will: conduct all follow-up data collection and help maintain study IRB regulatory procedures and documentation, follow and refine (as-needed) standard operating procedures for neurodevelopmental follow-up data collection, add developmental measures to the project REDCap database. Dr. Bradshaw will be responsible for clinical follow-up and referral (e.g., referral to BabyNet for Early Intervention) for all infants enrolled in this study and will oversee participant retention.

Additionally, Drs. Dail and Bradshaw will maintain communication with all investigators and key personnel through monthly meetings. Drs. Dail and Bradshaw will work collaboratively with Dr. O'Reilly to analyze data and interpret results, as they have done together for the past 2 years on a funded pilot study. Drs. Dail and Bradshaw have ongoing collaborative studies and will continue to communicate weekly through email, virtual meetings, or in-person meetings. Drs. Dail and Bradshaw have had ongoing communication for the past two years, including 2x/month meetings, to conduct an internally funded pilot study and design the proposed study, equally drafting the proposed grant application.

Publication Authorship: Authorship will be determined by Drs. Dail and Bradshaw, and with Co-Investigators based on the scientific contribution to the manuscript and writing of the manuscript. All manuscripts and presentations must be approved by both Drs. Dail & Bradshaw prior to dissemination. The MPIs will negotiate author order with study team members prior to drafting any manuscripts. This authorship plan has been effective in producing 2 conference presentations and 1 published manuscript, with another publication in-prep.

Conflict Resolution: If a conflict develops that cannot be solved amongst the MPIs, the Department Chair of Biobehavioral Health & Nursing Science and the Department Chair of Psychology, will serve as arbitrators to enable conflict resolution. Currently, both Chairs are successful NIH-funded investigators with notable experience in MPI studies and team science.

Change in PI Location: If one of the PIs moves to a new institution or cannot carry out their duties, a suitable PI will be recruited to take over the duties of Dr. Dail (neonatal enrollment and data collection) or Dr. Bradshaw (neurological follow up until age 3) and is subject to NIH and USC approval.

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12 June 2024

Institutional Letter of Agreement

Ms. Kristie Marshall
Sponsored Programs Administrator
Sponsored Awards Management
Office of the Vice President for Research
University of South Carolina
1600 Hampton Street, Suite 414
Columbia, SC 29208-0001

Dear Ms. Marshall:

The Pediatrix Medical Group of South Carolina will provide the necessary resources to assure that the scope of work for our Investigator participation in the NIH grant titled: *Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants*.

Institutional PI: Raviprasad Rao, MD

Pediatrix Medical Group of South Carolina, requests the budget support in the amount \$79,330 for the five-year period of
Start/End Dates – April 1, 2025 - March 31, 2030, for personnel cost.

PMG Services, Inc.
DUNS Number: 791497431
EIN Number: 65-0271219
UEI: HGG1PLL88294

Best Regards,

Reese H. Clark, MD

Reese H. Clark, M.D.
Sr. Vice President, Clinical Research

Authorized Institutional Official Signature

Date:



**Letter of Intent to Establish a Consortium
Agreement Prisma Health–Midlands as Subrecipient**

Prisma Health–Midlands PI: Ravi Rao, MD Prime Applicant Organization: University of South Carolina
 Department: Neonatal Intensive Care Prime Applicant PI: Robin Dail, PhD
 Sponsor Name: National Institutes of Health
 Proposal Title: Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants
 Proposed Project Period: 4/1/25 to 3/31/30 Proposed Project Amount: \$ 176,491.00

Attached Subcontract Proposal Documents

- ☒ Scope of Work (required)
☒ Budget and Budget Justification (required)
☒ Biosketches and Other Support for all Key Personnel – in agency required format (if applicable)
☒ Federally negotiated F&A Costs Rate Agreement (required for Federal sponsors)

Project/Performance Site Location

Organization Name: Prisma Health–Midlands County: Richland
 Street 1: Seven Medical Park Country: USA
 Street 2: UEI: KQ3QX9AHQY76
 City: Columbia Congressional District: SC-006
 State: South Carolina DUNS #: 009035853
 Zip/Postal Code: 29203-6808 EIN #: 58-2296052

Year 1 Subcontract Budget		Total Subcontract Budget	
Direct Costs:	\$ 52,309.00	Direct Costs:	\$ 130,773.00
F&A Costs:	\$ 18,287.00	F&A Costs:	\$ 45,718.00
Total Costs:	\$ 70,596.00	Total Costs:	\$ 176,491.00

Are Animals Applicable to this Proposed Project? ☐ Yes ☒ No
 Are Human Subjects Applicable to this Proposed Project? ☒ Yes ☐ No

The appropriate programmatic and administrative personnel of each institution involved in this grant application are aware of the pertinent Federal regulations and policies and are prepared to establish written inter-organizational agreements that will ensure compliance with all such policies.

Prisma Health–Upstate’s Conflict of Interest in Research Policy complies with the U.S. Department of Health and Human Services, “Objectivity in Research,” 42 CFR Part 50 and “Responsible Prospective Contractors,” 45 CFR Part 94 requirements.

This letter certifies that all project personnel are compliant with Prisma Health–Upstate’s Financial Conflict of Interest Policy.

Signature of Authorized Prisma Health–Midlands Personnel

04/11/2024

Date



OFFICE OF THE DEAN

June 12, 2024

Robin Dail PhD, RN, FAAN
Health Sciences Endowed Professor
University of South Carolina, College of Nursing
1601 Greene Street
Columbia, SC 29208

Dear Drs. Dail and Bradshaw:

As Dean of the College of Nursing, University of South Carolina, I am delighted to offer my full support of your study "Neonatal Autonomic Nervous System Dysfunction as a Predictor of Autism Spectrum Disorder in Preterm Infants" to be submitted to the National Institute of Health as an R01 application. I understand this study will examine body temperature, heart rate, respirations and morbidities in 260 preterm infants at Prisma Richland NICU in comparison to Autism Spectrum Disorder and Neurodevelopmental Impairments at 3 years of age.

I am fully supportive of your role as Principal Investigator at 9-14% effort, over the 5 years. Our extensive resources in the College of Nursing and University will be ample to assist you in successfully executing this proposed study. We have the personnel, technology and administrative resources to support you as you lead this study, with MPI Dr. Jessica Bradshaw from College of Arts & Sciences who also has ample resources from her home unit. Our college of nursing has received over \$18 million in extramural funding in the past three years, with the majority of funding received by the National Institutes of Health and other federal sources, and excellent infrastructure support to assist you in this important work.

In a region where premature infant outcomes are less than optimal, your study finding will provide important context and application in the clinical arena. I wish you the very best with your application, and you have my and the College of Nursing's full support with this initiative.

Sincerely,

A handwritten signature in cursive script that reads "Jeannette O. Andrews".

Jeannette O. Andrews, PhD, RN, FAAN
Dean and Helen Gurley Wolford Professor of Nursing
College of Nursing
University of South Carolina
803-777-3862
j.andrews@sc.edu



Carolina Autism and Neurodevelopment
(CAN) Research Center

June 27, 2024

To the NIH Study Section,

As the Steering Director of the Carolina Autism and Neurodevelopmental (CAN) Research Center at the University of South Carolina (USC), **I offer our enthusiastic support of MPIs Dr. Robin Dail (USC, Nursing) and Dr. Jessica Bradshaw (USC, Psychology)**. Their proposal, titled "*Neonatal autonomic nervous system dysfunction as a predictor of autism spectrum disorder in preterm infants*" reflects the collaborative, interdisciplinary philosophy on which CAN is built in support of bridging basic and clinical science in order to advance our understanding of ASD and improve the lives of autistic individuals.

To provide some context, CAN is a research center of excellence focused on autism and neurodevelopmental disorders that was founded in 2018 with almost \$5 million in support from the USC. Our mission is to increase scientific understanding that will improve health and well-being of families and individuals with neurodevelopmental disorders (NDDs), including autism, intellectual disability, learning disorders, and known genetic syndromes (e.g., fragile X, Down syndrome). This initiative has been extremely successful to date with 23 core faculty and an additional 70+ researchers across USC. Currently and collectively, CAN researchers hold over ~\$55M in active extramural funding from NIH, NSF, and other external sources.

I'd like to emphasize the support of the CAN Community Advisory Board for this project. The goal for our Board is to hear input about the alignment of (1) our research and community outreach activities with (2) the needs and values of the community. **Our Board is a very engaged group with insightful and valuable feedback** at all stages of research (i.e., grant applications, project setup, data collection, interpretation and impact of findings, dissemination). **Our Board is comprised of 8-10 members, including autistic adult self-advocates, family advocates, practitioners, and policymakers.** CAN co-organizers (myself and Dr. Bradshaw) attend all meetings, and we invite 1-2 researchers (including graduate students and postdoc trainees) to join each meeting to solicit feedback. We ensure a diverse membership across a range of experiences that include early development, intervention services, profound autism, and/or high-support needs. We meet at least 4 times per year as a group, as well as *ad hoc* consultations, and compensate all Board members for their time (\$50 per meeting or consultation).

Relevant to your project:

- We are committed to participatory research and engaging stakeholders as partners in research. The Board is intrigued by your proposal and impressed with your commitment to see a diverse, representative group of very preterm infants.
- The Board is committed to being a partner in this study in the following ways:
 - We will include it on the agenda at least 2x/year throughout the duration of the grant in order for the MPIs to raise questions, challenges, and barriers, and garner feedback from the board.
 - Individual members are also willing to have ad-hoc consultations with the research team as needed.
 - The PI will present annual progress reports and the Board will provide feedback on progress.

1800 Gervais Street • Columbia, South Carolina 29201 • can@sc.edu • <https://go.sc.edu/can>
An Equal Opportunity Institution

DIGITAL DOCUMENT

- Importantly, we are continually growing our Board. By Winter 2024, we expect to recruit another 2-4 members, which will include more parents and more autistic adults.
- CAN and the Board will help connect your research team to family resources across the state to ensure you meet your retention goals. We know your population often has limited resources and we would like to support your efforts to retain >90% of those enrolled by helping identify family-oriented strategies to combat attrition.

In addition, the CAN Research Center has other valuable resources that will support community dissemination of your research. Your findings can be disseminated through our **Neurodevelopment Research Registry**, currently comprised of 2,399 families with ~54.5% of families that have a child or individual with autism or a neurodevelopmental disorder. Our CAN Program Coordinator will be available to assist with dissemination via the registry, as well as advertise within our quarterly family newsletters, community tabling events, and larger community outreach events. As one example, we have begun annual partnership with the South Carolina Department of Disabilities and Special Needs for an event called AutismConnect – the first year (April 2024) was a tremendous success, with over 500 attendees from across the state, including families and providers. We expect over 700 attendees in 2025, including over 300 parents of autistic children. Lastly, you and your research team are encouraged to solicit additional study feedback during our **weekly CAN science meetings** (virtual) and **meet with top autism scholars** across the nation that come to USC to give talks 2-3 times a semester.

Your project is a true opportunity to help understand the etiology of ASD in very preterm infants and the role of ANS dysregulation, which will help identify infants with early needs for intervention. CAN and our Community Advisory Board recognize the urgent need for this work. We are thrilled to support this project!

Sincerely,

Caitlin M. Hudac, Ph.D.

Associate Professor, University of South Carolina

Department of Psychology



Email: chudac@mailbox.sc.edu

Director, Brain Research Across Development Lab, www.b-radlab.com

Steering Director, Carolina Autism and Neurodevelopment (CAN) Research Center, go.sc.edu/can

Co-Organizer of CAN Community Advisory Board

Executive Committee, Institute for Mind and Brain, go.sc.edu/imb



Medical Predictive Science Corporation
Letter of Support

June 19th, 2024

Dear Drs. Dail and Bradshaw,

We are pleased to collaborate with you on your upcoming grant submission, entitled, “Neonatal autonomic nervous dysfunction as a predictor of autism spectrum disorder in preterm infants.” Our company, Medical Predictive Science Corporation (MPSC), worked with you on your last study, “Central & peripheral body temperature in VLBW preterm infants during the neonatal period: Relationship to neonatal infection and necrotizing enterocolitis” to provide 16 HeRO monitors for purchase which were used in 5 NICUs in North and South Carolina to measure heart rate characteristic (HRC) scores to predict infection. For the proposed study, our engineers will develop and validate new functionality for your HeRO Solo monitors to measure and store R-R Intervals, in addition to the HRCs measured and stored in your last study. These data will be stored on a USB drive attached to the HeRO solo monitor. We presented a quote of \$22,500 for this re-engineering service to University of South Carolina (USC) which has been approved by USC. This upgrade will occur onsite at the College of Nursing at USC for all 16 HeRO Solo monitors you currently have in storage, by December 2024 so that the HeRO Solo monitors can be used in the Prisma Midlands NICU for your study, proposed to begin in April 2025. We have worked with yourself and Dr. Kayla Everhart to service and maintain functionality on your HeRO monitors between 2018 and 2023 and will continue to support your needs related to using the HeRO Solo for the proposed study.

This study presents exciting opportunities to examine dysregulated autonomic nervous system function in relationship to the onset of autism spectrum disorder in infants born preterm, using heart rate indices measured through HeRO solo monitors and standard cardiopulmonary monitoring. We look forward to this collaboration and the results from your study.

Sincerely,

A handwritten signature in black ink, appearing to read 'Will King'.

Will King, CEO



June 20, 2024

Dear Drs. Dail & Bradshaw,

I am excited to hear about your proposed study, "Neonatal autonomic nervous dysfunction as a predictor of autism spectrum disorder in preterm infants.," which will be the second study that you have done in our Prisma Health NICU since 2018. I understand that Dr. Ravi Rao will be the Site Principal Investigator for our unit as with your current study, and that you will hire a Research Nurse for 20 hours per week to conduct consent, study procedures and collect data. I also understand that you will enroll 260 preterm infants less than 32 weeks gestational age, and collect foot and abdominal temperatures, heart rate and respirations over the NICU stay. This study will enroll about 12 infants per month for just over 2 years.

When the grant has been funded, I will work with you, Drs. Everhart and Rao to hire a Research Nurse who has NICU experience for the study through Prisma Health. We have successfully hired research nurses for yours and Dr. Everhart's most recent study.

I am looking forward to collaborating with Drs. Rao and Everhart to implement the study in our NICU and know that we can support study enrollment and procedures. We look forward to hearing of successful funding for this study and will be delighted to continue your research studies in our Prisma Health NICU.

Thank You,

A handwritten signature in black ink that reads "Beth Green, NM".

Beth Green, RN, BSN

Nurse Manager for NICU
Prisma Health Richland Hospital
5 Medical Park
Columbia, SC 29203
803-434-7122 (office)



June 12, 2024

Dear Drs. Dail and Bradshaw,

As the NICU Medical Director at Prisma Health Richland, I am pleased to provide a letter of support for your novel proposal on the links between Autonomic Nervous System (ANS) dysfunction and the emergence of Autism Spectrum Disorder (ASD) in children born preterm. We are happy to continue the collaboration between your team and our NICU team, as we have done since 2018. This will be the first time we are able to follow continuous measures of ANS dysfunction, including body temperature and heart rate indices from the HeRO monitors attached to cardiopulmonary monitors throughout infants' NICU hospitalization. We understand the enrolled infants will be followed until they reach 3 years of age. The proposed study promises innovation and high impact in a longitudinal format.

Your scientific agenda is well aligned with my own clinical and scientific interests, which lie in optimizing health outcomes for infants born preterm. Preterm infants experience disrupted autonomic system function (thermoregulation, respiratory, and cardiac problems) and are at a significant risk for poor neurodevelopmental outcomes. The proposed focus on ANS dysfunction in very preterm infants will greatly add to our understanding of neurodevelopmental outcomes for infants born preterm and importantly, to the increased incidence of ASD in preterm infants.

I am committed to supporting this project through active recruitment of very preterm infants and oversight of the NICU research nurse, as well as medical expertise, as needed in neonatal development. In particular, the volume of preterm infants born at our hospital less than 32 weeks gestational age is sufficient to fulfill your recruitment goal of N = 260 in just over 2 years or an average of 12 infants per month.

The aims of your proposal are innovative, and results will be influential in the fields of neonatology, psychology, and ASD. I strongly support the present application and look forward to being a Co-Investigator and actively involved in this project.

Sincerely,

A handwritten signature in black ink that reads "Ravi Rao". The signature is written in a cursive, flowing style with a horizontal line underneath the name.

Ravi Rao, MD

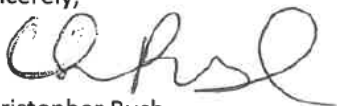
June 25, 2024

Dear Drs. Dail & Bradshaw,

It is with great enthusiasm that I agree to receive and clean data for your proposed study, investigating autonomic dysfunction in the NICU with neurodevelopmental follow-up assessments in children born preterm. I have enjoyed working with you Dr. Dail on your current NIH funded study since 2019, to receive body temperature and heart rate scores on 362 preterm infants. In your current study which has completed enrollment, I have received individual infant datafiles in Excel from your research coordinator by direct export in their study folders, have then cleaned data, and placed in study folders for the study statistician. I understand that I will do the same activities for your proposed study, with 260 enrolled infants over the first 2 ½ years of the study. Compensation for this activity will be \$500 a month, payable as a consultant to University of South Carolina College of Nursing, with successful grant funding.

I look forward to working with you on this innovative and exciting project and understand the grant is to be submitted by June 26, 2024, with the possibility of funding by April 2025.

Sincerely,



Christopher Bush

Resource Sharing Plan

There are no model organisms or other research tools generated from the proposed research to share.

Sharing Human Data via the NIMH Data Archives (NDA)

For this project, data will be collected from preterm infants, born 24-32 weeks gestation, from birth to 36 months of age. There is a high likelihood that some, but not all, children will be diagnosed with autism spectrum disorder. Data collected as part of this project will be disseminated in an open and timely manner in line with the objective of PA-21-201 to elucidate the etiology, epidemiology, diagnosis, and optimal means of service delivery in relation to autism spectrum disorders. Thus, raw data collected in support of this project will be shared biannually through the NIMH Data Archive (NDA), in accordance with the mandate set forth PA-21-201, to fully comply with the NIH initiative to accelerate and advance autism research. Shared datasets will include demographic, physiological (temperatures and R-R intervals), clinical, and behavioral data collected from infants. Demographic information will be collected to generate a GUID for each research participant, in accordance with other studies funded through NIMH. During the informed consent process, research staff will engage in specific discussions and provide documentation that participants' data will be shared for research purposes through the NIH NDA repository and allow for participants to provide or deny consent for such data to be shared. When sharing de-identified data, every effort will be taken to safeguard the privacy of participants. The MPIs will also share data upon request, under their auspices, through data-sharing agreements with other researchers. Moreover, the data collected and analyzed throughout this project will be disseminated at local, national, and international conferences, as well as published in high-impact journals. All final peer-reviewed manuscripts will be submitted to PubMed Central, adhering to NIH Public Access Policy guidelines.

Trained research staff will ensure the collection of high-value data in the NICU and following hospital discharge through the 36-month assessment. Two full-time Project Managers (one for MPI Dail and one for MPI Bradshaw) will dedicate 40% of their time to managing a quality-controlled database, which will bolster the feasibility of sharing accurate, comprehensive, and informative data. We have designed the budget to account for our data sharing efforts and have protocols in place to collect needed information.

A **Data Management and Sharing Plan** is located in **Other Documents** of this proposal.

NIH Generated message:

The Other Plan(s) attachment included with the application is not evaluated during the peer review process but will be evaluated prior to a funding decision. Although part of the official submission, the attachment is maintained as a separate document in eRA Commons viewable by authorized users and is not part of this assembled application.